

École Polytechnique Fédérale de Lausanne

Bachelor Thesis in Mathematics

From Rough to Lifted

A Comparative Study of Markovian Approximations
of Rough Volatility Models

with a Proposed Two-Factor Extension of the Lifted Heston Model

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Spring 2026

Abstract

Rough volatility models, in which the spot variance is driven by a fractional Brownian motion with Hurst index $H \in (0, 1/2)$, have become a standard tool in derivatives pricing since their introduction by Gatheral, Jaisson, and Rosenbaum [2018] and El Euch and Rosenbaum [2019]. Their key empirical advantage is reproducing the power-law explosion of the at-the-money (ATM) skew at short maturities. However, these models are non-Markovian and non-semimartingale, which makes simulation, hedging, and calibration substantially more difficult than for classical stochastic volatility models. Abi Jaber [2019] introduced the lifted Heston model, a finite-dimensional Markovian approximation of the rough Heston model that recovers the rough behaviour as the number of factors goes to infinity, while remaining tractable through a system of n ordinary Riccati ODEs. More recent empirical work [Cont and Das, 2023, Abi Jaber and Li, 2025] has cast doubt on whether rough models genuinely outperform their Markovian counterparts on real option data.

This thesis pursues three goals. **First**, we implement the lifted Heston model, reproduce its convergence to the rough Heston as $n \rightarrow \infty$, and study the role of the geometric ratio r_n controlling the spread of mean-reversion speeds. **Second**, we compare the lifted Heston with the affine Bergomi (aBergomi) model of Zhu, Loeper, Chen, and Langrené [2021], which provides an analogous Markovian approximation of the rough Bergomi model, on SPX implied volatility surfaces. **Third**, we propose an original extension: a two-factor lifted Heston model that combines a rough component (a sum of OU factors approximating a power-law kernel) with a slow exponential component, designed to reproduce the global short-versus-long maturity skew structure that single-kernel rough models cannot capture, as identified by Abi Jaber and Li [2025]. We treat in detail three sub-cases: independent Brownian motions, fully correlated Brownian motions, and partially correlated Brownian motions. We show that the case of independent Brownian motions retains the affine structure and admits a tractable Riccati system, while partial correlation introduces bilinear cross terms requiring more careful analysis.

Keywords: rough volatility, lifted Heston, affine Bergomi, Riccati equations, multi-factor stochastic volatility, ATM skew, SPX calibration.

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Chapter 1

Introduction

1.1 The implied volatility surface and its puzzles

A fundamental object in equity derivatives pricing is the implied volatility surface $\sigma_{\text{BS}}(K, T)$, defined for a strike K and maturity T as the unique value of constant volatility that, when plugged into the Black–Scholes formula, recovers the observed market price of a European call option. If the market followed Black–Scholes assumptions, the implied volatility surface would be flat. Empirically, it is not. Two stylised facts dominate the literature on equity options:

1. For each fixed maturity T , the function $K \mapsto \sigma_{\text{BS}}(K, T)$ is decreasing and convex; this is the volatility smile (or smirk, in equity markets).
2. The slope of the smile at the money, called the ATM skew,

$$\psi(T) := \left. \frac{\partial \sigma_{\text{BS}}}{\partial k} \right|_{k=0}, \quad k = \log(K/F_T), \quad (1.1)$$

is large in absolute value for short maturities and decays as $T \rightarrow \infty$. Empirically, $|\psi(T)|$ behaves approximately as a power law $\sim T^{-\alpha}$ with $\alpha \in (0, 1/2)$ for SPX options [Bayer, Friz, and Gatheral, 2016, Gatheral, Jaisson, and Rosenbaum, 2018].

The classical Heston model [Heston, 1993] is a one-factor stochastic volatility model that produces a smile, but cannot reproduce the short-maturity skew explosion: in classical Heston, $\psi(T) \rightarrow \text{constant}$ as $T \rightarrow 0$, whereas market data shows $|\psi(T)| \rightarrow \infty$. This empirical failure motivated the introduction of rough volatility models in 2014–2018.

1.2 Rough volatility and its critique

The rough Heston model of El Euch and Rosenbaum [2019] replaces the Brownian-driven CIR variance dynamics with a fractional Volterra dynamics of Hurst index $H \in (0, 1/2)$. The variance process becomes

$$V_t = V_0 + \frac{1}{\Gamma(H + 1/2)} \int_0^t (t-s)^{H-1/2} \lambda(\theta - V_s) ds + \frac{\nu}{\Gamma(H + 1/2)} \int_0^t (t-s)^{H-1/2} \sqrt{V_s} dW_s. \quad (1.2)$$

The fractional kernel $K(t) = t^{H-1/2}$ is singular at the origin when $H < 1/2$. This singularity is the mathematical source of the non-Markovian and non-semimartingale nature of the model. It is also responsible, at first order in the vol-of-vol, for the power-law ATM skew

$$\psi(T) \sim CT^{H-1/2}, \quad T \downarrow 0, \quad (1.3)$$

which matches the classical empirical observation that short-maturity equity skews are very steep.

Rough volatility is therefore motivated by two different empirical observations. The first is a statistical observation under the physical measure $\mathbb{P}$: realised volatility time series appear rough, with empirical estimates often close to $\hat{H} \approx 0.1$. The second is a pricing observation under the risk-neutral measure $\mathbb{Q}$: implied volatility surfaces exhibit a steep short-maturity ATM skew. These two observations are related but logically distinct. A model calibrated under $\mathbb{Q}$ to option prices does not automatically explain the statistical roughness of realised volatility under $\mathbb{P}$, and conversely a good $\mathbb{P}$ -model of volatility need not be the best $\mathbb{Q}$ -model for option pricing. The mathematical distinction between these two viewpoints will be made precise in section 2.1, where we explain how the physical measure $\mathbb{P}$ and the risk-neutral measure $\mathbb{Q}$ are related by Girsanov's theorem, and which structural properties of volatility models are preserved by this change of measure.

The statistical evidence for roughness has been challenged by Cont and Das [2023]. Their starting point is that spot volatility is not directly observed. What is observed is a realised volatility or realised variance estimator computed from high-frequency returns. For a trading day $[t, t + 1]$ sampled at times $t = t_0 < t_1 < \dots < t_n = t + 1$, realised variance is

$$RV_t^{(n)} := \sum_{k=1}^n (\log S_{t_k} - \log S_{t_{k-1}})^2. \quad (1.4)$$

In the ideal continuous semimartingale setting, $RV_t^{(n)}$ converges in probability to the integrated variance $\int_t^{t+1} \sigma_s^2 ds$ as the sampling frequency increases. In finite samples, however, $RV_t^{(n)}$ contains a non-negligible estimation error:

$$RV_t^{(n)} = \int_t^{t+1} \sigma_s^2 ds + \varepsilon_t^{(n)}. \quad (1.5)$$

The key point of Cont and Das [2023] is that this error is not an innocuous independent noise term. Because it is produced by high-frequency sampling, microstructure effects, and path-dependent estimation error, it can itself affect the roughness properties of the observed realised volatility signal.

Their methodology is deliberately non-parametric. Instead of assuming from the start that the volatility follows a fractional Brownian motion, they try to estimate the roughness directly from the observed path. The basic idea is the following. Suppose that a process X behaves locally like a process with Hurst index H , meaning that increments over a small time step Δ typically have size

$$|X_{t+\Delta} - X_t| \sim \Delta^H. \quad (1.6)$$

Then the p -th power of one increment is of order Δ^{pH} . On a fixed time interval, there are about $1/\Delta$ such increments, so the empirical p -variation behaves heuristically like

$$\sum_k |X_{t_{k+1}} - X_{t_k}|^p \approx \frac{1}{\Delta} \Delta^{pH} = \Delta^{pH-1}. \quad (1.7)$$

Therefore:

- if $pH > 1$, the sum tends to 0 as $\Delta \rightarrow 0$;
- if $pH < 1$, the sum explodes;
- if $pH = 1$, the sum remains of order one.

The critical value is thus

$$p^* = \frac{1}{H}, \quad \text{or equivalently} \quad H = \frac{1}{p^*}. \quad (1.8)$$

This is the intuition behind the p -variation roughness estimator. One observes a discretised path, computes sums of the form

$$V_p^\Delta(X) := \sum_k |X_{t_{k+1}} - X_{t_k}|^p, \quad (1.9)$$

for different values of p , and looks for the critical exponent p^* at which the behaviour changes from explosion to convergence. The estimated roughness is then

$$\hat{H} = \frac{1}{\hat{p}}. \quad (1.10)$$

For example, Brownian motion has $H = 1/2$, so the critical exponent is $p^* = 2$: this is the usual quadratic variation. A rougher process with $H = 0.1$ would have critical exponent $p^* = 10$.

Cont and Das refine this idea using a normalized version of p -variation, which is more stable on discrete data. But the interpretation remains the same: they are not fitting a full volatility model; they are asking which value of H is suggested by the scaling of the observed path. Their numerical experiments show that this method recovers the correct Hurst index on simulated fractional Brownian motion, but gives a much smaller apparent Hurst index when applied to realised volatility generated from ordinary diffusion-type stochastic volatility models. Hence a realised volatility series may look rough even when the latent spot volatility is a semimartingale with Brownian regularity. In particular, observing $\hat{H} \approx 0.1$ in realised volatility does not by itself prove that the underlying spot volatility is genuinely non-semimartingale.

This critique should be read together with Section 4 of [Abi Jaber \[2019\]](#). Abi Jaber performs a complementary experiment in the lifted Heston model. The point of the experiment is not that the lifted Heston model is genuinely rough. On the contrary, for every finite number of factors n , the lifted Heston variance process is finite-dimensional, Markovian, and a semimartingale. Therefore its true infinitesimal path regularity is Brownian-like:

$$H_{\text{true}} = \frac{1}{2}. \quad (1.11)$$

This must be distinguished from the parameter $H = 0.1$ used in the simulation. In the lifted Heston model, $H = 0.1$ is an input parameter used to choose the weights and mean-reversion speeds $(c_i^n, x_i^n)_{i=1}^n$ so that, when $n \rightarrow \infty$, the lifted model converges to a rough Heston model with Hurst index 0.1. For fixed n , however, the model remains a Markovian semimartingale, so its true pathwise regularity is still $1/2$.

Abi Jaber simulates the lifted volatility process with $n = 20$ factors and input parameter $H = 0.1$, then applies the standard roughness estimators used in the rough volatility literature. At daily time scales, the estimator returns an apparent Hurst index close to

$$\hat{H} \approx 0.19. \quad (1.12)$$

Thus the simulated path looks statistically similar to a rough volatility path with Hurst index around 0.19, even though it was generated by a finite-dimensional Markovian semimartingale. The number 0.19 is therefore not the true mathematical roughness of the model; it is the apparent roughness detected over the observed range of time scales.

This interpretation is confirmed by the intraday experiment in the same paper. For time scales larger than roughly ten minutes, the lifted Heston path still produces rough-looking estimates. Only at sufficiently fine time scales does the estimator start to recognise the semimartingale nature of the process and move back towards $1/2$. This is exactly what one should expect: a

finite-dimensional diffusion cannot be genuinely rough at infinitesimal scale, but a multi-factor diffusion with many fast mean-reverting components can mimic rough scaling on the time scales available in data.

This distinction is crucial for the present thesis. The empirical signature “ $\hat{H} < 1/2$ ” is not synonymous with true mathematical roughness. It may mean either:

1. genuine roughness of the latent volatility process, as in a fractional Volterra model;
2. apparent finite-scale roughness produced by a Markovian semimartingale with many time scales;
3. apparent roughness introduced by the estimation of realised volatility from high-frequency price data.

Therefore, the lifted Heston model provides a concrete warning: a Markovian model can look rough to standard statistical procedures while remaining non-rough in the strict pathwise sense.

A second, independent critique concerns option pricing. [Abi Jaber and Li \[2025\]](#) show, using SPX implied volatility surfaces from 2011 to 2022, that rough Bergomi models are structurally constrained by the universal skew scaling

$$\psi(T) \propto T^{H-1/2}. \quad (1.13)$$

The same value of H controls both the short-maturity explosion and the long-maturity decay. This makes the model too rigid: it cannot simultaneously fit the short end and the long end of the SPX ATM skew term structure. In their empirical study, simple Markovian Bergomi-type models, in particular a two-factor Markovian model, outperform the rough Bergomi model on the global SPX surface.

Taken together, these critiques motivate the central question of this thesis: does rough volatility provide a genuinely necessary modelling ingredient, or can its empirical and pricing advantages be reproduced by finite-dimensional multi-scale Markovian models? The lifted Heston model of [Abi Jaber \[2019\]](#) is precisely designed to test this question. It approximates the rough Heston model through a finite sum of mean-reverting factors, remains Markovian and semimartingale for every finite n , admits fast Fourier pricing through Riccati ODEs, and yet reproduces both rough-Heston-like implied volatility surfaces and rough-looking statistical behaviour over observable time scales.

1.3 Goals and contributions of this thesis

This thesis has three main goals.

Goal 1 (Implementation and convergence study). We implement the lifted Heston model from scratch in Python: solving the system of Riccati ODEs, computing the characteristic function via the affine formula, recovering option prices by Fourier inversion (cosine method), and inverting Black–Scholes to extract implied volatilities. We reproduce numerically the convergence of the lifted model to the rough Heston as n grows, study the role of r_n , and verify the empirical claims of [Abi Jaber \[2019\]](#).

Goal 2 (Comparison with aBergomi). We implement the affine Bergomi model of [Zhu, Loeper, Chen, and Langrené \[2021\]](#), which provides a Markovian approximation of the rough Bergomi model using a kernel-fitting procedure. We compare the lifted Heston and aBergomi on SPX-calibrated implied volatility surfaces, on three criteria:

1. Quality of the global smile fit (mean squared error in implied volatility);
2. Term structure of the ATM skew;
3. Speed of pricing and calibration.

This comparison has not been performed systematically in the literature: [Abi Jaber and Li \[2025\]](#) compare rough versus Markovian models within the Bergomi family, but never against the lifted Heston.

Goal 3 (Two-factor lifted Heston extension). We propose an original extension: a two-factor lifted Heston model in which the variance is the sum of two independent components,

$$V_t = V_t^{(1)} + V_t^{(2)}, \quad (1.14)$$

where $V^{(1)}$ is a standard lifted Heston driven by W^1 approximating a rough kernel $K_1(t) = t^{H_1-1/2}$, and $V^{(2)}$ is a single OU process driven by W^2 acting as a slow component. This structure is designed to reproduce the short-versus-long maturity crossover that single-kernel rough models cannot capture.

Chapter 2

Mathematical Background

2.1 The two measures, Girsanov's theorem, and stochastic volatility

Throughout this thesis, $(\Omega, \mathcal{F}, \mathbb{P})$ denotes a filtered probability space with filtration $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$ satisfying the usual conditions. We distinguish between two probability measures:

- the physical measure $\mathbb{P}$, under which one describes the real statistical dynamics observed in markets;
- the risk-neutral measure $\mathbb{Q}$, under which discounted traded asset prices are martingales and derivatives are priced by expectation.

This distinction is central in rough volatility. Realised volatility, roughness estimation, forecasting, and time-series calibration are $\mathbb{P}$ -questions. Option pricing, implied volatility surfaces, and calibration to market option prices are $\mathbb{Q}$ -questions. Therefore, an empirical Hurst estimate $\hat{H}$ obtained from realised volatility under $\mathbb{P}$ should not automatically be identified with a Hurst parameter H calibrated to option prices under $\mathbb{Q}$.

2.1.1 Risk-neutral pricing

Let $B^0 = (B_t^0)_{t \geq 0}$ denote the money-market account. If the interest rate is constant equal to r , then

$$B_t^0 = e^{rt}. \quad (2.1)$$

We use the notation B^0 for the bank account in order not to confuse it with the Brownian motion driving the stock price.

For a traded asset S , define the discounted price process

$$S_t^* := \frac{S_t}{B_t^0}. \quad (2.2)$$

A probability measure $\mathbb{Q}$ equivalent to $\mathbb{P}$ is called a risk-neutral measure if S^* is a $\mathbb{Q}$ -martingale. Equivalence means that

$$\mathbb{Q}(A) = 0 \iff \mathbb{P}(A) = 0, \quad A \in \mathcal{F}_T. \quad (2.3)$$

Thus $\mathbb{P}$ and $\mathbb{Q}$ agree on which events are impossible, but they may assign different probabilities to events that can occur.

Let X be an $\mathcal{F}_T$ -measurable payoff paid at maturity T . For example, a European call option with strike K has payoff

$$X = (S_T - K)^+ = (S_T - K) \vee 0. \quad (2.4)$$

The risk-neutral price process of X is

$$X_t = B_t^0 \mathbb{E}^{\mathbb{Q}}[X(B_T^0)^{-1} \mid \mathcal{F}_t], \quad 0 \leq t \leq T. \quad (2.5)$$

In particular, at time 0,

$$X_0 = \mathbb{E}^{\mathbb{Q}}[X(B_T^0)^{-1}]. \quad (2.6)$$

If r is constant, this becomes

$$X_t = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}}[X \mid \mathcal{F}_t]. \quad (2.7)$$

The reason for eq. (2.5) is simple. Dividing by B_t^0 gives

$$\frac{X_t}{B_t^0} = \mathbb{E}^{\mathbb{Q}}\left[\frac{X}{B_T^0} \mid \mathcal{F}_t\right], \quad (2.8)$$

which is a $\mathbb{Q}$ -martingale by the tower property of conditional expectation. Therefore the discounted price of the derivative is a martingale under the risk-neutral measure, exactly as required by no-arbitrage pricing.

2.1.2 Girsanov's theorem

We now recall the mathematical mechanism connecting $\mathbb{P}$ and $\mathbb{Q}$. The key tool is Girsanov's theorem. We state it in the one-dimensional Brownian setting, which is enough to understand the models used in this thesis. The multidimensional version is analogous.

Theorem 2.1 (Girsanov theorem). *Let $W^{\mathbb{P}} = (W_t^{\mathbb{P}})_{0 \leq t \leq T}$ be a Brownian motion under $\mathbb{P}$. Let $\Lambda = (\Lambda_t)_{0 \leq t \leq T}$ be an adapted process such that the exponential process*

$$Z_t = \exp\left(-\int_0^t \Lambda_s dW_s^{\mathbb{P}} - \frac{1}{2} \int_0^t \Lambda_s^2 ds\right) \quad (2.9)$$

is a martingale. A sufficient condition is Novikov's condition:

$$\mathbb{E}^{\mathbb{P}}\left[\exp\left(\frac{1}{2} \int_0^T \Lambda_s^2 ds\right)\right] < \infty. \quad (2.10)$$

Define a new probability measure $\mathbb{Q}$ on $\mathcal{F}_T$ by

$$\left.\frac{d\mathbb{Q}}{d\mathbb{P}}\right|_{\mathcal{F}_T} = Z_T. \quad (2.11)$$

Then the process

$$W_t^{\mathbb{Q}} := W_t^{\mathbb{P}} + \int_0^t \Lambda_s ds \quad (2.12)$$

is a Brownian motion under $\mathbb{Q}$.

The process Λ is called a market price of risk. Equivalently,

$$dW_t^{\mathbb{P}} = dW_t^{\mathbb{Q}} - \Lambda_t dt. \quad (2.13)$$

This is the most important practical form of Girsanov's theorem: when changing from $\mathbb{P}$ to $\mathbb{Q}$, Brownian motion is shifted by a drift.

The key consequence is that Girsanov changes drift terms but not quadratic variation. Indeed, from eq. (2.12),

$$W_t^{\mathbb{P}} = W_t^{\mathbb{Q}} - \int_0^t \Lambda_s ds. \quad (2.14)$$

The second term is of finite variation, hence it has zero quadratic variation. Therefore

$$d\langle W^{\mathbb{P}} \rangle_t = d\langle W^{\mathbb{Q}} \rangle_t = dt. \quad (2.15)$$

Similarly, in a multidimensional Brownian model, the instantaneous covariance matrix is preserved. Girsanov shifts Brownian motions by adapted drifts, but does not change the diffusion coefficients.

2.1.3 Black–Scholes as a first example

Consider the Black–Scholes model under the physical measure $\mathbb{P}$:

$$dS_t = \mu S_t dt + \sigma S_t dW_t^{\mathbb{P}}, \quad \sigma > 0, \quad (2.16)$$

$$dB_t^0 = r B_t^0 dt, \quad B_0^0 = 1. \quad (2.17)$$

The discounted stock price is $S_t^* = S_t/B_t^0$. By Itô's formula,

$$dS_t^* = S_t^* ((\mu - r) dt + \sigma dW_t^{\mathbb{P}}). \quad (2.18)$$

Define

$$\Lambda := \frac{\mu - r}{\sigma}. \quad (2.19)$$

Then

$$dS_t^* = \sigma S_t^* (dW_t^{\mathbb{P}} + \Lambda dt). \quad (2.20)$$

By Girsanov's theorem,

$$W_t^{\mathbb{Q}} = W_t^{\mathbb{P}} + \Lambda t \quad (2.21)$$

is a Brownian motion under an equivalent measure $\mathbb{Q}$. Therefore, under $\mathbb{Q}$,

$$dS_t^* = \sigma S_t^* dW_t^{\mathbb{Q}}. \quad (2.22)$$

Equivalently,

$$dS_t = r S_t dt + \sigma S_t dW_t^{\mathbb{Q}}. \quad (2.23)$$

The drift has changed from the physical expected return μ to the risk-free rate r . The volatility coefficient σ is unchanged.

For a European call option with payoff $X = (S_T - K)^+$, the arbitrage-free price is

$$X_0 = \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+(B_T^0)^{-1}] = e^{-rT} \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+]. \quad (2.24)$$

This is the continuous-time analogue of the discrete-time formula

$$X_n = B_n \mathbb{E}^{\mathbb{Q}}[X B_N^{-1} \mid \mathcal{F}_n], \quad (2.25)$$

used in the binomial model.

2.1.4 General stochastic volatility under $\mathbb{P}$ and $\mathbb{Q}$

Under $\mathbb{P}$, a general stochastic volatility model may be written as

$$\frac{dS_t}{S_t} = \mu_t^{\mathbb{P}} dt + \sqrt{V_t} dB_t^{\mathbb{P}}, \quad (2.26)$$

$$dV_t = b^{\mathbb{P}}(t, V_t) dt + a(t, V_t) dW_t^{\mathbb{P}}, \quad (2.27)$$

with

$$d\langle B^{\mathbb{P}}, W^{\mathbb{P}} \rangle_t = \rho dt. \quad (2.28)$$

Here $\mu_t^{\mathbb{P}}$ is the real-world expected return and $b^{\mathbb{P}}$ is the physical drift of the variance.

Suppose the volatility Brownian motion is changed by

$$dW_t^{\mathbb{P}} = dW_t^{\mathbb{Q}} - \Lambda_t^V dt, \quad (2.29)$$

where Λ^V is the market price of volatility risk. Substituting this into eq. (2.27) gives

$$\begin{aligned} dV_t &= b^{\mathbb{P}}(t, V_t) dt + a(t, V_t) (dW_t^{\mathbb{Q}} - \Lambda_t^V dt) \\ &= (b^{\mathbb{P}}(t, V_t) - a(t, V_t) \Lambda_t^V) dt + a(t, V_t) dW_t^{\mathbb{Q}}. \end{aligned} \quad (2.30)$$

Hence, under $\mathbb{Q}$,

$$b^{\mathbb{Q}}(t, V_t) = b^{\mathbb{P}}(t, V_t) - a(t, V_t)\Lambda_t^V. \quad (2.31)$$

The drift changes, but the diffusion coefficient $a(t, V_t)$ is unchanged.

The stock drift is also modified so that the discounted stock price is a $\mathbb{Q}$ -martingale:

$$\frac{dS_t}{S_t} = r dt + \sqrt{V_t} dB_t^{\mathbb{Q}}. \quad (2.32)$$

Remark 2.2 (Main lesson). The passage from $\mathbb{P}$ to $\mathbb{Q}$ changes risk premia and drift terms. It does not change the martingale volatility coefficients. Since local path regularity is determined by the noise part rather than by finite-variation drift terms, pathwise roughness properties are preserved under equivalent changes of measure.

2.1.5 Application to Heston

Under $\mathbb{P}$, a Heston-type variance process may be written as

$$dV_t = \kappa^{\mathbb{P}}(\theta^{\mathbb{P}} - V_t) dt + \nu\sqrt{V_t} dW_t^{\mathbb{P}}. \quad (2.33)$$

Using

$$dW_t^{\mathbb{P}} = dW_t^{\mathbb{Q}} - \Lambda_t^V dt, \quad (2.34)$$

we obtain

$$dV_t = \left[\kappa^{\mathbb{P}}(\theta^{\mathbb{P}} - V_t) - \nu\sqrt{V_t}\Lambda_t^V \right] dt + \nu\sqrt{V_t} dW_t^{\mathbb{Q}}. \quad (2.35)$$

For suitable choices of the volatility risk premium, this drift can be rewritten in mean-reverting affine form:

$$dV_t = \kappa^{\mathbb{Q}}(\theta^{\mathbb{Q}} - V_t) dt + \nu\sqrt{V_t} dW_t^{\mathbb{Q}}. \quad (2.36)$$

The parameters $\kappa^{\mathbb{P}}, \theta^{\mathbb{P}}$ and $\kappa^{\mathbb{Q}}, \theta^{\mathbb{Q}}$ need not be equal. They describe different drifts under different measures. However, the square-root diffusion coefficient $\nu\sqrt{V_t}$ is the same. Thus the semimartingale nature and Brownian-type local regularity of the Heston variance are preserved. To see this explicitly, compare the drift in both equations. We want

$$\kappa^{\mathbb{P}}(\theta^{\mathbb{P}} - V_t) - \nu\sqrt{V_t}\Lambda_t^V = \kappa^{\mathbb{Q}}(\theta^{\mathbb{Q}} - V_t). \quad (2.37)$$

Solving for the market price of volatility risk gives

$$\Lambda_t^V = \frac{\kappa^{\mathbb{P}}(\theta^{\mathbb{P}} - V_t) - \kappa^{\mathbb{Q}}(\theta^{\mathbb{Q}} - V_t)}{\nu\sqrt{V_t}}. \quad (2.38)$$

Equivalently,

$$\Lambda_t^V = \frac{\kappa^{\mathbb{P}}\theta^{\mathbb{P}} - \kappa^{\mathbb{Q}}\theta^{\mathbb{Q}}}{\nu\sqrt{V_t}} + \frac{\kappa^{\mathbb{Q}} - \kappa^{\mathbb{P}}}{\nu}\sqrt{V_t}. \quad (2.39)$$

With this choice, the drift becomes

$$\kappa^{\mathbb{Q}}(\theta^{\mathbb{Q}} - V_t), \quad (2.40)$$

and hence the variance dynamics under $\mathbb{Q}$ can be written as

$$dV_t = \kappa^{\mathbb{Q}}(\theta^{\mathbb{Q}} - V_t) dt + \nu\sqrt{V_t} dW_t^{\mathbb{Q}}. \quad (2.41)$$

Thus the change of measure modifies the mean-reversion parameters but preserves the square-root diffusion coefficient $\nu\sqrt{V_t}$.

2.1.6 Application to rough Heston

The same principle applies to rough Heston. Under $\mathbb{P}$, write

$$V_t = V_0 + \int_0^t K_H(t-s) \lambda^{\mathbb{P}}(\theta^{\mathbb{P}} - V_s) ds + \int_0^t K_H(t-s) \nu \sqrt{V_s} dW_s^{\mathbb{P}}, \quad (2.42)$$

where

$$K_H(t) = \frac{t^{H-1/2}}{\Gamma(H+1/2)}. \quad (2.43)$$

Using

$$dW_s^{\mathbb{P}} = dW_s^{\mathbb{Q}} - \Lambda_s^V ds, \quad (2.44)$$

we get

$$\begin{aligned} V_t = V_0 + \int_0^t K_H(t-s) \left[\lambda^{\mathbb{P}}(\theta^{\mathbb{P}} - V_s) - \nu \sqrt{V_s} \Lambda_s^V \right] ds \\ + \int_0^t K_H(t-s) \nu \sqrt{V_s} dW_s^{\mathbb{Q}}. \end{aligned} \quad (2.45)$$

Again, the drift has changed, but the stochastic convolution still contains the same deterministic kernel K_H . Therefore the singularity

$$K_H(t) \sim t^{H-1/2}, \quad t \downarrow 0, \quad (2.46)$$

is the same under $\mathbb{P}$ and $\mathbb{Q}$.

This matters because the roughness of rough Heston is generated by the singular kernel K_H , not by the drift. Girsanov changes the drift, but it does not remove or create the fractional kernel. Therefore, if the same rough Volterra model is used under both measures, the pathwise roughness index H is preserved.

2.1.7 Application to lifted Heston

For fixed n , the lifted Heston model is finite-dimensional. Under $\mathbb{Q}$, its variance is

$$V_t^n = g_0^n(t) + \sum_{i=1}^n c_i^n U_t^{n,i}, \quad (2.47)$$

with factor dynamics

$$dU_t^{n,i} = \left(-x_i^n U_t^{n,i} - \lambda^{\mathbb{Q}} V_t^n \right) dt + \nu \sqrt{V_t^n} dW_t^{\mathbb{Q}}. \quad (2.48)$$

A physical-measure version has the same diffusion term and the same finite factor architecture, but possibly a different drift:

$$dU_t^{n,i} = \left(-x_i^n U_t^{n,i} - \lambda^{\mathbb{P}} V_t^n \right) dt + \nu \sqrt{V_t^n} dW_t^{\mathbb{P}}. \quad (2.49)$$

The change from $\mathbb{P}$ to $\mathbb{Q}$ can be absorbed into the drift through the market price of volatility risk. What is not changed is the multi-factor kernel

$$K_n(t) = \sum_{i=1}^n c_i^n e^{-x_i^n t}. \quad (2.50)$$

This kernel approximates the fractional kernel K_H over a range of time scales:

$$K_H(t) \approx K_n(t). \quad (2.51)$$

Therefore the finite-scale rough-looking behaviour of the lifted Heston model comes from the multi-scale factor structure, not from the drift. For every fixed n , the lifted Heston variance remains a finite-dimensional semimartingale under both $\mathbb{P}$ and $\mathbb{Q}$.

2.1.8 What is preserved by the change of measure?

Proposition 2.3 (Properties preserved under an equivalent Brownian change of measure). *Let $\mathbb{Q}$ be equivalent to $\mathbb{P}$ on $\mathcal{F}_T$, and suppose the change of measure is given by Girsanov's theorem. Then:*

1. $\mathbb{P}$ and $\mathbb{Q}$ have the same null sets. Hence almost-sure path properties are preserved.
2. Quadratic variations and quadratic covariations are preserved.
3. The semimartingale property is preserved.
4. In rough Volterra models, the deterministic kernel K_H is unchanged by the change of measure.

Proof. Since $\mathbb{Q}$ is equivalent to $\mathbb{P}$, an event has probability zero under $\mathbb{P}$ if and only if it has probability zero under $\mathbb{Q}$. Therefore almost-sure path properties, such as Hölder regularity, are preserved.

The Brownian motions under the two measures differ only by a finite-variation process. Finite-variation processes have zero quadratic variation and zero quadratic covariation with continuous semimartingales. Hence quadratic variations and covariations are unchanged.

The invariance of the semimartingale property under equivalent changes of measure is a standard result of stochastic integration theory. Finally, in a Volterra model, the kernel is deterministic: Girsanov shifts the Brownian motion but does not alter the deterministic function K_H appearing in the stochastic convolution. $\square$

Remark 2.4 (What is not preserved). The full probability law is not preserved. Drifts, expectations, mean-reversion levels, and risk premia may change. Markovianity is preserved in standard affine models when the market price of risk is chosen as a function of the current state, but it should not be assumed automatically for arbitrary path-dependent changes of measure.

2.1.9 Consequence for rough volatility

The Hurst parameter must be interpreted carefully.

If the same rough volatility model is specified under $\mathbb{P}$ and $\mathbb{Q}$, then the pathwise roughness generated by the kernel is preserved by Girsanov. A change of measure cannot turn a Brownian-regular volatility diffusion into a genuinely rough Volterra process, nor can it smooth a genuinely rough Volterra process into a semimartingale.

However, this does not imply that an empirical estimate $\hat{H}$ from realised volatility must equal a risk-neutral parameter H calibrated to option prices. These are different objects:

1. the true pathwise roughness of the latent volatility process under $\mathbb{P}$;
2. the apparent roughness estimated from realised volatility data;
3. the risk-neutral parameter H calibrated to option prices under $\mathbb{Q}$.

They may coincide in an ideal correctly specified model, but they should not be identified automatically.

2.2 Implied volatility and ATM skew

The Black–Scholes price of a European call with strike K , maturity T , spot price S_0 , interest rate r , and volatility σ is

$$C^{\text{BS}}(S_0, K, T, \sigma) = S_0 \Phi(d_+) - K e^{-rT} \Phi(d_-), \quad (2.52)$$

where

$$d_{\pm} = \frac{\log(S_0/K) + \left(r \pm \frac{1}{2}\sigma^2\right) T}{\sigma\sqrt{T}}, \quad (2.53)$$

and Φ denotes the standard normal distribution function.

Given a market price $C^{\text{mkt}}(K, T)$, the implied volatility $\sigma_{\text{BS}}(K, T)$ is defined as the value of σ satisfying

$$C^{\text{BS}}(S_0, K, T, \sigma_{\text{BS}}(K, T)) = C^{\text{mkt}}(K, T). \quad (2.54)$$

In practice, the implied volatility surface is the function

$$(K, T) \mapsto \sigma_{\text{BS}}(K, T). \quad (2.55)$$

Let F_T denote the forward price of the stock at maturity T . Under constant interest rate r and no dividends,

$$F_T = S_0 e^{rT}. \quad (2.56)$$

We use log-moneyness

$$k = \log(K/F_T). \quad (2.57)$$

The at-the-money implied volatility skew is

$$\psi(T) = \left. \frac{\partial \sigma_{\text{BS}}(T, k)}{\partial k} \right|_{k=0}. \quad (2.58)$$

When $r = 0$, $F_T = S_0$, and this reduces to the convention used in most of our numerical sections.

Empirical estimation of the ATM skew. The theoretical definition of $\psi(T)$ involves a partial derivative $\partial_k \sigma_{\text{BS}}|_{k=0}$ that cannot be evaluated directly on discrete strike grids. Following [Abi Jaber \[2019\]](#) (Figure 1 and Section 4.1), we estimate the ATM skew by the symmetric finite difference

$$\hat{\psi}(T) := \frac{\sigma_{\text{BS}}(0.98 F_T, T) - \sigma_{\text{BS}}(1.02 F_T, T)}{\log(1.02/0.98)} \approx - \left. \frac{\partial \sigma_{\text{BS}}}{\partial k} \right|_{k=0}. \quad (2.59)$$

The sign is chosen so that empirical SPX values are positive. The implied volatilities at $K = 0.98 F_T$ and $K = 1.02 F_T$ are obtained by linear interpolation of $\sigma_{\text{BS}}(\cdot, T)$ on the available strike grid, since these wings are rarely traded exactly. The bandwidth $\log(1.02/0.98) \approx 0.0396$ is wide enough to be insensitive to bid-ask noise at individual strikes, and narrow enough that the higher-order Taylor remainder remains negligible for typical SPX convexities. The same convention is applied to both market and model surfaces, so any systematic bias of the finite-difference estimator cancels in the model-versus-market comparison.

2.3 The Bergomi–Guyon expansion

The Bergomi–Guyon expansion gives a first-order approximation of the ATM skew in terms of the covariation between the stock price and the forward variance curve. Define the forward variance curve by

$$\xi_t(T) := \mathbb{E}^{\mathbb{Q}}[V_T | \mathcal{F}_t], \quad t \leq T. \quad (2.60)$$

For each fixed maturity T , the process $(\xi_t(T))_{0 \leq t \leq T}$ is a $\mathbb{Q}$ -martingale.

Theorem 2.5 (Bergomi–Guyon first-order ATM skew expansion; [Bergomi and Guyon \[2012\]](#)). *Let $X_t = \log S_t$, let $\xi_t(u) := \mathbb{E}^{\mathbb{Q}}[V_u | \mathcal{F}_t]$ be the forward variance curve, and set $w_T := \int_0^T \xi_0(s) ds$. At first order in the vol-of-vol, the at-the-money implied-volatility skew is*

$$\psi(T) = \left. \frac{\partial \sigma_{\text{BS}}(T, k)}{\partial k} \right|_{k=0} = \frac{1}{2\sqrt{T} w_T^{3/2}} \int_0^T \int_t^T C(t, u) du dt + O(\text{vol-of-vol}^2), \quad C(t, u) := \frac{d\langle X, \xi(\cdot)(u) \rangle_t}{dt}, \quad (2.61)$$

where $C(t, u)$ is the instantaneous spot/forward-variance covariation density.

Heston-type corollary. If, at first order in ν , $d\xi_t(u) = K(u-t)\nu\sqrt{V_t}dW_t + O(\nu^2)$ with $d\langle B, W \rangle_t = \rho dt$, then $C(t, u) = \rho\nu V_t K(u-t)$. Localising $V_t \approx V_0$ and assuming a flat initial forward variance $\xi_0 \equiv V_0$ (so $w_T = V_0 T$),

$$\psi(T) \approx \frac{\rho\nu}{2\sqrt{V_0}T^2} \mathcal{I}_K(T) + O(\nu^2), \quad \mathcal{I}_K(T) := \int_0^T \int_t^T K(u-t) du dt = \int_0^T \int_0^r K(s) ds dr. \quad (2.62)$$

Remark 2.6 (Interpretation). The ATM skew is governed by the twice-integrated kernel $\mathcal{I}_K(T)$ of eq. (2.62) over the interval $[0, T]$. This explains why the kernel shape is so important.

For a classical Heston-type exponential kernel $K(t) = e^{-\kappa t}$,

$$\mathcal{I}_{e^{-\kappa \cdot}}(T) = \frac{1}{\kappa} \left(T - \frac{1 - e^{-\kappa T}}{\kappa} \right), \quad (2.63)$$

so the skew remains bounded as $T \downarrow 0$.

For a rough kernel

$$K_H(t) = \frac{t^{H-1/2}}{\Gamma(H+1/2)}, \quad (2.64)$$

one obtains

$$\mathcal{I}_{K_H}(T) = \frac{T^{H+3/2}}{\Gamma(H+5/2)}. \quad (2.65)$$

Therefore

$$\psi(T) \sim \frac{\rho\nu}{2\sqrt{V_0}\Gamma(H+5/2)} T^{H-1/2}. \quad (2.66)$$

If $H < 1/2$, this explodes as $T \downarrow 0$.

For a lifted Heston kernel

$$K_n(t) = \sum_{i=1}^n c_i^n e^{-x_i^n t}, \quad (2.67)$$

one has

$$\mathcal{I}_{K_n}(T) = \sum_{i=1}^n \frac{c_i^n}{x_i^n} \left(T - \frac{1 - e^{-x_i^n T}}{x_i^n} \right). \quad (2.68)$$

For fixed n , this gives a bounded short-maturity skew. However, for suitable choices of (c_i^n, x_i^n) , it can mimic the rough power-law behaviour over observable maturities.

The often-quoted “skew $\approx$ time-average of the kernel” rule $\psi(T) \propto \frac{1}{T} \int_0^T K(T-t) dt$ captures the correct power law and sign but not the constant; the exact first-order coefficient requires the twice-integrated kernel $\mathcal{I}_K(T)$ above (for K_H this replaces $\Gamma(H+3/2)$ by $\Gamma(H+5/2)$).

2.4 Variance swaps and the forward variance curve

The forward variance curve can be linked to variance swaps. A variance swap with maturity T pays

$$\frac{1}{T} \int_0^T V_t dt - K_{\text{var}}(T) \quad (2.69)$$

at maturity. The fair variance swap strike is therefore

$$K_{\text{var}}(T) = \frac{1}{T} \mathbb{E}^{\mathbb{Q}} \left[\int_0^T V_t dt \right] = \frac{1}{T} \int_0^T \xi_0(t) dt. \quad (2.70)$$

Thus,

$$TK_{\text{var}}(T) = \int_0^T \xi_0(t) dt. \quad (2.71)$$

Formally differentiating gives

$$\xi_0(T) = \frac{d}{dT} [TK_{\text{var}}(T)]. \quad (2.72)$$

This is why the initial forward variance curve can be inferred from market variance swap data. In models such as rough Bergomi or lifted Heston, the deterministic input curve is used to match this initial forward variance curve.

Chapter 3

The Heston Model: Classical and Rough

3.1 The classical Heston model

The Heston model [Heston, 1993] is the canonical one-factor stochastic volatility model under $\mathbb{Q}$:

$$\begin{aligned} dS_t &= S_t \sqrt{V_t} dB_t, & S_0 &> 0, \\ dV_t &= \lambda(\theta - V_t) dt + \nu \sqrt{V_t} dW_t, & V_0 &> 0, \end{aligned} \quad (3.1)$$

with $B = \rho W + \sqrt{1 - \rho^2} W^\perp$ and $\rho \in [-1, 1]$, $\lambda, \theta, \nu > 0$.

Theorem 3.1 (Heston characteristic function). *For $u \in \mathbb{C}$ with $\text{Re}(u) \in [0, 1]$,*

$$\mathbb{E}^{\mathbb{Q}}[e^{u \log S_T}] = \exp(\phi(T, u) + u \log S_0 + \psi(T, u) V_0), \quad (3.2)$$

where (ϕ, ψ) solve the Riccati ODE system

$$\begin{aligned} \psi'(t, u) &= F(u, \psi(t, u)), & \psi(0, u) &= 0, \\ \phi'(t, u) &= \lambda \theta \psi(t, u), & \phi(0, u) &= 0, \end{aligned} \quad (3.3)$$

with $F(u, v) = \frac{1}{2}(u^2 - u) + (\rho \nu u - \lambda)v + \frac{\nu^2}{2}v^2$.

The Riccati equation is scalar and can be solved in closed form, giving the celebrated Heston pricing formula. The model is Markovian: the state space is two-dimensional (S_t, V_t) .

Proposition 3.2 (Expected variance in Heston). *The expected variance $m(t) := \mathbb{E}^{\mathbb{Q}}[V_t]$ satisfies*

$$m'(t) = \lambda(\theta - m(t)), \quad m(0) = V_0, \quad (3.4)$$

with explicit solution $m(t) = \theta + (V_0 - \theta)e^{-\lambda t}$.

Proof. Take expectation in the SDE for V_t . The stochastic integral $\int_0^t \nu \sqrt{V_s} dW_s$ is a true martingale on $[0, T]$ provided V is integrable (which holds for Heston by standard arguments), so its expectation is zero. The remaining ODE is solved by integrating factor $e^{\lambda t}$. $\square$

The Heston ATM skew failure. Applied to the Heston model, the Bergomi–Guyon formula eq. (2.62) with kernel $K(t) = e^{-\lambda t}$ gives

$$\psi^{\text{Heston}}(T) \approx \frac{\rho \nu}{2\sqrt{V_0} T^2} \cdot \frac{1}{\lambda} \left(T - \frac{1 - e^{-\lambda T}}{\lambda} \right) \xrightarrow{T \rightarrow 0} \frac{\rho \nu}{4\sqrt{V_0}}. \quad (3.5)$$

The skew is bounded as $T \rightarrow 0$. This contradicts the empirical SPX behaviour and is the central failure of the Heston model.

3.2 The rough Heston model

To remedy the Heston skew failure, [El Euch and Rosenbaum \[2019\]](#) introduced the rough Heston model:

$$\begin{aligned} dS_t &= S_t \sqrt{V_t} dB_t, \\ V_t &= V_0 + \frac{1}{\Gamma(H + 1/2)} \int_0^t (t-s)^{H-1/2} \lambda(\theta - V_s) ds + \frac{\nu}{\Gamma(H + 1/2)} \int_0^t (t-s)^{H-1/2} \sqrt{V_s} dW_s, \end{aligned} \quad (3.6)$$

with $H \in (0, 1/2)$.

Remark 3.3 (Loss of structure). The variance process eq. (3.6) is:

- **Not Markovian:** V_t depends on the entire path $(V_s)_{s \leq t}$ via the kernel.
- **Not a semimartingale:** the kernel $(t-s)^{H-1/2}$ has a non-integrable singularity at $s = t$ when $H < 1/2$, in a sense that breaks the semimartingale decomposition.

The model nevertheless retains affine Volterra structure [[Abi Jaber, Larsson, and Pulido, 2019](#)], which yields a Fourier–Laplace transform that, while not closed-form, satisfies a fractional Riccati equation:

$$\psi(t, u) = \frac{1}{\Gamma(H + 1/2)} \int_0^t (t-s)^{H-1/2} F(u, \psi(s, u)) ds. \quad (3.7)$$

Pricing requires solving eq. (3.7) numerically (e.g. via the Adams predictor–corrector scheme) and then performing Fourier inversion.

Rough Heston ATM skew. The Bergomi–Guyon formula with the rough kernel gives

$$\psi^{\text{rough}}(T) \approx \frac{\rho\nu}{2\sqrt{V_0}\Gamma(H + 5/2)} T^{H-1/2}. \quad (3.8)$$

For $H < 1/2$, this diverges as $T \rightarrow 0$, matching the empirical SPX behaviour.

The price of roughness. Despite producing the correct ATM skew, the rough Heston model is computationally expensive: solving eq. (3.7) for each u in the Fourier grid is roughly twenty times slower than the equivalent computation in classical Heston, simulation requires special schemes, and hedging becomes substantially more delicate due to the loss of the Markov property.

Chapter 4

The Lifted Heston Model

4.1 Definition

Definition 4.1 (Lifted Heston model). Fix $n \in \mathbb{N}$, parameters $\lambda, \nu \in \mathbb{R}_+$, $\rho \in [-1, 1]$, weights $c_1^n, \dots, c_n^n \geq 0$, mean-reversion speeds $0 \leq x_1^n < x_2^n < \dots < x_n^n$, and an admissible deterministic input curve g_0^n . The lifted Heston model of order n under $\mathbb{Q}$ is the system

$$dS_t^n = S_t^n \sqrt{V_t^n} dB_t, \quad S_0^n > 0, \quad (4.1)$$

$$V_t^n = g_0^n(t) + \sum_{i=1}^n c_i^n U_t^{n,i}, \quad (4.2)$$

$$dU_t^{n,i} = (-x_i^n U_t^{n,i} - \lambda V_t^n) dt + \nu \sqrt{V_t^n} dW_t, \quad U_0^{n,i} = 0, \quad i = 1, \dots, n, \quad (4.3)$$

with $B = \rho W + \sqrt{1 - \rho^2} W^\perp$.

The numbering eq. (4.1)–eq. (4.3) corresponds to equations (2.1)–(2.3) of [Abi Jaber \[2019\]](#).

Key structural features.

- The variance V_t^n is a deterministic function g_0^n plus a weighted sum of the n factors $U^{n,i}$. The factors all start at zero.
- All factors share the same driving Brownian motion W ; only the speeds x_i^n differ. This keeps the model effectively driven by a single source of randomness from the variance side.
- The drift of each $U^{n,i}$ contains $-\lambda V_t^n$, which depends on all factors. The system is therefore coupled, but linearly so.
- For $n = 1$, $c_1 = 1$, $x_1 = 0$, and $g_0(t) = V_0 + \lambda \theta t$, the model reduces to classical Heston (verified in proposition 4.4).

4.2 The forward variance curve and the choice of g_0^n

The function g_0^n encodes the initial term structure of forward variance.

Proposition 4.2 (Volterra equation for the expected variance). *Let $m_n(t) := \mathbb{E}^\mathbb{Q}[V_t^n]$. Then*

$$m_n(t) + \lambda \sum_{i=1}^n c_i^n \int_0^t e^{-x_i^n(t-s)} m_n(s) ds = g_0^n(t), \quad t \geq 0. \quad (4.4)$$

Proof. We use the mild form of the factor process. Starting from eq. (4.3),

$$dU_t^{n,i} = \left(-x_i^n U_t^{n,i} - \lambda V_t^n\right) dt + \nu \sqrt{V_t^n} dW_t, \quad U_0^{n,i} = 0, \quad (4.5)$$

multiply by the integrating factor $e^{x_i^n t}$. Since $e^{x_i^n t}$ has finite variation, Itô's product rule gives

$$\begin{aligned} d\left(e^{x_i^n t} U_t^{n,i}\right) &= e^{x_i^n t} dU_t^{n,i} + x_i^n e^{x_i^n t} U_t^{n,i} dt \\ &= e^{x_i^n t} \left(-\lambda V_t^n dt + \nu \sqrt{V_t^n} dW_t\right). \end{aligned} \quad (4.6)$$

Integrating from 0 to t and using $U_0^{n,i} = 0$, we obtain

$$e^{x_i^n t} U_t^{n,i} = -\lambda \int_0^t e^{x_i^n s} V_s^n ds + \nu \int_0^t e^{x_i^n s} \sqrt{V_s^n} dW_s. \quad (4.7)$$

Hence

$$U_t^{n,i} = -\lambda \int_0^t e^{-x_i^n(t-s)} V_s^n ds + \nu \int_0^t e^{-x_i^n(t-s)} \sqrt{V_s^n} dW_s. \quad (4.8)$$

Now take expectations. Under the integrability assumptions of the lifted Heston model, the stochastic integral in (4.8) is a true martingale with expectation zero. Therefore

$$\mathbb{E}^\mathbb{Q}[U_t^{n,i}] = -\lambda \int_0^t e^{-x_i^n(t-s)} \mathbb{E}^\mathbb{Q}[V_s^n] ds. \quad (4.9)$$

By definition, $m_n(s) := \mathbb{E}^\mathbb{Q}[V_s^n]$, so

$$\mathbb{E}^\mathbb{Q}[U_t^{n,i}] = -\lambda \int_0^t e^{-x_i^n(t-s)} m_n(s) ds. \quad (4.10)$$

Finally, using

$$V_t^n = g_0^n(t) + \sum_{i=1}^n c_i^n U_t^{n,i}, \quad (4.11)$$

and taking expectations, we get

$$\begin{aligned} m_n(t) &= g_0^n(t) + \sum_{i=1}^n c_i^n \mathbb{E}^\mathbb{Q}[U_t^{n,i}] \\ &= g_0^n(t) - \lambda \sum_{i=1}^n c_i^n \int_0^t e^{-x_i^n(t-s)} m_n(s) ds. \end{aligned} \quad (4.12)$$

Rearranging gives

$$m_n(t) + \lambda \sum_{i=1}^n c_i^n \int_0^t e^{-x_i^n(t-s)} m_n(s) ds = g_0^n(t), \quad (4.13)$$

which is the desired Volterra equation. $\square$

The canonical choice (equation 2.6 of [Abi Jaber \[2019\]](#)). The previous proposition shows that g_0^n determines the expected variance curve $m_n(t) = \mathbb{E}^\mathbb{Q}[V_t^n]$. In full generality, one could choose g_0^n so as to match an observed initial forward variance curve extracted from variance swaps. For comparison with classical Heston and rough Heston, however, [Abi Jaber \[2019\]](#) uses the canonical specification

$$g_0^n(t) = V_0 + \lambda \theta \sum_{i=1}^n c_i^n \int_0^t e^{-x_i^n(t-s)} ds. \quad (4.14)$$

Equivalently, if

$$K_n(t) := \sum_{i=1}^n c_i^n e^{-x_i^n t}, \quad (4.15)$$

then

$$g_0^n(t) = V_0 + \lambda \theta \int_0^t K_n(t-s) ds. \quad (4.16)$$

With this choice, the expected variance equation becomes

$$m_n(t) = V_0 + \lambda \int_0^t K_n(t-s)(\theta - m_n(s)) ds. \quad (4.17)$$

Thus V_0 plays the role of the initial variance level, θ the long-run variance level, and λ the mean-reversion strength. The difference from classical Heston is that mean reversion is not local in time: it is filtered through the memory kernel K_n .

In the special case $n = 1$, $c_1^1 = 1$, and $x_1^1 = 0$, one has $K_1(t) = 1$, and eq. (4.17) reduces to

$$m_1(t) = V_0 + \lambda \int_0^t (\theta - m_1(s)) ds. \quad (4.18)$$

Differentiating gives

$$m_1'(t) = \lambda(\theta - m_1(t)), \quad m_1(0) = V_0, \quad (4.19)$$

so

$$m_1(t) = \theta + (V_0 - \theta)e^{-\lambda t}. \quad (4.20)$$

Therefore the canonical choice eq. (4.14) recovers the classical Heston expected variance only in the one-factor case. For general n , it gives the lifted analogue of Heston mean reversion, with the exponential kernel replaced by the multi-factor kernel K_n .

4.3 Change of variable and reduction to a closed-form SDE system

The deterministic curve g_0^n in the variance equation eq. (4.2) can be absorbed into a translated factor system, yielding a cleaner formulation.

Proposition 4.3 (Change of variable). *Assume that g_0^n is chosen as in eq. (4.14), namely*

$$g_0^n(t) = V_0 + \lambda \theta \sum_{i=1}^n c_i^n \int_0^t e^{-x_i^n(t-s)} ds. \quad (4.21)$$

For each $i = 1, \dots, n$, define

$$\tilde{U}_t^{n,i} := U_t^{n,i} + \lambda \theta \int_0^t e^{-x_i^n(t-s)} ds. \quad (4.22)$$

Then $\tilde{U}_0^{n,i} = 0$ and the lifted Heston model can be rewritten as

$$V_t^n = V_0 + \sum_{i=1}^n c_i^n \tilde{U}_t^{n,i}, \quad (4.23)$$

$$d\tilde{U}_t^{n,i} = \left(-x_i^n \tilde{U}_t^{n,i} + \lambda(\theta - V_t^n) \right) dt + \nu \sqrt{V_t^n} dW_t. \quad (4.24)$$

Proof. The point of the change of variable is to absorb the deterministic curve g_0^n into the factors. Let

$$h_i(t) := \int_0^t e^{-x_i^n(t-s)} ds. \quad (4.25)$$

Then $h_i(0) = 0$ and

$$h_i'(t) = 1 - x_i^n h_i(t). \quad (4.26)$$

For $x_i^n > 0$, one has explicitly

$$h_i(t) = \frac{1 - e^{-x_i^n t}}{x_i^n},$$

while for $x_i^n = 0$, $h_i(t) = t$. This explains why the translation is not simply $U_t^{n,i} + \lambda\theta$. The term $\lambda\theta$ appears as a constant input in the drift of an OU equation, and the solution of an OU equation filters every input through the kernel $e^{-x_i^n(t-s)}$. Therefore the accumulated contribution of this constant input up to time t is

$$\lambda\theta \int_0^t e^{-x_i^n(t-s)} ds = \lambda\theta h_i(t),$$

not the constant $\lambda\theta$.

By the definition of g_0^n ,

$$\begin{aligned} V_t^n &= g_0^n(t) + \sum_{i=1}^n c_i^n U_t^{n,i} \\ &= V_0 + \lambda\theta \sum_{i=1}^n c_i^n h_i(t) + \sum_{i=1}^n c_i^n U_t^{n,i} \\ &= V_0 + \sum_{i=1}^n c_i^n \left(U_t^{n,i} + \lambda\theta h_i(t) \right) \\ &= V_0 + \sum_{i=1}^n c_i^n \tilde{U}_t^{n,i}. \end{aligned} \quad (4.27)$$

This proves eq. (4.23).

It remains to derive the SDE for $\tilde{U}^{n,i}$. From eq. (4.22),

$$d\tilde{U}_t^{n,i} = dU_t^{n,i} + \lambda\theta h_i'(t) dt. \quad (4.28)$$

Using the original factor equation

$$dU_t^{n,i} = \left(-x_i^n U_t^{n,i} - \lambda V_t^n \right) dt + \nu \sqrt{V_t^n} dW_t, \quad (4.29)$$

we obtain

$$d\tilde{U}_t^{n,i} = \left(-x_i^n U_t^{n,i} - \lambda V_t^n \right) dt + \nu \sqrt{V_t^n} dW_t + \lambda\theta (1 - x_i^n h_i(t)) dt. \quad (4.30)$$

Since

$$U_t^{n,i} = \tilde{U}_t^{n,i} - \lambda\theta h_i(t), \quad (4.31)$$

we have

$$-x_i^n U_t^{n,i} = -x_i^n \tilde{U}_t^{n,i} + x_i^n \lambda\theta h_i(t). \quad (4.32)$$

Substituting this into the drift gives

$$\begin{aligned} d\tilde{U}_t^{n,i} &= \left(-x_i^n \tilde{U}_t^{n,i} + x_i^n \lambda\theta h_i(t) - \lambda V_t^n + \lambda\theta - x_i^n \lambda\theta h_i(t) \right) dt + \nu \sqrt{V_t^n} dW_t \\ &= \left(-x_i^n \tilde{U}_t^{n,i} + \lambda(\theta - V_t^n) \right) dt + \nu \sqrt{V_t^n} dW_t. \end{aligned} \quad (4.33)$$

This proves eq. (4.24). $\square$

Proposition 4.4 (Recovery of classical Heston). *Setting $n = 1$, $c_1^1 = 1$, $x_1^1 = 0$ in the system eq. (4.23)–eq. (4.24) recovers the classical Heston SDE for V .*

Proof. With $x_1^1 = 0$, the drift becomes $\lambda(\theta - V_t^1)$ and $V_t^1 = V_0 + \tilde{U}_t^{1,1}$, so $dV_t^1 = d\tilde{U}_t^{1,1} = \lambda(\theta - V_t^1)dt + \nu\sqrt{V_t^1}dW_t$. $\square$

4.4 Fourier–Laplace transform and Riccati system

The lifted Heston model is an affine process, with characteristic function in closed form up to the solution of n ordinary Riccati ODEs.

Theorem 4.5 (Affine characteristic function, equation (2.9) of [Abi Jaber \[2019\]](#)). *For $u \in \mathbb{C}$ with $\text{Re}(u) \in [0, 1]$,*

$$\mathbb{E}^{\mathbb{Q}}[\exp(u \log S_T^n) | \mathcal{F}_t] = \exp\left(\phi^n(t, T) + u \log S_t^n + \sum_{i=1}^n c_i^n \psi^{n,i}(T-t) U_t^{n,i}\right), \quad (4.34)$$

where the functions $\psi^{n,i} : [0, T] \rightarrow \mathbb{C}$ solve the system of Riccati ODEs

$$\frac{d\psi^{n,i}}{dt}(t) = -x_i^n \psi^{n,i}(t) + F\left(u, \sum_{j=1}^n c_j^n \psi^{n,j}(t)\right), \quad \psi^{n,i}(0) = 0, \quad (4.35)$$

with

$$F(u, v) = \frac{1}{2}(u^2 - u) + (\rho\nu u - \lambda)v + \frac{\nu^2}{2}v^2, \quad (4.36)$$

and

$$\phi^n(t, T) = \int_0^{T-t} F\left(u, \sum_{j=1}^n c_j^n \psi^{n,j}(s)\right) g_0^n(T-s) ds. \quad (4.37)$$

Derivation of the Riccati system. We give the derivation because it is the key affine computation behind the lifted Heston model. To simplify notation, write

$$X_t^n := \log S_t^n.$$

From

$$dS_t^n = S_t^n \sqrt{V_t^n} dB_t,$$

Itô's formula gives

$$dX_t^n = -\frac{1}{2}V_t^n dt + \sqrt{V_t^n} dB_t. \quad (4.38)$$

We look for the conditional Fourier–Laplace transform in exponential-affine form. Define, for $t \leq T$,

$$M_t := \exp(Y_t), \quad (4.39)$$

where

$$Y_t := \phi^n(t, T) + uX_t^n + \sum_{i=1}^n c_i^n \psi^{n,i}(T-t) U_t^{n,i}. \quad (4.40)$$

At maturity $t = T$, we want $M_T = e^{uX_T^n}$. Hence the terminal conditions are

$$\phi^n(T, T) = 0, \quad \psi^{n,i}(0) = 0, \quad i = 1, \dots, n. \quad (4.41)$$

The goal is to choose ϕ^n and $\psi^{n,i}$ so that M is a true martingale. Since the terminal conditions give

$$M_T = e^{uX_T^n},$$

the martingale property implies

$$M_t = \mathbb{E}^{\mathbb{Q}}[M_T \mid \mathcal{F}_t] = \mathbb{E}^{\mathbb{Q}}[e^{uX_T^n} \mid \mathcal{F}_t].$$

Thus identifying the drift-free exponential-affine martingale M gives the desired conditional Fourier–Laplace transform.

Set

$$\Psi(t) := \sum_{j=1}^n c_j^n \psi^{n,j}(t). \quad (4.42)$$

Differentiating eq. (4.40), and using

$$dU_t^{n,i} = \left(-x_i^n U_t^{n,i} - \lambda V_t^n \right) dt + \nu \sqrt{V_t^n} dW_t,$$

we get

$$\begin{aligned} dY_t &= \partial_t \phi^n(t, T) dt + u dX_t^n \\ &\quad + \sum_{i=1}^n c_i^n \left[-(\psi^{n,i})'(T-t) U_t^{n,i} dt + \psi^{n,i}(T-t) dU_t^{n,i} \right]. \end{aligned} \quad (4.43)$$

Substituting the dynamics of X^n and $U^{n,i}$ gives

$$\begin{aligned} dY_t &= \left[\partial_t \phi^n(t, T) - \frac{u}{2} V_t^n + \sum_{i=1}^n c_i^n \left(-(\psi^{n,i})'(T-t) - x_i^n \psi^{n,i}(T-t) \right) U_t^{n,i} \right. \\ &\quad \left. - \lambda \sum_{i=1}^n c_i^n \psi^{n,i}(T-t) V_t^n \right] dt \\ &\quad + u \sqrt{V_t^n} dB_t + \nu \sqrt{V_t^n} \sum_{i=1}^n c_i^n \psi^{n,i}(T-t) dW_t. \end{aligned} \quad (4.44)$$

Using the notation eq. (4.42), this becomes

$$\begin{aligned} dY_t &= \left[\partial_t \phi^n(t, T) - \frac{u}{2} V_t^n + \sum_{i=1}^n c_i^n \left(-(\psi^{n,i})'(T-t) - x_i^n \psi^{n,i}(T-t) \right) U_t^{n,i} \right. \\ &\quad \left. - \lambda \Psi(T-t) V_t^n \right] dt + \sqrt{V_t^n} (u dB_t + \nu \Psi(T-t) dW_t). \end{aligned} \quad (4.45)$$

Next, since $d\langle B, W \rangle_t = \rho dt$, the quadratic variation of Y is

$$d\langle Y \rangle_t = V_t^n \left[u^2 + 2\rho\nu u \Psi(T-t) + \nu^2 \Psi(T-t)^2 \right] dt. \quad (4.46)$$

By Itô's formula for the exponential,

$$\frac{dM_t}{M_t} = dY_t + \frac{1}{2} d\langle Y \rangle_t. \quad (4.47)$$

Therefore the drift of dM_t/M_t is

$$\begin{aligned} &\partial_t \phi^n(t, T) + \sum_{i=1}^n c_i^n \left(-(\psi^{n,i})'(T-t) - x_i^n \psi^{n,i}(T-t) \right) U_t^{n,i} \\ &\quad + V_t^n \left[\frac{1}{2} (u^2 - u) + (\rho\nu u - \lambda) \Psi(T-t) + \frac{\nu^2}{2} \Psi(T-t)^2 \right]. \end{aligned} \quad (4.48)$$

This motivates the definition

$$F(u, v) := \frac{1}{2}(u^2 - u) + (\rho\nu u - \lambda)v + \frac{\nu^2}{2}v^2. \quad (4.49)$$

The drift can now be written as

$$\partial_t \phi^n(t, T) + \sum_{i=1}^n c_i^n \left(-(\psi^{n,i})'(T-t) - x_i^n \psi^{n,i}(T-t) \right) U_t^{n,i} + F(u, \Psi(T-t)) V_t^n. \quad (4.50)$$

Using

$$V_t^n = g_0^n(t) + \sum_{i=1}^n c_i^n U_t^{n,i},$$

we obtain

$$\begin{aligned} \text{drift} &= \partial_t \phi^n(t, T) + F(u, \Psi(T-t)) g_0^n(t) \\ &\quad + \sum_{i=1}^n c_i^n \left[-(\psi^{n,i})'(T-t) - x_i^n \psi^{n,i}(T-t) + F(u, \Psi(T-t)) \right] U_t^{n,i}. \end{aligned} \quad (4.51)$$

For M to be a local martingale, this drift must vanish. Since the expression is affine in the state variables $U_t^{n,i}$, we set separately the coefficients of each $U_t^{n,i}$ and the deterministic part equal to zero.

First, for each $i = 1, \dots, n$,

$$-(\psi^{n,i})'(T-t) - x_i^n \psi^{n,i}(T-t) + F(u, \Psi(T-t)) = 0. \quad (4.52)$$

Writing $s = T - t$, this is equivalent to

$$\frac{d\psi^{n,i}}{ds}(s) = -x_i^n \psi^{n,i}(s) + F\left(u, \sum_{j=1}^n c_j^n \psi^{n,j}(s)\right), \quad \psi^{n,i}(0) = 0. \quad (4.53)$$

This is the Riccati system.

Second, the remaining deterministic term gives

$$\partial_t \phi^n(t, T) + F(u, \Psi(T-t)) g_0^n(t) = 0, \quad \phi^n(T, T) = 0. \quad (4.54)$$

Integrating backward from T to t yields

$$\phi^n(t, T) = \int_t^T F(u, \Psi(T-r)) g_0^n(r) dr. \quad (4.55)$$

Changing variables $s = T - r$, so that $r = T - s$, gives

$$\phi^n(t, T) = \int_0^{T-t} F(u, \Psi(s)) g_0^n(T-s) ds. \quad (4.56)$$

This is eq. (4.37).

With these choices, the drift of M is zero. Thus M is a local martingale. Under the integrability conditions established in Abi Jaber's affine transform theorem, it is a true martingale, and therefore

$$M_t = \mathbb{E}^{\mathbb{Q}} \left[e^{u X_T^n} \mid \mathcal{F}_t \right]. \quad (4.57)$$

Substituting back $X_t^n = \log S_t^n$ proves the affine transform formula. $\square$

Remark 4.6 (Integral form of the Riccati system). The Riccati ODE system can also be written in Volterra form. Indeed, from

$$(\psi^{n,i})'(t) + x_i^n \psi^{n,i}(t) = F\left(u, \sum_{j=1}^n c_j^n \psi^{n,j}(t)\right), \quad \psi^{n,i}(0) = 0, \quad (4.58)$$

multiplying by $e^{x_i^n t}$ gives

$$\frac{d}{dt} \left(e^{x_i^n t} \psi^{n,i}(t) \right) = e^{x_i^n t} F\left(u, \sum_{j=1}^n c_j^n \psi^{n,j}(t)\right). \quad (4.59)$$

Hence

$$\psi^{n,i}(t) = \int_0^t e^{-x_i^n(t-s)} F\left(u, \sum_{j=1}^n c_j^n \psi^{n,j}(s)\right) ds. \quad (4.60)$$

If we define

$$\Psi^n(t) := \sum_{i=1}^n c_i^n \psi^{n,i}(t), \quad (4.61)$$

then summing over i gives

$$\Psi^n(t) = \int_0^t K_n(t-s) F(u, \Psi^n(s)) ds, \quad K_n(t) := \sum_{i=1}^n c_i^n e^{-x_i^n t}. \quad (4.62)$$

Thus the n -dimensional Riccati ODE system is equivalent to a Riccati–Volterra equation with kernel K_n . This is the exact finite-dimensional analogue of the rough Heston fractional Riccati equation, where K_n is replaced by the rough kernel K_H .

Remark 4.7 (Why this is fast). Compared to the rough Heston eq. (3.7), which is a fractional Volterra equation, eq. (4.35) is a system of standard ODEs of dimension n . For $n = 20$, this is solvable in milliseconds with any standard ODE integrator (e.g. Runge–Kutta, or the explicit-implicit scheme of [Abi Jaber \[2019, Appendix C\]](#)). The Fourier inversion can then be performed efficiently via the COS method of [Fang and Oosterlee \[2009\]](#).

4.5 Parametrisation in terms of the Hurst index H

The lifted Heston model in its raw form has $2n$ “shape” parameters (c_i^n, x_i^n) , which is impractical for calibration. [Abi Jaber \[2019\]](#) reduce this to a single parameter H (plus the geometric ratio r_n) by exploiting the Laplace representation of the fractional kernel.

Lemma 4.8 (Laplace representation of the rough kernel; Remark 3.1 of [Abi Jaber \[2019\]](#)). *For $H \in (0, 1/2)$ and $t > 0$,*

$$\frac{t^{H-1/2}}{\Gamma(H+1/2)} = \int_0^\infty e^{-xt} \mu(dx), \quad \mu(dx) = \frac{x^{-H-1/2}}{\Gamma(1/2-H)\Gamma(H+1/2)} dx. \quad (4.63)$$

Proof. Compute the right-hand side directly:

$$\int_0^\infty e^{-xt} \frac{x^{-H-1/2}}{\Gamma(1/2-H)\Gamma(H+1/2)} dx = \frac{1}{\Gamma(1/2-H)\Gamma(H+1/2)} \int_0^\infty e^{-xt} x^{-H-1/2} dx. \quad (4.64)$$

Substituting $u = xt$ (so $x = u/t$, $dx = du/t$):

$$= \frac{t^{H-1/2}}{\Gamma(1/2-H)\Gamma(H+1/2)} \int_0^\infty e^{-u} u^{-H-1/2} du = \frac{t^{H-1/2} \Gamma(1/2-H)}{\Gamma(1/2-H)\Gamma(H+1/2)} = \frac{t^{H-1/2}}{\Gamma(H+1/2)}, \quad (4.65)$$

where we used the Gamma function identity $\Gamma(1/2-H) = \int_0^\infty e^{-u} u^{(1/2-H)-1} du$ (convergent for $1/2-H > 0$, i.e. $H < 1/2$). $\square$

Remark 4.9 (Interpretation: rough volatility as a continuum of OU factors). Equation eq. (4.63) is the central structural identity behind the lifted Heston model. It says that the rough kernel

$$K_H(t) = \frac{t^{H-1/2}}{\Gamma(H+1/2)} \quad (4.66)$$

can be written as

$$K_H(t) = \int_0^\infty e^{-xt} \mu(dx). \quad (4.67)$$

In other words, the power-law kernel is a continuous superposition of exponential kernels.

This is important because an exponential kernel is exactly the kernel generated by an Ornstein–Uhlenbeck-type factor. Indeed, if

$$dU_t(x) = (-xU_t(x) + \lambda(\theta - V_t)) dt + \nu\sqrt{V_t} dW_t, \quad U_0(x) = 0, \quad (4.68)$$

then its mild form is

$$U_t(x) = \int_0^t e^{-x(t-s)} \left[\lambda(\theta - V_s) ds + \nu\sqrt{V_s} dW_s \right]. \quad (4.69)$$

The parameter x is the mean-reversion speed. Large values of x correspond to very fast factors, which mainly affect very short time scales. Small values of x correspond to slow factors, which affect longer maturities.

Therefore the rough Heston variance can be interpreted formally as a continuum of OU factors indexed by their mean-reversion speed x :

$$V_t = \int_0^\infty U_t(x) \mu(dx), \quad (4.70)$$

where

$$dU_t(x) = (-xU_t(x) + \lambda(\theta - V_t)) dt + \nu\sqrt{V_t} dW_t. \quad (4.71)$$

The same Brownian motion W drives all factors. The non-Markovian memory of the rough model is therefore encoded in the fact that one needs a continuum of factors to represent the fractional kernel exactly.

The lifted Heston model is obtained by replacing this continuous measure μ by a finite discrete measure:

$$\mu(dx) \approx \sum_{i=1}^n c_i^n \delta_{x_i^n}(dx). \quad (4.72)$$

Consequently,

$$K_H(t) = \int_0^\infty e^{-xt} \mu(dx) \approx \sum_{i=1}^n c_i^n e^{-x_i^n t} = K_n(t). \quad (4.73)$$

This is the whole lifting idea. Instead of working with a non-Markovian rough Volterra process, one stores the memory in finitely many Markovian factors $U^{n,1}, \dots, U^{n,n}$. For every finite n , the model is a semimartingale and finite-dimensional Markovian. As n increases and the discrete measure $\sum_i c_i^n \delta_{x_i^n}$ approximates μ , the lifted kernel K_n approaches the rough kernel K_H , and the lifted Heston model approaches rough Heston.

Geometric grid (equation 3.3 of [Abi Jaber \[2019\]](#)). For a geometric ratio $r_n > 1$, partition $(0, \infty)$ into intervals with endpoints $\eta_i^n = r_n^{i-1-n/2}$, $i = 0, 1, \dots, n$ (with n even for the symmetry $\eta_{n/2}^n = r_n^{-1/2}$ to be well-defined). The weights and speeds are derived as follows:

$$c_i^n = \int_{\eta_{i-1}^n}^{\eta_i^n} \mu(dx) = \frac{(r_n^{1-\alpha} - 1) r_n^{(\alpha-1)(1+n/2)}}{\Gamma(\alpha)\Gamma(2-\alpha)} r_n^{(1-\alpha)i}, \quad (4.74)$$

$$x_i^n = \frac{\int_{\eta_{i-1}^n}^{\eta_i^n} x \mu(dx)}{\int_{\eta_{i-1}^n}^{\eta_i^n} \mu(dx)} = \frac{1-\alpha}{2-\alpha} \cdot \frac{r_n^{2-\alpha} - 1}{r_n^{1-\alpha} - 1} \cdot r_n^{i-1-n/2}, \quad (4.75)$$

where $\alpha := H + 1/2 \in (1/2, 1)$.

Theorem 4.10 (Convergence to rough Heston; Theorem A.2 of [Abi Jaber \[2019\]](#)). *If the sequence $(r_n)_{n \geq 1}$ satisfies*

$$r_n \downarrow 1 \quad \text{and} \quad n \log r_n \rightarrow \infty \quad \text{as } n \rightarrow \infty, \quad (4.76)$$

then the lifted Heston model with parameters (c_i^n, x_i^n) given by eq. (4.74)–eq. (4.75) converges in finite-dimensional distributions to the rough Heston model with Hurst index H as $n \rightarrow \infty$.

Reading the conditions. The condition $r_n \downarrow 1$ ensures the partition becomes finer (the discrete approximation $\sum_i c_i^n \delta_{x_i^n}$ converges weakly to μ). The condition $n \log r_n \rightarrow \infty$ ensures that the largest mean-reversion speed $x_n^n \sim r_n^{n/2}$ grows to infinity, capturing the singularity of the rough kernel at zero. A natural choice satisfying both is $r_n = 1 + 10 n^{-0.9}$, used numerically in [Abi Jaber \[2019\]](#) (their equation 3.9).

4.6 Practical choice of n and r_n

Section 3.2 of [Abi Jaber \[2019\]](#) establishes that, in practice, $n = 20$ and $r_{20} = 2.5$ provide an excellent trade-off between accuracy and speed. The ATM skew of the lifted model is then almost indistinguishable from that of the rough Heston for maturities ranging from one week to two years, and pricing is approximately twenty times faster.

The role of r_n for fixed $n = 20$ is the following. From eq. (4.75), the ratio of largest to smallest mean-reversion speeds is $x_n^n / x_1^n \propto r_n^{n-1}$. With $r_{20} = 2.5$, this gives speeds spanning approximately eight orders of magnitude (from 10^{-4} to 10^4 in suitable units), covering all timescales relevant for option maturities from minutes to multiple years. Increasing r_n beyond 2.5 yields diminishing returns in fit quality but does not harm tractability.

4.7 ATM skew of the lifted model

Applying eq. (2.62) to the lifted Heston with kernel $K_n(t) = \sum_i c_i^n e^{-x_i^n t}$:

$$\psi^{\text{lifted}}(T) \approx \frac{\rho\nu}{2\sqrt{V_0} T^2} \sum_{i=1}^n \frac{c_i^n}{x_i^n} \left(T - \frac{1 - e^{-x_i^n T}}{x_i^n} \right). \quad (4.77)$$

Comparison with rough Heston.

- As $T \rightarrow 0$, $K_n(0) = \sum_i c_i^n < \infty$, so $\psi^{\text{lifted}}(T) \rightarrow \rho\nu/(4\sqrt{V_0}) \cdot K_n(0)$, a finite limit. The lifted skew does not blow up for any finite n .
- For intermediate maturities T such that $1/x_n^n \ll T \ll 1/x_1^n$, the sum approximates the integral $\int_0^\infty (1 - e^{-xT})/x \mu(dx)$ which scales as $T^{H-1/2}$. The lifted skew tracks the rough power law over this range.
- As $n \rightarrow \infty$ (with r_n satisfying eq. (4.76)), $\psi^{\text{lifted}}(T) \rightarrow \psi^{\text{rough}}(T)$ uniformly on compact subsets of $(0, \infty)$.

This is a key qualitative observation for our comparison with rough Bergomi: the lifted Heston automatically regularises the short-maturity singularity, since exponentials are bounded at zero. In light of [Abi Jaber and Li \[2025\]](#), who show that rough Bergomi overshoots SPX skew at very short maturities, the lifted Heston may avoid this failure mode by construction.

Chapter 5

The aBergomi Model

The affine Bergomi (aBergomi) model of [Zhu, Loeper, Chen, and Langrené \[2021\]](#) provides a Markovian approximation of the rough Bergomi model of [Bayer, Friz, and Gatheral \[2016\]](#). We first recall the classical Bergomi model from which the rough version is built, then summarise the aBergomi structure and contrast it with the lifted Heston.

5.1 The classical Bergomi model

The Bergomi model [[Bergomi, 2016](#)] approaches stochastic volatility from the opposite end to the Heston family. Rather than prescribing dynamics for the spot variance V_t , it models the entire forward variance curve $\xi_t(T) := \mathbb{E}^{\mathbb{Q}}[V_T \mid \mathcal{F}_t]$ of section [2.3](#) directly, requiring only that each forward variance be a positive $\mathbb{Q}$ -martingale. In its one-factor form the curve evolves lognormally,

$$\frac{d\xi_t(T)}{\xi_t(T)} = \eta e^{-k(T-t)} dW_t, \quad t \leq T, \quad (5.1)$$

with vol-of-vol $\eta > 0$, mean-reversion speed $k > 0$, and a deterministic initial curve $\xi_0(\cdot)$ fitted to the market term structure of variance swaps. Integrating eq. (5.1) and evaluating at $T = t$ gives the spot variance

$$V_t = \xi_t(t) = \xi_0(t) \exp\left(\eta \int_0^t e^{-k(t-s)} dW_s - \frac{\eta^2}{2} \int_0^t e^{-2k(t-s)} ds\right), \quad (5.2)$$

where the second term in the exponential is the Itô correction that enforces $\mathbb{E}^{\mathbb{Q}}[V_t] = \xi_0(t)$, so the model is calibrated to the variance term structure by construction. The stock obeys $dS_t = S_t \sqrt{V_t} dB_t$ with $B = \rho W + \sqrt{1 - \rho^2} W^\perp$ and $\rho \in [-1, 1]$.

Markovian structure. The driving factor $Y_t := \int_0^t e^{-k(t-s)} dW_s$ is an Ornstein–Uhlenbeck process, $dY_t = -k Y_t dt + dW_t$, so eq. (5.2) is Markovian in the finite-dimensional state (S_t, Y_t) , exactly as classical Heston is Markovian in (S_t, V_t) . The general n -factor Bergomi model superposes several such factors, replacing the single kernel $e^{-k(T-t)}$ by $\sum_{i=1}^n \omega_i e^{-k_i(T-t)}$, while keeping a finite-dimensional Markovian state. This is precisely the structural template that the aBergomi model exploits below: it chooses its factors $(\tilde{c}_i, \tilde{x}_i)$ so that the exponential superposition approximates a rough power-law kernel.

The Bergomi ATM skew failure. Applying the Bergomi–Guyon formula eq. (2.62) with the exponential kernel $e^{-k(T-t)}$ yields

$$\psi^{\text{Bergomi}}(T) \approx \frac{\rho\eta}{2T^2} \cdot \frac{1}{k} \left(T - \frac{1 - e^{-kT}}{k} \right) \xrightarrow{T \rightarrow 0} \frac{\rho\eta}{4}. \quad (5.3)$$

As in classical Heston, the short-maturity skew converges to a finite limit, in contradiction with the empirical SPX power law $|\psi(T)| \sim T^{-\alpha}$, $\alpha \in (0, 1/2)$. Replacing the bounded exponential kernel $e^{-k(T-t)}$ by the singular power law $(T-t)^{H-1/2}$ cures this defect and produces the rough Bergomi model, to which we now turn.

5.2 Rough Bergomi: motivation

The rough Bergomi model is, under $\mathbb{Q}$,

$$dS_t = S_t \sqrt{V_t} dB_t, \quad V_t = \xi_0(t) \exp\left(\eta \widehat{W}_t^H - \frac{\eta^2}{2} t^{2H}\right), \quad (5.4)$$

where $\widehat{W}_t^H = \int_0^t (t-s)^{H-1/2} dW_s$ is a Riemann–Liouville fractional Brownian motion with Hurst index H , $\xi_0(\cdot)$ is the initial forward variance curve, and $\eta > 0$ is a vol-of-vol parameter.

The model is lognormal in the variance and produces a clean ATM skew formula at first order:

$$\psi^{\text{rBergomi}}(T) \approx \frac{\rho\eta}{2(H+1/2)(H+3/2)} T^{H-1/2}. \quad (5.5)$$

Like rough Heston, however, it is non-Markovian and does not admit a closed-form characteristic function: pricing requires Monte Carlo.

5.3 The aBergomi model

Zhu, Loeper, Chen, and Langrené [2021] replace the fractional kernel by a sum of exponential kernels:

$$\frac{t^{H-1/2}}{\Gamma(H+1/2)} \approx \sum_{i=1}^n \tilde{c}_i e^{-\tilde{x}_i t}, \quad (5.6)$$

where the weights and speeds $(\tilde{c}_i, \tilde{x}_i)$ are obtained by minimising an L^2 kernel-fitting objective:

$$(\tilde{c}_i, \tilde{x}_i)_i = \arg \min_{(c, x)} \int_{\varepsilon}^T \left(\frac{t^{H-1/2}}{\Gamma(H+1/2)} - \sum_{i=1}^n c_i e^{-x_i t} \right)^2 w(t) dt, \quad (5.7)$$

where $\varepsilon = 1/365$ (one day) is a short-maturity cutoff, T the maturity range, and we take $w \equiv 1$. The resulting variance process is

$$V_t = \xi_0(t) \exp\left(\sum_{i=1}^n \tilde{c}_i X_t^{(i)} - \frac{1}{2} \text{Var}\left(\sum_{i=1}^n \tilde{c}_i X_t^{(i)}\right)\right), \quad X_t^{(i)} = \int_0^t e^{-\tilde{x}_i(t-s)} \eta dW_s. \quad (5.8)$$

	Lifted Heston	aBergomi
Variance dynamics	Square-root (CIR-type)	Lognormal
Approximation method	Geometric grid + Lévy measure	L^2 kernel fit
Parametrisation by H	Canonical (eq. 4.74–4.75)	Optimisation-based
Characteristic function	Closed-form via Riccati ODEs	None
Pricing	Fourier inversion (cosine method)	Monte Carlo
Speed	Fast	Slower
Recovers $H = 0.5$ as special case	Yes (classical Heston)	Yes (classical Bergomi)

Table 5.1: Structural comparison of the lifted Heston and the aBergomi.

Key differences from the lifted Heston.

ATM skew of aBergomi. By the same Bergomi–Guyon argument with kernel eq. (5.6),

$$\psi^{\text{aBergomi}}(T) \approx \frac{\rho\eta}{2T^2} \sum_i \frac{\tilde{c}_i}{\tilde{x}_i} \left(T - \frac{1 - e^{-\tilde{x}_i T}}{\tilde{x}_i} \right), \quad (5.9)$$

which has the same qualitative form as the lifted Heston skew eq. (4.77), but with weights determined by an L^2 fit rather than the canonical Lévy-measure parametrisation.

5.4 Why this comparison has not been done

To our knowledge, no published work systematically compares the lifted Heston (Heston-family Markovian approximation) and the aBergomi (Bergomi-family Markovian approximation) on SPX market data. [Abi Jaber and Li \[2025\]](#) compare rough versus Markovian models within the Bergomi family, but never against the lifted Heston. [Abi Jaber \[2019\]](#) compare lifted Heston only to rough Heston and to classical Heston, on synthetic data. The contribution of this thesis is to fill this gap. We do not have the resources to reproduce the full 2011–2022 comparison of [Abi Jaber and Li \[2025\]](#), so ours is a smaller-scale study.

Chapter 6

The Two-Factor Lifted Heston: Theory

This chapter develops an original extension of the lifted Heston model: a two-factor variant designed to reproduce the global short-versus-long maturity ATM skew structure that single-kernel rough models cannot capture, as documented by [Abi Jaber and Li \[2025\]](#). The chapter is written under the risk-neutral measure $\mathbb{Q}$, since the object of interest is option pricing and calibration to implied volatility surfaces.

6.1 Motivation

[Abi Jaber and Li \[2025\]](#) establish empirically that the universal single-kernel rough scaling

$$\psi(T) \propto T^{H-1/2} \quad (6.1)$$

is too rigid to reproduce the global SPX ATM skew term structure. A single Hurst exponent H controls both the short-maturity explosion and the long-maturity decay. A value of H that fits the short end may misrepresent the long end, and conversely. Their empirical solution is a two-factor Bergomi-type model with one fast component and one slow component, which they show outperforms rough Bergomi on more than ten years of daily SPX implied volatility surfaces.

The two-factor Bergomi model is, however, lognormal in the variance and admits no closed-form characteristic function: pricing requires Monte Carlo. The purpose of the present chapter is to propose an analogous two-factor mechanism inside the Heston family, where the affine structure can be preserved and pricing remains based on ordinary Riccati ODEs and Fourier inversion.

The guiding idea is simple. Write the instantaneous variance as

$$V_t = V_t^{(1)} + V_t^{(2)}, \quad (6.2)$$

where $V^{(1)}$ is a lifted Heston rough block, designed to reproduce the steep short-maturity skew via a multi-scale exponential kernel approximation, and $V^{(2)}$ is a slow CIR block, designed to add flexibility at medium and long maturities.

Here “slow” means a small mean-reversion speed λ_2 , i.e. a long timescale $1/\lambda_2$, in contrast to the fast multi-scale speeds $x_i^{n_1}$ of the rough block. The role of $V^{(2)}$ is clearest at the level of the leading-order ATM skew: because the two variance drivers are independent, $W^1 \perp W^2$, the skew separates additively, $\psi(T) \approx \psi^{(1)}(T) + \psi^{(2)}(T)$, with the strength of each term set by its own leverage ρ_j . A single rough block is governed by one Hurst index H_1 , and hence over its multi-scale window by a single power law $\psi^{(1)}(T) \propto T^{H_1-1/2}$, a near-straight line in log-log, too rigid to match the curved SPX skew term structure with one exponent. The slow block adds an independent term whose sign and size are set by $(\rho_2, \nu_2, V_0^{(2)})$ separately from the rough-block

parameters; with λ_2 small enough that $1/\lambda_2$ exceeds the longest traded maturity, it contributes a roughly constant additive shift across the calibration window (one week to two years). This extra, independently-signed degree of freedom enlarges the family of attainable skew shapes to include the curvature that a single H_1 cannot produce. The skew is given explicitly in proposition 6.12, and remark 6.14 shows why the sum of the two blocks is curved in log–log, whereas a single block is straight.

The construction is inspired by two sources:

- the lifted Heston construction of [Abi Jaber \[2019\]](#), where a rough kernel is approximated by a finite sum of exponentials;
- the two-factor classical Heston specification originally proposed by Christoffersen, Heston and Jacobs (2009), where the stock-return variance is the sum of two independent CIR processes.

Design choices. We adopt the following two-factor specification: each block carries its own square-root diffusion $\nu_j \sqrt{V_t^{(j)}}$ on its own state $V_t^{(j)}$, and the price is written in the double-Heston form of [Christoffersen, Heston and Jacobs \[2009\]](#),

$$\frac{dS_t}{S_t} = \sqrt{V_t^{(1)}} dB_t^{(1)} + \sqrt{V_t^{(2)}} dB_t^{(2)}, \quad (6.3)$$

pairing each price-noise component with its own variance block. The tempting alternative, a single shared Brownian driver (one B correlated with both W^1 and W^2 , on top of a shared diffusion $\sqrt{V_t}$ with $V_t = V_t^{(1)} + V_t^{(2)}$), is not affine: the cross-variation with a rough-block factor,

$$d\langle \log S, \tilde{U}^{1,i} \rangle_t = \rho_1 \nu_1 \sqrt{V_t} \sqrt{V_t^{(1)}} dt = \rho_1 \nu_1 \sqrt{V_t^{(1)}(V_t^{(1)} + V_t^{(2)})} dt,$$

is not affine in the state $(\tilde{U}^{1,i}, V^{(2)})$, so the exponential-affine transform ansatz does not close. Our specification removes this term and, with each block diffusing on its own state, makes V positive by construction (a sum of two non-negative processes) and the model affine in the state, hence tractable via a fully decoupled Riccati system. The discarded alternatives, and the variants we explored, are discussed briefly in remark 6.3 and remark 6.4.

6.2 Definition of the two-factor lifted Heston

Definition 6.1 (Two-factor lifted Heston model). Fix an integer $n_1 \in \mathbb{N}$, weights $c_1^{n_1}, \dots, c_{n_1}^{n_1} \geq 0$, and mean-reversion speeds $0 \leq x_1^{n_1} < \dots < x_{n_1}^{n_1}$ associated with a Hurst parameter $H_1 \in (0, 1/2)$ via the canonical parametrisation eq. (4.74)–eq. (4.75). Fix further parameters

$$\nu_1, \nu_2 > 0, \quad \lambda_1, \theta_1, \lambda_2, \theta_2 \geq 0, \quad \rho_1, \rho_2 \in [-1, 1],$$

and initial values $S_0 > 0$, $V_0^{(1)} \geq 0$, $V_0^{(2)} \geq 0$.

Let $(W^1, W^{1,\perp}, W^2, W^{2,\perp})$ be four mutually independent standard Brownian motions on a filtered probability space $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{Q})$ satisfying the usual conditions. Define, for $j \in \{1, 2\}$,

$$B_t^{(j)} := \rho_j W_t^j + \sqrt{1 - \rho_j^2} W_t^{j,\perp}. \quad (6.4)$$

Then $B^{(1)}$ and $B^{(2)}$ are independent Brownian motions, and each $B^{(j)}$ has instantaneous correlation ρ_j with the variance driver W^j of block j .

The two-factor lifted Heston model is the system

$$\frac{dS_t}{S_t} = \sqrt{V_t^{(1)}} dB_t^{(1)} + \sqrt{V_t^{(2)}} dB_t^{(2)}, \quad (6.5)$$

$$V_t := V_t^{(1)} + V_t^{(2)}, \quad (6.6)$$

$$V_t^{(1)} = V_0^{(1)} + \sum_{i=1}^{n_1} c_i^{n_1} \tilde{U}_t^{1,i}, \quad (6.7)$$

$$d\tilde{U}_t^{1,i} = \left(-x_i^{n_1} \tilde{U}_t^{1,i} + \lambda_1(\theta_1 - V_t^{(1)}) \right) dt + \nu_1 \sqrt{V_t^{(1)}} dW_t^1, \quad \tilde{U}_0^{1,i} = 0, \quad (6.8)$$

$$dV_t^{(2)} = \lambda_2(\theta_2 - V_t^{(2)}) dt + \nu_2 \sqrt{V_t^{(2)}} dW_t^2. \quad (6.9)$$

Reading the definition. The model decomposes the variance into two structurally independent blocks:

- $V_t^{(1)}$ is a lifted Heston variance in the sense of definition 4.1, approximating a rough kernel $K_{H_1}(t) = t^{H_1-1/2}/\Gamma(H_1+1/2)$ via the multi-exponential kernel $K_{n_1}(t) = \sum_i c_i^{n_1} e^{-x_i^{n_1}t}$. It carries its own square-root diffusion $\nu_1 \sqrt{V_t^{(1)}}$ and is driven by W^1 . This is the rough block, responsible for the steep short-maturity skew via its multi-scale structure.
- $V_t^{(2)}$ is a classical CIR process with mean-reversion speed λ_2 , long-run level θ_2 , vol-of-vol ν_2 , and diffusion $\nu_2 \sqrt{V_t^{(2)}}$, driven by W^2 .
- The two blocks are completely independent at the SDE level: W^1 and W^2 are independent, and the diffusion of each block depends only on its own variance level $V_t^{(j)}$. This is the structural difference with a shared $\sqrt{V_t}$ formulation.
- The price-volatility correlations ρ_1 and ρ_2 are independent risk-neutral parameters. They control how strongly each variance block contributes to the implied volatility skew. The independence of ρ_1 and ρ_2 is the crucial degree of freedom that breaks the single-kernel skew rigidity.

Remark 6.2 (Log-price dynamics). The log-price $X_t := \log S_t$ admits the expanded representation by itô formula :

$$\begin{aligned} dX_t = & -\frac{1}{2}(V_t^{(1)} + V_t^{(2)}) dt \\ & + \rho_1 \sqrt{V_t^{(1)}} dW_t^1 + \sqrt{1 - \rho_1^2} \sqrt{V_t^{(1)}} dW_t^{1,\perp} \\ & + \rho_2 \sqrt{V_t^{(2)}} dW_t^2 + \sqrt{1 - \rho_2^2} \sqrt{V_t^{(2)}} dW_t^{2,\perp}. \end{aligned} \quad (6.10)$$

This is the form used in the proof of theorem 6.7 below. The four mutually independent Brownian motions appear explicitly, which simplifies the computation of the quadratic variation.

Remark 6.3 (Why the price equation is not single-noise). One could also have thought of the two-factor model as being defined through the single-noise price specification $dS_t/S_t = \sqrt{V_t} dB_t$, with B correlated with both W^1 and W^2 . This is natural because V_t is the total variance, but it destroys affine tractability.

Indeed, with $X_t = \log S_t$ in this specification, the cross-variation between X and a rough-block factor is

$$d\langle X, \tilde{U}^{1,i} \rangle_t = \rho_1 \nu_1 \sqrt{V_t} \sqrt{V_t^{(1)}} dt = \rho_1 \nu_1 \sqrt{V_t^{(1)}(V_t^{(1)} + V_t^{(2)})} dt.$$

The right-hand side is not affine in the state variables, and the exponential-affine transform ansatz does not close.

The double-Heston specification eq. (6.5) avoids this problem by pairing each price-noise component with its own variance block. The instantaneous total variance is preserved:

$$\frac{d\langle \log S \rangle_t}{dt} = V_t^{(1)} + V_t^{(2)} = V_t,$$

but all covariation terms become affine in the state. This is the standard construction of Christoffersen, Heston and Jacobs (2009) for two-factor classical Heston models, adapted here by replacing one CIR block with a lifted Heston block.

Remark 6.4 (Why not a shared $\sqrt{V}$ diffusion?). A second natural alternative is the model

$$V_t = \xi_0(t) + \tilde{U}_t + Z_t, \quad d\tilde{U}_t^{1,i} = \dots dt + \nu_1 \sqrt{V_t} dW_t^1, \quad dZ_t = -\kappa Z_t dt + \nu_2 \sqrt{V_t} dW_t^2,$$

where both factor families share the diffusion coefficient $\sqrt{V_t}$ of the aggregate variance.

If the two Brownian motions are in fact the same, $W^1 = W^2$, then all factors are driven by a single noise source. In that degenerate case the model can be viewed as a standard lifted-Heston-type system with one additional factor, and the usual scalar Volterra representation is recovered.

The situation is different when W^1 and W^2 are independent. The aggregate variance then has two distinct stochastic convolution components: one rough/lifted component driven by W^1 and one slow exponential component driven by W^2 . Hence V cannot be written in the scalar lifted-Heston form

$$V_t = g_0(t) + \int_0^t K(t-s)b(V_s)ds + \int_0^t K(t-s)\sigma(V_s)dW_s,$$

with one completely monotone kernel K and one Brownian driver W . Equivalently, the stochastic part of V is not obtained by applying a single kernel to a single square-root noise. It is a sum of two different kernels applied to two independent Brownian directions. Therefore the model falls outside the scalar affine-Volterra/lifted-Heston framework used in Abi Jaber [Abi Jaber \[2019\]](#) and Abi Jaber–Larsson–Pulido [Abi Jaber, Larsson, and Pulido \[2019\]](#) to obtain positivity and affine tractability.

Since the state constraint is the half-space $\{V \geq 0\}$ rather than a coordinatewise CIR state space, positivity is no longer inherited from the standard square-root argument. A possible way to investigate failure of positivity would be to rewrite the system in Stratonovich form and apply a Stroock-Varadhan support-theorem argument to construct controlled paths crossing the boundary $V = 0$. We do not pursue this strategy here.

For this reason we adopt the structurally cleaner formulation of definition 6.1, in which each block has its own square-root diffusion. Positivity then follows from the positivity of the individual blocks, and the total variance is simply a sum of non-negative components.

Reduced market-calibration parametrisation. For the empirical comparison with aBergomi (see section 7.2), we use the market forward variance curve $\xi_0(\cdot)$ as an exogenous input, following the standard practical calibration convention for stochastic volatility models. Strictly speaking, in the reduced market-calibration parametrisation used below, where we set $\lambda_1 = 0$, the scalar mean-reverting form cannot match an arbitrary market forward variance curve exactly for all maturities. Indeed, when $\lambda_1 = 0$,

$$\mathbb{E}^\mathbb{Q}[V_t^{(1)}] = V_0^{(1)},$$

whereas the CIR block satisfies

$$\mathbb{E}^\mathbb{Q}[V_t^{(2)}] = \theta_2 + (V_0^{(2)} - \theta_2)e^{-\lambda_2 t}.$$

Thus

$$\mathbb{E}^{\mathbb{Q}}[V_t] = V_0^{(1)} + \theta_2 + (V_0^{(2)} - \theta_2)e^{-\lambda_2 t},$$

that is, a constant plus one exponential. This cannot reproduce a general market forward variance curve $\xi_0(t)$.

More generally, if $\lambda_1 \neq 0$, the mean of the lifted block is no longer constant, but it is still generated by a finite-dimensional linear Volterra system driven by the finite exponential kernel $K_1^n(t) = \sum_i c_i e^{-x_i t}$. Hence the scalar parametrisation still gives only a restricted parametric family of forward-variance curves. To match a general market curve exactly, one must instead use deterministic input curves $g_0^{(1)}$ and/or $g_0^{(2)}$ chosen so that

$$\mathbb{E}^{\mathbb{Q}}[V_t^{(1)}] + \mathbb{E}^{\mathbb{Q}}[V_t^{(2)}] = \xi_0(t), \quad t \geq 0.$$

This is the deterministic-input formulation used in practical lifted Heston calibration.

The reduced calibration vector is

$$(H_1, \nu_1, \rho_1, \lambda_2, \theta_2, \nu_2, \rho_2), \quad (6.11)$$

with $n_1 = 20$ and $r_{n_1} = 2.5$ fixed. The slow block is initialised at its stationary level, $V_0^{(2)} = \theta_2$, so its mean is constant, $\mathbb{E}^{\mathbb{Q}}[V_t^{(2)}] = \theta_2$ for all t . The block-1 forward-variance input is then set to

$$g_0^{(1)}(t) = \xi_0(t) - \theta_2,$$

and since block 1 runs in forward-variance form ($\lambda_1 = 0$, hence $\mathbb{E}^{\mathbb{Q}}[V_t^{(1)}] = g_0^{(1)}(t)$), the total expected variance reproduces the market curve exactly at every maturity:

$$\mathbb{E}^{\mathbb{Q}}[V_t^{(1)}] + \mathbb{E}^{\mathbb{Q}}[V_t^{(2)}] = (\xi_0(t) - \theta_2) + \theta_2 = \xi_0(t), \quad t \geq 0.$$

Thus block 1 carries the full term structure of ξ_0 , while θ_2 fixes only the constant level absorbed by the slow block, equivalently the initial split of variance between the two blocks. The split fraction β of the motivating discussion corresponds to $\beta = 1 - \theta_2/\xi_0(0)$, so that $V_0^{(1)} = \beta \xi_0(0)$ and $V_0^{(2)} = (1 - \beta) \xi_0(0)$; we calibrate θ_2 directly.

The instantaneous variance is the sum of two blocks, $V_t = V_t^{(1)} + V_t^{(2)}$, and the model is required to reproduce the market forward-variance curve exactly, $\mathbb{E}^{\mathbb{Q}}[V_t] = \xi_0(t)$ for all t . Block 2 is a CIR factor with known mean $\mathbb{E}^{\mathbb{Q}}[V_t^{(2)}] = \theta_2 + (V_0^{(2)} - \theta_2)e^{-\lambda_2 t}$, and block 1 absorbs the remainder through the deterministic input

$$g_0^{(1)}(t) = \xi_0(t) - \mathbb{E}^{\mathbb{Q}}[V_t^{(2)}],$$

so that the two block means sum to $\xi_0(t)$ identically; at inception this splits the total initial variance as $V_0^{(1)} + V_0^{(2)} = \xi_0(0)$.

Since $g_0^{(1)}(t)$ is itself a variance, it must satisfy $g_0^{(1)}(t) \geq 0$ at every t in the window. As $\mathbb{E}^{\mathbb{Q}}[V_t^{(2)}]$ lies between $V_0^{(2)}$ and θ_2 , requiring $\xi_0(t) \geq \mathbb{E}^{\mathbb{Q}}[V_t^{(2)}]$ at all maturities forces

$$V_0^{(2)} \leq \min_{t \in [0, T_{\max}]} \xi_0(t),$$

so block 2's initial variance is capped not by the total $\xi_0(0)$ but by the minimum of the forward-variance curve over the whole window.

This cap is what renders block 2 inert. Its leading-order ATM-skew contribution scales with its initial variance, $\psi^{(2)}(T) \propto \rho_2 \nu_2 V_0^{(2)} / \lambda_2$, and on SPX the forward-variance curve falls steeply with maturity, so $\min_t \xi_0(t)$ is small: on 2024-08-05, $\xi_0(0) = 0.109$ while $\min_t \xi_0(t) = 0.005$. The positivity cap then confines block 2 to at most about 5% of the initial variance ($\beta \approx 0.95$),

making $V_0^{(2)}$, and hence $\psi^{(2)}$, negligible irrespective of ρ_2 and ν_2 . This is a structural property of the additive shared- ξ_0 construction on a downward-sloping variance curve, not a numerical accident.

A natural remedy for this structural limitation is to make the two blocks symmetric, each a lifted-Heston-type block with its own non-negative deterministic forcing,

$$V_t = \left(g_0^{(1)}(t) + \sum_{i=1}^{n_1} c_i^{(1)} U_t^{(1),i} \right) + \left(g_0^{(2)}(t) + \sum_{j=1}^{n_2} c_j^{(2)} U_t^{(2),j} \right), \quad g_0^{(1)} + g_0^{(2)} = \xi_0,$$

with $g_0^{(1)}, g_0^{(2)} \geq 0$ (for instance $g_0^{(2)} = w \xi_0$, $g_0^{(1)} = (1-w)\xi_0$, $w \in [0, 1]$), block 2 reverting around $g_0^{(2)}(t)$ rather than a constant θ_2 . Because the forcing curves are partitioned by addition rather than subtraction, positivity holds by construction for any w , and the cap $V_0^{(2)} \leq \min_t \xi_0$ disappears: the optimiser is then free to allocate a genuine share of variance to the second block and to determine whether it earns that weight. The smooth block may be taken as a single factor ($n_2 = 1$) for parsimony. The square-root diffusion is retained, so the model stays affine (each forcing enters the deterministic ϕ -ODE rather than the Riccati system) and pricing remains Fourier-based. Re-deriving the second block's characteristic-function contribution for a time-dependent forcing, and recalibrating, is left to future work.

6.3 Existence, uniqueness, and positivity

The main analytical advantage of definition 6.1 is that positivity becomes transparent: the two variance blocks are separately non-negative, and the total variance is their sum.

Proposition 6.5 (Existence, uniqueness and positivity). *Under the assumptions of definition 6.1, there exists a unique continuous strong solution*

$$\left(S_t, V_t^{(1)}, V_t^{(2)}, (\tilde{U}_t^{1,i})_{1 \leq i \leq n_1} \right)_{t \geq 0}$$

to the system eq. (6.5)–eq. (6.9) such that

$$V_t^{(1)} \geq 0 \quad \text{and} \quad V_t^{(2)} \geq 0 \quad \text{a.s. for all } t \geq 0.$$

In particular, $V_t = V_t^{(1)} + V_t^{(2)} \geq 0$ a.s. for all $t \geq 0$, and S is a strictly positive process which is a true $\mathbb{Q}$ -martingale under the integrability condition stated in the proof of theorem 6.7.

Proof. The two blocks are completely decoupled at the SDE level. The system $(V_t^{(1)}, (\tilde{U}_t^{1,i})_{1 \leq i \leq n_1})$ depends only on W^1 and on its own state, not on $V^{(2)}$ or W^2 . It is therefore a standard lifted Heston model in the sense of definition 4.1, with deterministic input $g_0^{n_1}(t) = V_0^{(1)} + \lambda_1 \theta_1 \sum_i c_i^{n_1} \int_0^t e^{-x_i^{n_1}(t-s)} ds$. By the existence theorem of [Abi Jaber \[2019, Theorem A.1\]](#), this system admits a unique continuous strong solution with $V_t^{(1)} \geq 0$ a.s. for all $t \geq 0$. The process $V^{(2)}$ depends only on W^2 and on its own state. It is the classical Cox–Ingersoll–Ross process (CIR, 1985). For every initial condition $V_0^{(2)} \geq 0$, the CIR equation admits a pathwise unique non-negative strong solution by standard results on square-root diffusions. Since the Brownian motions $(W^1, W^{1,\perp}, W^2, W^{2,\perp})$ are jointly independent, the two block solutions can be defined on a common filtered probability space without inducing any coupling between them. Hence $V_t = V_t^{(1)} + V_t^{(2)} \geq 0$ a.s. for all $t \geq 0$. The stock price satisfies

$$S_t = S_0 \mathcal{E} \left(\int_0^t \sqrt{V_s^{(1)}} dB_s^{(1)} + \int_0^t \sqrt{V_s^{(2)}} dB_s^{(2)} \right)_t,$$

where $\mathcal{E}$ denotes the Doléans–Dade exponential. Since $V^{(1)}, V^{(2)}$ are continuous and non-negative, the stochastic integrals are well-defined on every finite horizon, and S is strictly positive. The

true martingale property of S follows from the affine transform formula in theorem 6.7 evaluated at $u = 1$: the Riccati system has the trivial solution $\psi^{(1),i} \equiv 0$, $\psi^{(2)} \equiv 0$ at $u = 1$, giving $\mathbb{E}^\mathbb{Q}[S_T \mid \mathcal{F}_t] = S_t$. □

Remark 6.6 (Contrast with the shared- $\sqrt{V}$ model). The contrast with the formulation of remark 6.4 is striking. By giving each block its own diffusion coefficient, we have transformed a non-trivial question of stochastic invariance on a half-space (which fails) into a trivial sum-of-positives argument. The price is that the cross-block dependence through the diffusion is removed; we show in proposition 6.12 below that this restriction does not reduce the empirical fitting power of the model at first order in vol-of-vol.

6.4 Affine characteristic function and Riccati system

The independence of the two blocks at the SDE level translates into an exact decoupling of the Riccati system at the affine-transform level. The result is a model whose conditional Fourier–Laplace transform is given explicitly in terms of the solution of two independent Riccati ODE systems: one n_1 -dimensional lifted Heston system, and one scalar Heston system.

Theorem 6.7 (Affine characteristic function). *Let $T > 0$ be a finite maturity and let $u \in \mathbb{C}$ with $\Re(u) \in [0, 1]$. Suppose there exist absolutely continuous functions $\psi^{(1),i} : [0, T] \rightarrow \mathbb{C}$ for $i = 1, \dots, n_1$ and $\psi^{(2)} : [0, T] \rightarrow \mathbb{C}$ solving the decoupled Riccati system*

$$\frac{d\psi^{(1),i}}{ds}(s) = -x_i^{n_1} \psi^{(1),i}(s) + F_1(u, \Psi^{(1)}(s)), \quad \psi^{(1),i}(0) = 0, \quad (6.12)$$

$$\frac{d\psi^{(2)}}{ds}(s) = F_2(u, \psi^{(2)}(s)), \quad \psi^{(2)}(0) = 0, \quad (6.13)$$

on $[0, T]$, where

$$\Psi^{(1)}(s) := \sum_{j=1}^{n_1} c_j^{n_1} \psi^{(1),j}(s), \quad (6.14)$$

and

$$F_1(u, v) := \frac{1}{2}(u^2 - u) + (\rho_1 \nu_1 u - \lambda_1) v + \frac{\nu_1^2}{2} v^2, \quad (6.15)$$

$$F_2(u, w) := \frac{1}{2}(u^2 - u) + (\rho_2 \nu_2 u - \lambda_2) w + \frac{\nu_2^2}{2} w^2. \quad (6.16)$$

Define

$$\phi(t, T) := \int_0^{T-t} \left[V_0^{(1)} F_1(u, \Psi^{(1)}(s)) + \lambda_1 \theta_1 \Psi^{(1)}(s) + \lambda_2 \theta_2 \psi^{(2)}(s) \right] ds. \quad (6.17)$$

Then, for all $0 \leq t \leq T$,

$$\begin{aligned} \mathbb{E}^\mathbb{Q} \left[e^{u \log S_T} \mid \mathcal{F}_t \right] = & \exp \left(\phi(t, T) + u \log S_t + \sum_{i=1}^{n_1} c_i^{n_1} \psi^{(1),i}(T-t) \tilde{U}_t^{1,i} \right. \\ & \left. + \psi^{(2)}(T-t) V_t^{(2)} \right). \end{aligned} \quad (6.18)$$

Proof. The proof proceeds in three steps: (i) check the terminal condition; (ii) compute the drift of the candidate exponential martingale via Itô's formula and verify that it vanishes; (iii) verify integrability so that the candidate is a true martingale.

Throughout the proof, write

$$Y_t := \phi(t, T) + u X_t + \sum_{i=1}^{n_1} c_i^{n_1} \psi^{(1),i}(T-t) \tilde{U}_t^{1,i} + \psi^{(2)}(T-t) V_t^{(2)}, \quad (6.19)$$

with $X_t := \log S_t$, and set $M_t := e^{Y_t}$. We use the shorthand $\Psi_t^{(1)} := \Psi^{(1)}(T - t)$ and $\psi_t^{(2)} := \psi^{(2)}(T - t)$.

First, at $t = T$, $\psi^{(1),i}(0) = 0$, $\psi^{(2)}(0) = 0$, and $\phi(T, T) = 0$ (the integral in eq. (6.17) is over an empty interval). Therefore $M_T = e^{uX_T} = e^{u \log S_T}$.

Next, we compute the finite-variation part of dY_t term by term, using eq. (6.10) for the log-price.

Contribution of $\phi(t, T)$: $\partial_t \phi(t, T) dt$.

Contribution of uX_t : $-\frac{u}{2}(V_t^{(1)} + V_t^{(2)}) dt$.

Contribution of $\sum_i c_i^{n_1} \psi^{(1),i}(T - t) \tilde{U}_t^{1,i}$: by the product rule and eq. (6.8),

$$\begin{aligned} & \sum_i c_i^{n_1} \left[-(\psi^{(1),i})'(T - t) \tilde{U}_t^{1,i} + \psi^{(1),i}(T - t) (-x_i^{n_1} \tilde{U}_t^{1,i} + \lambda_1 \theta_1 - \lambda_1 V_t^{(1)}) \right] \\ &= - \sum_i c_i^{n_1} \left[(\psi^{(1),i})'(T - t) + x_i^{n_1} \psi^{(1),i}(T - t) \right] \tilde{U}_t^{1,i} + \lambda_1 \theta_1 \Psi_t^{(1)} - \lambda_1 \Psi_t^{(1)} V_t^{(1)}. \end{aligned}$$

Contribution of $\psi^{(2)}(T - t) V_t^{(2)}$: by the product rule and eq. (6.9),

$$-(\psi^{(2)})'(T - t) V_t^{(2)} + \psi_t^{(2)} (\lambda_2 \theta_2 - \lambda_2 V_t^{(2)}).$$

Turning to the quadratic variation, the martingale part of dY_t is

$$\begin{aligned} dY_t^{\text{mart}} &= u \left(\rho_1 \sqrt{V_t^{(1)}} dW_t^1 + \sqrt{1 - \rho_1^2} \sqrt{V_t^{(1)}} dW_t^{1,\perp} + \rho_2 \sqrt{V_t^{(2)}} dW_t^2 + \sqrt{1 - \rho_2^2} \sqrt{V_t^{(2)}} dW_t^{2,\perp} \right) \\ &\quad + \nu_1 \Psi_t^{(1)} \sqrt{V_t^{(1)}} dW_t^1 + \nu_2 \psi_t^{(2)} \sqrt{V_t^{(2)}} dW_t^2. \end{aligned}$$

Grouping by Brownian motion:

$$\begin{aligned} dY_t^{\text{mart}} &= (u\rho_1 + \nu_1 \Psi_t^{(1)}) \sqrt{V_t^{(1)}} dW_t^1 + u \sqrt{1 - \rho_1^2} \sqrt{V_t^{(1)}} dW_t^{1,\perp} \\ &\quad + (u\rho_2 + \nu_2 \psi_t^{(2)}) \sqrt{V_t^{(2)}} dW_t^2 + u \sqrt{1 - \rho_2^2} \sqrt{V_t^{(2)}} dW_t^{2,\perp}. \end{aligned}$$

Since the four Brownian motions are mutually independent, the quadratic variation is the sum of squares of the coefficients:

$$\begin{aligned} \frac{d\langle Y \rangle_t}{dt} &= V_t^{(1)} \left[(u\rho_1 + \nu_1 \Psi_t^{(1)})^2 + u^2(1 - \rho_1^2) \right] \\ &\quad + V_t^{(2)} \left[(u\rho_2 + \nu_2 \psi_t^{(2)})^2 + u^2(1 - \rho_2^2) \right]. \end{aligned}$$

Expanding $(u\rho_j + \nu_j \cdot)^2 = u^2 \rho_j^2 + 2u\rho_j \nu_j \cdot + \nu_j^2 (\cdot)^2$ and adding $u^2(1 - \rho_j^2)$:

$$\frac{d\langle Y \rangle_t}{dt} = V_t^{(1)} [u^2 + 2u\rho_1 \nu_1 \Psi_t^{(1)} + \nu_1^2 (\Psi_t^{(1)})^2] + V_t^{(2)} [u^2 + 2u\rho_2 \nu_2 \psi_t^{(2)} + \nu_2^2 (\psi_t^{(2)})^2]. \quad (6.20)$$

Note that eq. (6.20) is affine in $(V_t^{(1)}, V_t^{(2)})$.

By Itô's formula,

$$\frac{dM_t}{M_t} = dY_t + \frac{1}{2} d\langle Y \rangle_t.$$

Substituting $V_t^{(1)} = V_0^{(1)} + \sum_j c_j^{n_1} \tilde{U}_t^{1,j}$ in the drift contributions proportional to $V_t^{(1)}$, the drift of dM_t/M_t becomes a linear combination of:

- the state variables $\tilde{U}_t^{1,i}$ for each i ;
- the state variable $V_t^{(2)}$;

- state-independent terms (functions of t only, plus constants involving $V_0^{(1)}$ and $\lambda_j \theta_j$).

For the drift to vanish identically (i.e., for all values of the state variables), each coefficient must vanish separately.

Coefficient of $\tilde{U}_t^{1,i}$. Dividing by $c_i^{n_1}$. If some $c_i^{n_1} = 0$, the corresponding factor can be removed from the state, so without loss of generality we assume $c_i^{n_1} > 0$.

$$-(\psi^{(1),i})'(T-t) - x_i^{n_1} \psi^{(1),i}(T-t) - \frac{u}{2} - \lambda_1 \Psi_t^{(1)} + \frac{1}{2}[u^2 + 2u\rho_1\nu_1\Psi_t^{(1)} + \nu_1^2(\Psi_t^{(1)})^2] = 0.$$

Rearranging:

$$(\psi^{(1),i})'(T-t) = -x_i^{n_1} \psi^{(1),i}(T-t) + \underbrace{\frac{1}{2}(u^2 - u) + (\rho_1\nu_1 u - \lambda_1)\Psi_t^{(1)} + \frac{\nu_1^2}{2}(\Psi_t^{(1)})^2}_{=F_1(u, \Psi_t^{(1)})}.$$

In the variable $s = T - t$, using $\frac{d}{dt}\psi^{(1),i}(T-t) = -(\psi^{(1),i})'(T-t)$,

$$\frac{d\psi^{(1),i}}{ds}(s) = -x_i^{n_1} \psi^{(1),i}(s) + F_1(u, \Psi^{(1)}(s)),$$

which is precisely eq. (6.12).

Coefficient of $V_t^{(2)}$.

$$-\frac{u}{2} - (\psi^{(2)})'(T-t) - \lambda_2 \psi_t^{(2)} + \frac{1}{2}[u^2 + 2u\rho_2\nu_2\psi_t^{(2)} + \nu_2^2(\psi_t^{(2)})^2] = 0,$$

i.e., $(\psi^{(2)})'(T-t) = F_2(u, \psi_t^{(2)})$. In the s variable, $\frac{d\psi^{(2)}}{ds}(s) = F_2(u, \psi^{(2)}(s))$, which is precisely eq. (6.13).

State-independent coefficient. Collecting terms not multiplied by any $\tilde{U}_t^{1,i}$ or $V_t^{(2)}$, and using that $V_t^{(1)}$ contributes a $V_0^{(1)}$ piece in addition to the $\tilde{U}_t^{1,j}$ pieces,

$$\partial_t \phi(t, T) + V_0^{(1)} F_1(u, \Psi_t^{(1)}) + \lambda_1 \theta_1 \Psi_t^{(1)} + \lambda_2 \theta_2 \psi_t^{(2)} = 0,$$

with terminal condition $\phi(T, T) = 0$. Integrating from t to T and changing variable $s = T - r$ yields exactly eq. (6.17).

All three coefficient equations are simultaneously satisfied by the system eq. (6.12)–eq. (6.13) together with eq. (6.17). The drift of dM_t/M_t therefore vanishes, and M is a local martingale.

It remains to verify that M is a true martingale. For real $u \in [0, 1]$, the exponential-affine local martingale M is non-negative. Hence it is a supermartingale by Fatou's lemma and

$$\mathbb{E}^{\mathbb{Q}}[M_T] \leq M_0.$$

It remains to show that no mass is lost, i.e.

$$\mathbb{E}^{\mathbb{Q}}[M_T] = M_0.$$

This follows from the independent-block structure. Indeed, write

$$X_t = \log S_0 + X_t^{(1)} + X_t^{(2)},$$

where $X^{(1)}$ is driven only by $(W^1, W^{1,\perp})$ and $X^{(2)}$ only by $(W^2, W^{2,\perp})$. The Brownian pairs are independent. The exponential-affine process therefore factorises as

$$M_t = e^{u \log S_0} M_t^{(1)} M_t^{(2)},$$

where $M^{(1)}$ is the lifted-Heston exponential-affine martingale and $M^{(2)}$ is the classical Heston/CIR exponential-affine martingale. By the standard affine-transform results for the lifted Heston and CIR blocks,

$$\mathbb{E}^{\mathbb{Q}}[M_T^{(1)}] = M_0^{(1)}, \quad \mathbb{E}^{\mathbb{Q}}[M_T^{(2)}] = M_0^{(2)}.$$

Using independence,

$$\mathbb{E}^{\mathbb{Q}}[M_T] = e^{u \log S_0} \mathbb{E}^{\mathbb{Q}}[M_T^{(1)}] \mathbb{E}^{\mathbb{Q}}[M_T^{(2)}] = M_0.$$

Thus M is a true martingale. Since $M_T = e^{uX_T}$, we obtain

$$M_t = \mathbb{E}^{\mathbb{Q}}[e^{uX_T} \mid \mathcal{F}_t],$$

which proves the affine transform formula for real values of u . The argument above proves the transform formula for real $u \in [0, 1]$. The argument above proves the transform formula for real $u \in [0, 1]$. We now extend it to complex u with $0 < \Re(u) < 1$ by analytic continuation.

Let $A \in \mathcal{F}_t$ and define

$$F_A(u) := \mathbb{E}^{\mathbb{Q}}[\mathbf{1}_A e^{uX_T}], \quad G_A(u) := \mathbb{E}^{\mathbb{Q}}[\mathbf{1}_A M_t(u)],$$

where $M_t(u)$ denotes the exponential-affine expression obtained above.

The reason for testing against all $A \in \mathcal{F}_t$ is the standard characterisation of conditional expectation: proving

$$\mathbb{E}^{\mathbb{Q}}[\mathbf{1}_A e^{uX_T}] = \mathbb{E}^{\mathbb{Q}}[\mathbf{1}_A M_t(u)] \quad \text{for all } A \in \mathcal{F}_t$$

is equivalent to proving

$$M_t(u) = \mathbb{E}^{\mathbb{Q}}[e^{uX_T} \mid \mathcal{F}_t].$$

We claim that both F_A and G_A are holomorphic on the open strip

$$\mathcal{S} := \{u \in \mathbb{C} : 0 < \Re(u) < 1\}.$$

For F_A , this follows from dominated holomorphy. Indeed, for every fixed ω , the map

$$u \longmapsto \mathbf{1}_A(\omega) e^{uX_T(\omega)}$$

is entire. Moreover, if $u \in \mathcal{S}$, then

$$|e^{uX_T}| = e^{\Re(u)X_T} = S_T^{\Re(u)}.$$

Since $0 < \Re(u) < 1$, and since for every $x \geq 0$ and every $a \in [0, 1]$ one has $x^a \leq 1 + x$, we obtain

$$|e^{uX_T}| = S_T^{\Re(u)} \leq 1 + S_T.$$

The real $u = 1$ case proved above gives

$$\mathbb{E}^{\mathbb{Q}}[S_T \mid \mathcal{F}_t] = S_t,$$

so S is a true $\mathbb{Q}$ -martingale. In particular,

$$\mathbb{E}^{\mathbb{Q}}[S_T] = S_0 < \infty.$$

Hence $1 + S_T$ is integrable. Therefore the family

$$\left(\mathbf{1}_A e^{uX_T} \right)_{u \in \mathcal{S}}$$

is locally dominated by an integrable random variable, and the expectation

$$F_A(u) = \mathbb{E}^{\mathbb{Q}}[\mathbf{1}_A e^{uX_T}]$$

is holomorphic on $\mathcal{S}$.

For G_A , recall that

$$M_t(u) = \exp\left(\phi(t, T; u) + uX_t + \sum_i c_i^{n_1} \psi^{(1),i}(T-t; u) \tilde{U}_t^{1,i} + \psi^{(2)}(T-t; u) V_t^{(2)}\right).$$

The Riccati equations have coefficients which are polynomial in u . By the standard analytic-dependence theorem for ODEs, the solutions

$$u \mapsto \psi^{(1),i}(\cdot; u), \quad u \mapsto \psi^{(2)}(\cdot; u), \quad u \mapsto \phi(\cdot, T; u)$$

depend holomorphically on u on the strip where the Riccati solutions exist. Thus $u \mapsto M_t(u)$ is holomorphic pointwise, and the finite-horizon moment bounds used above justify taking expectation. Hence G_A is holomorphic on $\mathcal{S}$.

For real $u \in (0, 1)$, the martingale argument above gives

$$F_A(u) = G_A(u).$$

The interval $(0, 1)$ has accumulation points inside the connected domain $\mathcal{S}$. Therefore, by the identity theorem for holomorphic functions,

$$F_A(u) = G_A(u), \quad u \in \mathcal{S}.$$

Since this equality holds for every $A \in \mathcal{F}_t$, the conditional expectation characterisation gives

$$M_t(u) = \mathbb{E}^{\mathbb{Q}}[e^{uX_T} \mid \mathcal{F}_t], \quad 0 < \Re(u) < 1.$$

It remains to pass to the boundary of the strip. If $\Re(u) = 0$, choose $u_n = u + \frac{1}{n}$. If $\Re(u) = 1$, choose $u_n = u - \frac{1}{n}$. In both cases $u_n \in \mathcal{S}$ and $u_n \rightarrow u$. Moreover,

$$|e^{u_n X_T}| = S_T^{\Re(u_n)} \leq 1 + S_T,$$

so dominated convergence gives

$$\mathbb{E}^{\mathbb{Q}}[\mathbf{1}_A e^{u_n X_T}] \longrightarrow \mathbb{E}^{\mathbb{Q}}[\mathbf{1}_A e^{u X_T}].$$

On the right-hand side, $M_t(u_n) \rightarrow M_t(u)$ by continuity of the Riccati solutions in the parameter u . Therefore the identity extends to $\Re(u) = 0$ and $\Re(u) = 1$, whenever the corresponding boundary moments are finite. Hence the affine transform formula holds for all

$$u \in \mathbb{C} \quad \text{with} \quad \Re(u) \in [0, 1].$$

□

Proof. Set $X_t = \log S_t$. By the double-Heston price specification,

$$X_t = \log S_0 + X_t^{(1)} + X_t^{(2)},$$

where

$$dX_t^{(1)} = -\frac{1}{2}V_t^{(1)}dt + \sqrt{V_t^{(1)}}dB_t^{(1)}, \quad dX_t^{(2)} = -\frac{1}{2}V_t^{(2)}dt + \sqrt{V_t^{(2)}}dB_t^{(2)}.$$

The first block is driven only by $(W^1, W^{1,\perp})$, while the second block is driven only by $(W^2, W^{2,\perp})$. Since these Brownian pairs are independent, the two blocks are independent.

The first block is a lifted Heston model, so Abi Jaber's affine transform theorem applies. The second block is a classical Heston/CIR block, so the standard Heston affine transform applies. Hence, for $u \in \mathbb{C}$ in the common admissible strip,

$$\mathbb{E}^{\mathbb{Q}}[e^{uX_T} \mid \mathcal{F}_t] = e^{u \log S_0} \mathbb{E}^{\mathbb{Q}}[e^{uX_T^{(1)}} \mid \mathcal{F}_t^{(1)}] \mathbb{E}^{\mathbb{Q}}[e^{uX_T^{(2)}} \mid \mathcal{F}_t^{(2)}].$$

Substituting the two affine transform formulas gives

$$\mathbb{E}^{\mathbb{Q}}[e^{uX_T} \mid \mathcal{F}_t] = \exp \left(uX_t + \phi(t, T) + \sum_i c_i \psi_i^{(1)}(T-t) \tilde{U}_t^{1,i} + \psi^{(2)}(T-t) V_t^{(2)} \right),$$

with

$$\phi(t, T) = \phi^{(1)}(t, T) + \phi^{(2)}(t, T).$$

The Riccati equations are precisely the Riccati equations of the two blocks. $\square$

Remark 6.8 (Why both F_1 and F_2 contain $\frac{1}{2}(u^2 - u)$). It might seem surprising that both F_1 and F_2 in eq. (6.15)–eq. (6.16) contain the full term $\frac{1}{2}(u^2 - u)$, without any partition weight α_j summing to one. The structural reason is the following.

The log-price contribution $-\frac{u}{2} V_t dt + \frac{u^2}{2} V_t dt = \frac{1}{2}(u^2 - u) V_t dt$ splits affinely as

$$\frac{1}{2}(u^2 - u) V_t^{(1)} + \frac{1}{2}(u^2 - u) V_t^{(2)},$$

since $V_t = V_t^{(1)} + V_t^{(2)}$ and $V^{(1)}, V^{(2)}$ are independent affine variables in the state. The full $\frac{1}{2}(u^2 - u)$ piece thus appears separately in the coefficient of $V_t^{(1)}$ and in the coefficient of $V_t^{(2)}$. Since the two Riccati equations are obtained from the cancellation of these two distinct coefficients, each inherits the full $\frac{1}{2}(u^2 - u)$. There is no double-counting.

An earlier attempt to handle the non-affine cross-term arising from a single-noise price specification (see remark 6.3) led to artificial weights $\alpha_1 + \alpha_2 = 1$ allocating the variance contribution between blocks. The correctly affine double-noise specification eq. (6.5) eliminates that cross-term entirely and removes any need for such partition weights.

Corollary 6.9 (Existence and uniqueness of the Riccati solutions). *For $u \in \mathbb{C}$ with $\Re(u) \in [0, 1]$, the Riccati system eq. (6.12)–eq. (6.13) admits a unique global solution on $[0, T]$ for any $T < \infty$.*

Proof. Define, as above,

$$\Psi^{(1)}(t) := \sum_{i=1}^{n_1} c_i^{n_1} \psi^{(1),i}(t).$$

For each $i = 1, \dots, n_1$, equation eq. (6.12) is the linear inhomogeneous ODE

$$(\psi^{(1),i})'(t) = -x_i^{n_1} \psi^{(1),i}(t) + F_1(u, \Psi^{(1)}(t)), \quad \psi^{(1),i}(0) = 0.$$

Solving this equation by variation of constants gives

$$\psi^{(1),i}(t) = \int_0^t e^{-x_i^{n_1}(t-s)} F_1(u, \Psi^{(1)}(s)) ds.$$

Multiplying by $c_i^{n_1}$ and summing over i yields

$$\Psi^{(1)}(t) = \sum_{i=1}^{n_1} c_i^{n_1} \int_0^t e^{-x_i^{n_1}(t-s)} F_1(u, \Psi^{(1)}(s)) ds.$$

Since the sum is finite, we may interchange the sum and the integral. Hence

$$\Psi^{(1)}(t) = \int_0^t \left(\sum_{i=1}^{n_1} c_i^{n_1} e^{-x_i^{n_1}(t-s)} \right) F_1(u, \Psi^{(1)}(s)) ds.$$

With

$$K^{n_1}(t) := \sum_{i=1}^{n_1} c_i^{n_1} e^{-x_i^{n_1}t},$$

we obtain the scalar Riccati–Volterra equation

$$\Psi^{(1)}(t) = \int_0^t K^{n_1}(t-s) F_1(u, \Psi^{(1)}(s)) ds.$$

This is precisely the Riccati–Volterra equation associated with the standard lifted Heston model with kernel K^{n_1} .

Conversely, once $\Psi^{(1)}$ is known, each component $\psi^{(1),i}$ is recovered uniquely from

$$\psi^{(1),i}(t) = \int_0^t e^{-x_i^{n_1}(t-s)} F_1(u, \Psi^{(1)}(s)) ds.$$

Thus solving the n_1 -dimensional lifted Riccati system is equivalent to solving the scalar Riccati–Volterra equation for $\Psi^{(1)}$.

The kernel K^{n_1} is a finite positive sum of exponentials, hence it is completely monotone and belongs to the class treated in Abi Jaber, Larsson and Pulido [Abi Jaber, Larsson, and Pulido \[2019\]](#). For $u \in \mathbb{C}$ with $\Re(u) \in [0, 1]$, the admissibility condition is satisfied since

$$\Re\left(\frac{1}{2}(u^2 - u)\right) \leq 0.$$

Indeed, writing $u = a + ib$ with $a = \Re(u) \in [0, 1]$, we have

$$\Re(u^2 - u) = a^2 - a - b^2 = a(a-1) - b^2 \leq 0.$$

Therefore the global existence and uniqueness result for Riccati–Volterra equations applies to the lifted block.

The second equation (6.13) is the classical Heston Riccati equation for the CIR block. Equivalently, it is the one-factor affine Riccati equation corresponding to the exponential kernel $e^{-\lambda_2 t}$. The same admissibility condition $\Re(u) \in [0, 1]$ ensures that the scalar Riccati equation admits a unique global solution on every finite interval $[0, T]$.

Combining the two independent blocks, the full Riccati system (6.12)–(6.13) admits a unique global solution on $[0, T]$ for every $T < \infty$. $\square$

Corollary 6.10 (Unconditional characteristic function). *At time $t = 0$, since $\tilde{U}_0^{1,i} = 0$, the characteristic function is*

$$\mathbb{E}^{\mathbb{Q}}[e^{u \log S_T}] = \exp\left(\phi(0, T) + u \log S_0 + \psi^{(2)}(T) V_0^{(2)}\right). \quad (6.21)$$

If a general lifted-Heston input curve $g_0^{(1)}(\cdot)$ is used instead of the mean-reverting input in definition 6.1, the term $V_0^{(1)} F_1(u, \Psi^{(1)}) + \lambda_1 \theta_1 \Psi^{(1)}$ in ϕ is replaced by the standard lifted-Heston input term $F_1(u, \Psi^{(1)}(s)) g_0^{(1)}(T-s)$, exactly as in eq. (4.37).

Remark 6.11 (Block-by-block implementation). The two blocks of eq. (6.12)–eq. (6.13) are strictly independent ODE systems:

- Block 1 is exactly the standard lifted Heston Riccati system of theorem 4.5 with parameters $(\nu_1, \rho_1, \lambda_1, x_i^{n_1}, c_i^{n_1})$. The existing Phase 1 solver (section 7.3) can be reused unmodified.
- Block 2 is the scalar Heston Riccati equation with parameters $(\nu_2, \rho_2, \lambda_2)$, which admits a closed-form solution via the standard Heston ansatz of [Heston \[1993\]](#).

The two solvers can be run in parallel.

6.5 ATM skew and short-versus-long maturity behaviour

We now compute the leading-order ATM skew. The argument is heuristic in the same sense as the Bergomi–Guyon expansion (theorem 2.5): it is a first-order expansion in the vol-of-vol parameters (ν_1, ν_2) . It is nevertheless useful because it reveals the precise mechanism by which the two-factor model breaks the single-kernel skew rigidity identified by [Abi Jaber and Li \[2025\]](#).

6.5.1 Forward variance dynamics of each block

Recall the forward variance curve $\xi_t(T) := \mathbb{E}^\mathbb{Q}[V_T \mid \mathcal{F}_t]$ for $t \leq T$. By the additive decomposition $V = V^{(1)} + V^{(2)}$,

$$\xi_t(T) = \xi_t^{(1)}(T) + \xi_t^{(2)}(T), \quad \xi_t^{(j)}(T) := \mathbb{E}^\mathbb{Q}[V_T^{(j)} \mid \mathcal{F}_t].$$

A direct computation using the mild form of each block, analogous to the one-factor case in proposition 4.2, gives at first order in (ν_1, ν_2) :

$$d\xi_t^{(1)}(T) = \nu_1 K_{n_1}(T-t) \sqrt{V_t^{(1)}} dW_t^1 + O(\nu_1^2), \quad (6.22)$$

$$d\xi_t^{(2)}(T) = \nu_2 e^{-\lambda_2(T-t)} \sqrt{V_t^{(2)}} dW_t^2 + O(\nu_2^2). \quad (6.23)$$

6.5.2 Bergomi–Guyon expansion under the two-factor model

The covariation between $\log S$ and the forward variance curve splits additively because $W^1 \perp W^2$:

$$\frac{d\langle \log S, \xi_t(T) \rangle_t}{dt} = \rho_1 \nu_1 V_t^{(1)} K_{n_1}(T-t) + \rho_2 \nu_2 V_t^{(2)} e^{-\lambda_2(T-t)}. \quad (6.24)$$

The key point is that the coefficient of each kernel is $V_t^{(j)}$, not $\sqrt{V_t} \sqrt{V_t^{(j)}}$. This is a direct consequence of the double-Heston price specification eq. (6.5): each block of $\log S$ is paired with its own variance block, so the covariation is automatically affine.

Applying the Bergomi–Guyon ATM skew formula eq. (2.61) with the covariation eq. (6.24), localising at $V_t^{(j)} \approx V_0^{(j)}$ to leading order in (ν_1, ν_2) , and computing the two time integrals explicitly:

Proposition 6.12 (Two-factor lifted Heston ATM skew). *Under the model of definition 6.1, the at-the-money implied volatility skew at maturity T satisfies, at first order in (ν_1, ν_2) ,*

$$\psi(T) \approx \frac{\rho_1 \nu_1}{2\sqrt{V_0} T^2} \frac{V_0^{(1)}}{V_0} \sum_{i=1}^{n_1} \frac{c_i^{n_1}}{x_i^{n_1}} \left(T - \frac{1 - e^{-x_i^{n_1} T}}{x_i^{n_1}} \right) + \frac{\rho_2 \nu_2}{2\sqrt{V_0} T^2} \frac{V_0^{(2)}}{V_0} \cdot \frac{1}{\lambda_2} \left(T - \frac{1 - e^{-\lambda_2 T}}{\lambda_2} \right) \quad (6.25)$$

with $V_0 := V_0^{(1)} + V_0^{(2)}$. When $x_i^{n_1} = 0$ or $\lambda_2 = 0$, the corresponding term is interpreted by continuity as $T^2/2$.

Remark 6.13 (Idealised rough block: the clean power law). It is clarifying to record the leading-order skew of the idealised two-factor model in which block 1 carries the genuine fractional kernel $K_{H_1}(t) = t^{H_1-1/2}/\Gamma(H_1+1/2)$, $H_1 \in (0, 1/2)$, in place of its lifted approximation. The covariation eq. (6.24) splits additively in the same way (with K_{n_1} replaced by K_{H_1}), and since $\mathcal{I}_{K_{H_1}}(T) = T^{H_1+3/2}/\Gamma(H_1+5/2)$, the Bergomi–Guyon corollary eq. (2.62) gives

$$\psi(T) \approx \frac{\rho_1 \nu_1}{2\sqrt{V_0} \Gamma(H_1+5/2)} \frac{V_0^{(1)}}{V_0} T^{H_1-1/2} + \frac{\rho_2 \nu_2}{2\sqrt{V_0} T^2} \frac{V_0^{(2)}}{V_0} \frac{1}{\lambda_2} \left(T - \frac{1 - e^{-\lambda_2 T}}{\lambda_2} \right).$$

The first term is exactly the rough power law $\propto T^{H_1-1/2}$ driving the short-maturity explosion; the second is an independently-signed contribution whose size is set by $(\rho_2, \nu_2, V_0^{(2)}, \lambda_2)$ separately from H_1 . This is the cleanest statement of the mechanism of proposition 6.12: a single rough kernel fixes one exponent for the entire term structure, whereas the slow block adds a degree of freedom that a single H_1 cannot supply.

The idealised block 1 is genuinely rough (non-Markovian and non-semimartingale) and therefore does not admit the finite affine/Riccati structure. For tractability we replace K_{H_1} by its lifted multi-exponential approximation $K_{n_1}(t) = \sum_i c_i^{n_1} e^{-x_i^{n_1} t} \approx K_{H_1}(t)$ (chapter 4), which reproduces $T^{H_1-1/2}$ over the multi-scale window $1/x_{n_1}^{n_1} \ll T \ll 1/x_1^{n_1}$ and returns the tractable skew eq. (6.25). The genuinely-rough block is used only as a motivating device; definition 6.1 and theorem 6.7 use the lifted block throughout.

Remark 6.14 (Why the two-block skew is curved in log-log). The skew eq. (6.25) is a sum of two terms, $\psi(T) = \psi^{(1)}(T) + \psi^{(2)}(T)$, coming from the rough and the slow block respectively. A log-log plot draws $\log \psi$ against $\log T$. A single rough block alone is a pure power law, $\psi^{(1)}(T) \propto T^{H_1-1/2}$, so $\log \psi^{(1)} = \text{const} + (H_1 - \frac{1}{2}) \log T$ is an exact straight line of slope $H_1 - \frac{1}{2}$. The sum is not, because the logarithm does not distribute over a sum,

$$\log(\psi^{(1)} + \psi^{(2)}) \neq \log \psi^{(1)} + \log \psi^{(2)},$$

and $\psi^{(1)} + \psi^{(2)}$ is not itself a pure power law, the two blocks having different mean-reversion speeds and hence different shapes in T . Where the steep block 1 dominates, the curve follows slope $H_1 - \frac{1}{2}$; where the slow block 2 dominates, it follows a different slope; and in between it bends from one to the other. This bending is the log-log curvature that a single H_1 cannot produce, and it is what lets the two-factor model match the curved SPX skew term structure.

6.5.3 Why this fits short and long maturities simultaneously

We now explain precisely why eq. (6.25) resolves the fitting problem identified by [Abi Jaber and Li \[2025\]](#).

1. The single-kernel constraint. A single-kernel rough model produces, via the Bergomi-Guyon formula and eq. (2.62), an ATM skew of the form

$$\psi^{\text{rough}}(T) \approx \frac{C}{T^2} \int_0^T \int_t^T K_H(u-t) du dt = \frac{C}{\Gamma(H+5/2)} T^{H-1/2}. \quad (6.26)$$

A single parameter $H \in (0, 1/2)$ controls both:

- the short-maturity explosion rate ($\psi \rightarrow \infty$ as $T^{H-1/2}$ when $T \rightarrow 0$);
- the long-maturity decay rate (same exponent $H - 1/2$).

This is a rigid constraint: empirically on SPX, the skew at one week and the skew at two years cannot be fit simultaneously by a single value of H . A best-fit H for the short end overshoots the long end, and conversely.

2. The two-block additive structure. In eq. (6.25), the skew is the sum of two qualitatively different contributions.

Block 1 (rough, fast multi-scale). The function

$$T \mapsto \frac{1}{T^2} \sum_{i=1}^{n_1} \frac{c_i^{n_1}}{x_i^{n_1}} \left(T - \frac{1 - e^{-x_i^{n_1} T}}{x_i^{n_1}} \right)$$

behaves as follows:

- For T in the multi-scale window $1/x_1^{n_1} \ll T \ll 1/x_1^{n_1}$, this sum approximates the rough power law $T^{H_1-1/2}$. This is the design property of the geometric Laplace grid (theorem 4.10).
- For $T \rightarrow 0$ with $T < 1/x_1^{n_1}$, expanding $1 - e^{-x_1^{n_1}T}$ to second order gives a function tending to the finite value $\frac{1}{2} \sum_i c_i^{n_1}$. The skew is regularised to a finite limit.
- For $T \gg 1/x_1^{n_1}$, all exponentials decay and the contribution behaves like $\frac{1}{T} \sum_i c_i^{n_1}/x_i^{n_1}$, decaying as $1/T$.

Block 2 (slow, single OU). The function

$$T \mapsto \frac{1}{T^2} \cdot \frac{1}{\lambda_2} \left(T - \frac{1 - e^{-\lambda_2 T}}{\lambda_2} \right)$$

behaves as follows:

- For $T \rightarrow 0$, $\frac{1}{\lambda_2 T^2} \left(T - \frac{1 - e^{-\lambda_2 T}}{\lambda_2} \right) \rightarrow \frac{1}{2}$. The contribution tends to the finite constant $\frac{\rho_2 \nu_2}{4\sqrt{V_0}} \frac{V_0^{(2)}}{V_0}$.
- For $T \gg 1/\lambda_2$ (long maturities, with λ_2 small), $\frac{1 - e^{-\lambda_2 T}}{\lambda_2 T} \rightarrow \frac{1}{\lambda_2 T}$. The contribution decays as $1/T$ but with a coefficient $\frac{\rho_2 \nu_2}{2\lambda_2 \sqrt{V_0}} \frac{V_0^{(2)}}{V_0}$ that is independent of the rough-block parameters.

3. The decoupling is in the coefficients, not the time profile. Both blocks decay as $1/T$ at very long maturities. What gives the model its flexibility is not that one block “handles long maturities” intrinsically. It is that the coefficient of the long-maturity $1/T$ tail is the sum of two independently controlled contributions:

$$\psi(T) \xrightarrow{T \gg 1/\lambda_2} \frac{1}{2\sqrt{V_0} T} \left[\frac{V_0^{(1)}}{V_0} \rho_1 \nu_1 \sum_i \frac{c_i^{n_1}}{x_i^{n_1}} + \frac{V_0^{(2)}}{V_0} \frac{\rho_2 \nu_2}{\lambda_2} \right]. \quad (6.27)$$

By choosing λ_2 small enough that $1/\lambda_2$ exceeds the longest maturity of interest (e.g., $\lambda_2 = 0.1$ gives $1/\lambda_2 = 10$ years), the second term remains in its short-maturity regime over the calibration window: it contributes a roughly constant additive shift to the skew throughout the SPX maturity range, with magnitude controlled by $(\rho_2, \nu_2, V_0^{(2)})$.

4. Operational summary. In practical calibration:

- H_1 is chosen to fit the short-maturity slope (steepness of the skew at $T \in [1 \text{ week}, 3 \text{ months}]$);
- $(\rho_1, \nu_1, V_0^{(1)})$ control the magnitude of the short-end skew;
- λ_2 is chosen small (calibrated to the long-end flattening of the skew);
- $(\rho_2, \nu_2, V_0^{(2)})$ control an additive long-end correction.

The crucial point is that ρ_1 and ρ_2 are independent parameters. Without this independence, the long-end correction would inherit the same sign and relative magnitude as the short end, which is exactly the structural rigidity that single-kernel models cannot escape.

Remark 6.15 (Effect of the double-Heston correction on the skew formula). If one had used the single-noise price specification $dS/S = \sqrt{V} dB$ (which fails affinity, see remark 6.3), the corresponding first-order skew would contain factors $\sqrt{V_0^{(j)} V_0}$ instead of $V_0^{(j)}$ (both under the common $V_0^{3/2}$ normalisation). The two formulas have identical functional form in T ; only the magnitudes of the two contributions differ. In calibration, this difference is absorbed entirely into the calibrated values of (ν_1, ν_2) . The empirical fitting power of the two formulations is therefore equivalent at first order in vol-of-vol; the double-Heston specification is preferred because it is the only one that is mathematically well-posed.

6.6 Conclusion of the chapter

The two-factor lifted Heston model combines a rough lifted Heston block and a slow CIR block in a way that simultaneously preserves the non-negativity of the variance and the affine tractability of the model. Two design choices are essential: each block must carry its own square-root diffusion (otherwise positivity fails on independent driving BMs), and the price equation must be written in double-Heston form (otherwise the model loses its exponential-affine characteristic function). Under these two choices, the Riccati system decouples exactly into one standard lifted Heston block and one scalar Heston block.

The model is therefore a natural Heston-family analogue of the two-factor Bergomi mechanism studied by [Abi Jaber and Li \[2025\]](#): it preserves the practical advantage of Riccati–Fourier pricing inherited from [Abi Jaber \[2019\]](#), while adding the flexibility needed to decouple short- and long-maturity skew behaviour through two independent price-volatility correlations ρ_1 and ρ_2 .

The in-sample improvement that the two-factor model initially appeared to show in [section 8.9](#) is largely an artifact of a forward-variance clipping bug in the calibration; under a corrected positivity-respecting comparison on the same ξ_0 the model does not robustly beat the best single-factor baseline (fair $\lambda = 0$ deltas +15.8%, –2.0%, –8.6%), consistent with the structural inertness of the second block established above. The one empirically surprising finding is that the calibrated λ_2 consistently sits in $[1.5, 10]$ rather than in the $[0.05, 0.3]$ range we anticipated for a genuine “slow” block; the optimiser exploits the additional flexibility through a two-fast-factor configuration rather than a fast–slow decoupling. This refines, without contradicting, the theoretical mechanism of [proposition 6.12](#): the additive skew decomposition with two independently-controlled coefficients is the structural degree of freedom the two-factor model adds; as noted above, however, the empirical gain it delivers does not survive the corrected, positivity-respecting comparison.

Chapter 7

Implementation and Comparison Protocol

7.1 Code architecture

The implementation is structured in modular Python packages:

1. `common/`: shared utilities.
 - `black_scholes.py`: Black–Scholes call and put prices, Vega, and implied-volatility inversion by Brent’s method.
 - `forward_variance.py`: flat and piecewise-constant forward variance curves $\xi_0(\cdot)$, together with numerical integration of the curve.
 - `cos_method.py`: COS pricing method for European call options from a characteristic function.
2. `lifted_heston/`: implements the lifted Heston model.
 - `params.py`: computes (c_i^n, x_i^n) from (H, n, r_n) via eq. (4.74)–eq. (4.75).
 - `kernel.py`: evaluation of the rough kernel K_H and the lifted kernel K_n .
 - `riccati.py`: vectorised explicit–implicit solver for the Riccati system eq. (4.35) (Appendix C of [Abi Jaber \[2019\]](#)).
 - `characteristic_function.py`: assembles the characteristic function eq. (4.34).
 - `pricing.py`: COS method (Fang–Oosterlee) for European call prices from the characteristic function.
 - `implied_vol.py`: Black–Scholes inversion via Brent’s method.
3. `abergomi/`: implements the aBergomi model.
 - `kernel_fit.py`: L^2 optimisation to compute $(\tilde{c}_i, \tilde{x}_i)$ via non-negative least squares and local refinement.
 - `simulation.py`: vectorised Monte Carlo simulator for $X^{(i)}$ via exact OU update.
 - `pricing.py`: Monte Carlo European call pricing with antithetic variates and control variate.
4. `rough_heston/`: benchmark implementation.
 - `fractional_riccati.py`: Adams predictor–corrector scheme for eq. (3.7).
 - `pricing.py`: COS method using the resulting characteristic function.

5. `calibration/`: SPX surface calibration.

- `loss.py`: vega-weighted IV-RMSE on the surface plus the ATM-skew term.
- `optimizer.py`: differential evolution + L-BFGS refinement.

6. `data/`: SPX option data preparation (forward extraction, no-arbitrage filtering).

7. `tests/`: unit tests verifying $n = 1$ recovers Heston, convergence to rough Heston for $n = 500$, etc.

7.2 Fair comparison protocol

The empirical comparison between lifted Heston and aBergomi must be designed carefully, because the two models differ along several dimensions:

- lifted Heston has square-root affine variance dynamics, whereas aBergomi has lognormal variance dynamics;
- lifted Heston admits Riccati–Fourier pricing, whereas aBergomi is priced by Monte Carlo;
- the natural lifted Heston kernel is the geometric Laplace grid, whereas the natural aBergomi kernel is obtained by L^2 optimisation.

A direct comparison without controlling for these differences would be difficult to interpret. We therefore impose the following common structure.

Same market inputs. Both models are calibrated to the same SPX implied volatility surface, on the same maturity-strike grid. Both models use the same initial forward variance curve $\xi_0(\cdot)$ extracted from market data. For aBergomi, this is natural because the variance is written as

$$V_t = \xi_0(t) \exp(\cdots).$$

For lifted Heston, we use the forward-variance input version. In the reduced calibration convention, following the practical simplification discussed by [Abi Jaber \[2019\]](#), we set $\lambda = 0$. The reason is that, in the market-calibration setting, the level and term structure of variance are imposed externally by the market forward variance curve $\xi_0(\cdot)$, while the multi-scale mean-reversion structure is already encoded by the lifted speeds $(x_i^{n_1})_{i=1}^{n_1}$. So that the deterministic input is simply

$$g_0^n(t) = \xi_0(t).$$

Thus the level and term structure of variance are fixed by the same market input in both models, and the calibrated parameters control the smile dynamics.

Same calibration objective. For both models, calibration is performed by minimising an implied-volatility loss on the same set of quotes:

$$\text{RMSE}_{\text{IV}}(\theta) = \left(\frac{\sum_{m,j} w_{m,j} (\sigma_{\text{model}}(K_j, T_m; \theta) - \sigma_{\text{mkt}}(K_j, T_m))^2}{\sum_{m,j} w_{m,j}} \right)^{1/2}. \quad (7.1)$$

In the empirical chapter we use vega weighting, $w_{m,j} = \text{Vega}_{m,j}^{\text{BS}}$, for the $\lambda = 0$ calibration on the full surface. For the $\lambda = \infty$ pure skew calibration we restrict to the five-strike grid $\{0.96 F_T, 0.98 F_T, F_T, 1.02 F_T, 1.04 F_T\}$ per maturity, because the loss enters via only the two interpolation endpoints $0.98 F_T$ and $1.02 F_T$ of the 98/102 finite-difference ATM-skew estimator

$s_{98/102}(T)$ (see section 7.4). The $0.96 F_T$, F_T and $1.04 F_T$ strikes are priced because the optimiser pipeline expects a full smile, but they do not enter the loss value. The $\lambda = \infty$ loss is then

$$L_{\lambda=\infty}(\theta) = \sqrt{\frac{1}{|\mathcal{T}|} \sum_{T \in \mathcal{T}} (s_{98/102}^{\text{model}}(T; \theta) - s_{98/102}^{\text{market}}(T))^2},$$

the RMSE across maturities $\mathcal{T}$ of the per-maturity scalar $s_{98/102}$ between model and market.

Two comparison layers. We run two complementary comparisons.

1. **Paper-faithful comparison.** Lifted Heston is implemented with the geometric Laplace grid of [Abi Jaber \[2019\]](#), while aBergomi is implemented with the L^2 -optimised kernel of [Zhu, Loeper, Chen, and Langrené \[2021\]](#). This answers the practical question: which model performs better when each is implemented according to its natural construction?
2. **Kernel-controlled comparison.** Both models are run with the same exponential kernel approximation. We run both the geometric grid and the L^2 -optimised kernel in each model. This isolates the effect of the variance dynamics from the effect of the kernel approximation.

Models compared. With $n = 20$ fixed, the target comparison table is

Model	Kernel	Parameters	Pricing method
Lifted Heston	geometric	(ν, ρ, H)	Riccati–Fourier
Lifted Heston	L^2 -optimised	(ν, ρ, H)	Riccati–Fourier
aBergomi	geometric	(η, ρ, H)	Monte Carlo
aBergomi	L^2 -optimised	(η, ρ, H)	Monte Carlo

The first and fourth rows are the paper-faithful versions. The second and third rows are included to control for the kernel approximation.

Monte Carlo accuracy for aBergomi. Since aBergomi is priced by Monte Carlo, while lifted Heston is priced by deterministic Fourier inversion, the aBergomi prices are reported with Monte Carlo standard errors. During calibration, common random numbers are used to stabilise the objective. Final reported prices and implied volatilities are recomputed with a larger number of paths.

7.2.1 Monte Carlo simulation of the aBergomi model

We simulate the aBergomi model under the risk-neutral measure $\mathbb{Q}$. The stock price satisfies

$$\frac{dS_t}{S_t} = \sqrt{V_t} dB_t,$$

where the Brownian motion driving the stock is correlated with the Brownian motion driving volatility:

$$B_t = \rho W_t + \sqrt{1 - \rho^2} W_t^\perp,$$

with $W^\perp$ independent of W . The parameter $\rho < 0$ produces the equity skew: negative stock shocks tend to coincide with positive volatility shocks, so downside moves are associated with higher future variance.

The rough Bergomi driver is approximated by a finite sum of Ornstein–Uhlenbeck factors. Starting from the exponential approximation

$$K_H(t) \approx K_n(t) := \sum_{i=1}^n c_i e^{-x_i t},$$

we define

$$X_t^i = \int_0^t e^{-x_i(t-s)} dW_s, \quad i = 1, \dots, n.$$

Each factor satisfies

$$dX_t^i = -x_i X_t^i dt + dW_t.$$

The Gaussian volatility driver is therefore

$$Y_t = \sum_{i=1}^n c_i X_t^i.$$

The aBergomi variance is defined by

$$V_t = \xi_0(t) \exp \left(\eta Y_t - \frac{\eta^2}{2} \text{Var}(Y_t) \right).$$

The correction term ensures

$$\mathbb{E}^{\mathbb{Q}}[V_t] = \xi_0(t),$$

so the model matches the initial forward variance curve.

Let $0 = t_0 < t_1 < \dots < t_N = T$ be a time grid, with $\Delta t_k = t_{k+1} - t_k$. The OU factors can be simulated exactly over one time step:

$$X_{t_{k+1}}^i = e^{-x_i \Delta t_k} X_{t_k}^i + G_k^i,$$

where

$$G_k^i = \int_{t_k}^{t_{k+1}} e^{-x_i(t_{k+1}-s)} dW_s.$$

The random vector

$$Z_k = \begin{pmatrix} \Delta W_k \\ G_k^1 \\ \vdots \\ G_k^n \end{pmatrix}, \quad \Delta W_k := W_{t_{k+1}} - W_{t_k},$$

is centered Gaussian. Its covariance matrix is known explicitly:

$$\text{Var}(\Delta W_k) = \Delta t_k,$$

$$\text{Cov}(\Delta W_k, G_k^i) = \frac{1 - e^{-x_i \Delta t_k}}{x_i},$$

and

$$\text{Cov}(G_k^i, G_k^j) = \frac{1 - e^{-(x_i + x_j) \Delta t_k}}{x_i + x_j}.$$

To simulate this correlated Gaussian vector, we use Cholesky decomposition. If Σ_k denotes the covariance matrix of Z_k , we compute L_k such that

$$\Sigma_k = L_k L_k^\top.$$

If $\varepsilon_k \sim N(0, I_{n+1})$, then

$$Z_k = L_k \varepsilon_k$$

has covariance Σ_k . Thus Cholesky transforms independent standard Gaussian variables into Gaussian variables with the correct correlations.

Once the factors have been updated, we compute

$$Y_{t_k} = \sum_{i=1}^n c_i X_{t_k}^i, \quad V_{t_k} = \xi_0(t_k) \exp \left(\eta Y_{t_k} - \frac{\eta^2}{2} \text{Var}(Y_{t_k}) \right).$$

The OU factors are simulated exactly. The stock equation, however, is discretized by freezing the variance on each interval. Since

$$d \log S_t = -\frac{1}{2} V_t dt + \sqrt{V_t} dB_t,$$

we approximate V_s by V_{t_k} on $[t_k, t_{k+1}]$. This gives the log-Euler update

$$\log S_{t_{k+1}} = \log S_{t_k} - \frac{1}{2} V_{t_k} \Delta t_k + \sqrt{V_{t_k}} \Delta B_k,$$

where

$$\Delta B_k = \rho \Delta W_k + \sqrt{1 - \rho^2} \Delta W_k^\perp, \quad \Delta W_k^\perp \sim N(0, \Delta t_k),$$

with $\Delta W_k^\perp$ independent of W . Equivalently,

$$S_{t_{k+1}} = S_{t_k} \exp \left(-\frac{1}{2} V_{t_k} \Delta t_k + \sqrt{V_{t_k}} \Delta B_k \right).$$

The term $-\frac{1}{2} V_{t_k} \Delta t_k$ is the Itô correction. Under the frozen-variance approximation, it preserves the conditional martingale property

$$\mathbb{E}^\mathbb{Q} [S_{t_{k+1}} | \mathcal{F}_{t_k}] = S_{t_k}.$$

After simulating M paths, the Monte Carlo estimator of a call option with maturity T and strike K is

$$\hat{C}(K, T) = \frac{1}{M} \sum_{m=1}^M \left(S_T^{(m)} - K \right)^+.$$

Finally, the corresponding implied volatility $\hat{\sigma}_{\text{IV}}(K, T)$ is obtained by solving

$$C^{BS}(S_0, K, T, \hat{\sigma}_{\text{IV}}) = \hat{C}(K, T).$$

7.2.2 Sobol quasi-Monte Carlo

In standard Monte Carlo, the Gaussian variables are generated from pseudo-random uniforms

$$U \sim \mathcal{U}([0, 1]), \quad \varepsilon = \Phi^{-1}(U),$$

where Φ^{-1} is the inverse standard normal distribution function. Quasi-Monte Carlo replaces pseudo-random points by low-discrepancy Sobol points. The purpose is to fill the unit cube more evenly, reducing simulation noise.

For one full aBergomi path, the simulation needs Gaussian variables for every time step and every factor. If the grid has N time steps and n OU factors, then one time step requires $n + 2$ Gaussian inputs: $n + 1$ variables to generate

$$(\Delta W_k, G_k^1, \dots, G_k^n)$$

through the Cholesky matrix, and one additional independent variable for $\Delta W_k^\perp$. Therefore one complete path requires dimension

$$d = N(n + 2).$$

A Sobol point in $[0, 1]^d$ is therefore interpreted as the complete collection of random numbers needed to generate one full path. After applying the inverse Gaussian CDF componentwise, the vector is reshaped into

$$(\varepsilon_k^1, \dots, \varepsilon_k^{n+2})_{k=0}^{N-1}.$$

It is important to generate the Sobol variables for the whole path at once. Generating separate Sobol points time step by time step may give good marginal distributions at each date,

but it does not correctly represent the joint distribution of the full path. Since the payoff depends on the entire variance path, the correct integration problem is high-dimensional. Thus one Sobol point should correspond to one full path.

In practice, the implementation uses scrambled Sobol points. Scrambling randomizes the Sobol sequence while preserving its low-discrepancy structure. This keeps the variance reduction benefit of quasi-Monte Carlo while allowing statistical error estimates.

Computational cost of aBergomi calibration. A practical difficulty in the comparison is that the aBergomi model is priced by Monte Carlo simulation, whereas the lifted Heston model is priced by deterministic Riccati–Fourier methods. Hence, for aBergomi, each evaluation of the calibration loss requires a full Monte Carlo pricing of the option surface. Combining a global optimiser such as differential evolution with a local refinement such as L-BFGS can therefore be computationally expensive.

To make the calibration stable and feasible, we use common random numbers (CRN): a single bundle of Brownian increments ω^* is drawn once and reused for every parameter value θ the optimiser visits during the search. Writing the Monte Carlo estimator of the loss as

$$\hat{L}(\theta, \omega) = L(\theta) + \varepsilon(\theta, \omega), \quad \varepsilon(\theta, \omega) = O_{\mathbb{P}}(M^{-1/2}), \quad (7.2)$$

where $L(\theta)$ is the true (infeasible) loss and $\varepsilon(\theta, \omega)$ is the Monte Carlo error of order $M^{-1/2}$, the effect of CRN can be described through the difference at two nearby parameters θ_1, θ_2 . With independent draws ω_1, ω_2 , the noise terms $\varepsilon(\theta_1, \omega_1)$ and $\varepsilon(\theta_2, \omega_2)$ are independent, so they do not cancel in the difference $\hat{L}(\theta_2, \omega_2) - \hat{L}(\theta_1, \omega_1)$: small variations in θ are dominated by Monte Carlo noise and the loss surface appears jagged. With a shared draw ω^* , by contrast, the simulated trajectories vary continuously in θ , so $\varepsilon(\theta_1, \omega^*)$ and $\varepsilon(\theta_2, \omega^*)$ are highly positively correlated and almost cancel in the difference. The Monte Carlo objective $\theta \mapsto \hat{L}(\theta, \omega^*)$ then becomes a smooth function of θ , which is essential for both the differential-evolution global search and the L-BFGS gradient-based refinement: without CRN, finite-difference gradients of the loss have noise of order $M^{-1/2}/\delta$, which dominates the signal for any reasonable step δ .

The price to pay is that the optimiser solves

$$\hat{\theta}^*(\omega^*) = \arg \min_{\theta} \hat{L}(\theta, \omega^*), \quad (7.3)$$

rather than the population minimiser $\theta^* = \arg \min L(\theta)$. The difference $\hat{\theta}^*(\omega^*) - \theta^*$ is itself a random variable of order $M^{-1/2}$ depending on the realisation ω^* . In addition, the in-sample loss attained at the optimum, $\hat{L}(\hat{\theta}^*, \omega^*)$, is biased downward relative to $L(\hat{\theta}^*)$: the optimiser selects parameters that exploit the specific noise realisation ω^* , an effect analogous to training-set optimism in statistical learning. Reporting this in-sample value as the model’s pricing error would overstate the quality of the fit.

To avoid this bias, the final reported prices and implied-volatility surface are recomputed at the calibrated parameters $\hat{\theta}^*$ using a fresh, independent Brownian bundle ω^{**} and a larger sample size $M' > M$. Concretely we use $M = 50\,000$ paths during the optimisation and $M' = 100\,000$ paths for the final pricing, with seeds chosen to guarantee ω^{**} and ω^* are independent. Since ω^{**} is drawn after the calibrated parameter $\hat{\theta}^*$ is fixed, the new estimator $\hat{L}(\hat{\theta}^*, \omega^{**})$ is an unbiased estimate of the true loss $L(\hat{\theta}^*)$, and the larger sample size reduces its standard error by a factor $\sqrt{M'/M} \approx 1.41$. All implied-volatility RMSEs, ATM-skew RMSEs, and out-of-sample errors reported in chapter 8 are computed from this independent simulation, not from the CRN-frozen value at which the optimiser terminated.

When possible, we also use Sobol quasi-Monte Carlo points instead of standard pseudo-random numbers. This improves the efficiency of the simulation and reduces the number of paths needed for a stable implied-volatility surface.

Because of this computational cost, the main empirical comparison is performed on one carefully cleaned SPX implied volatility surface rather than on a long daily panel. This is

sufficient for the objective of this thesis, which is to compare the mechanisms and numerical behaviour of lifted Heston and aBergomi under a common calibration protocol, rather than to reproduce a full out-of-sample panel study.

Interpretation of results. If aBergomi with an L^2 -optimised kernel outperforms lifted Heston with a geometric grid, this does not immediately prove that lognormal Bergomi dynamics are superior to square-root Heston dynamics. The improvement may come from the kernel approximation. The kernel-controlled comparison is therefore essential. If aBergomi still outperforms lifted Heston when both use the same kernel, then the difference is more likely due to the variance specification. Conversely, if lifted Heston catches up when using the L^2 -optimised kernel, then the kernel approximation was a major driver of the difference.

Following Abi Jaber and Li (2025), we do not use a fixed log-moneyness range for all maturities. Instead, we use a maturity-dependent filter. Very short-dated options are kept only in a narrow range around the money, while longer-dated options are allowed to cover a wider range of log-moneyness. This reflects the fact that deep out-of-the-money short-dated options are often illiquid and lead to unstable implied-volatility estimates, whereas long-dated out-of-the-money options may still contain meaningful information about the downside tail of the risk-neutral distribution.

More precisely, with $k = \log(K/F_T)$, we use the following ranges:

T	accepted range for k
$T < 2$ weeks	$[-0.15, 0.03]$
$T < 1$ month	$[-0.25, 0.03]$
$T < 2$ months	$[-0.30, 0.04]$
$T < 3$ months	$[-0.40, 0.15]$
$T < 6$ months	$[-0.60, 0.15]$
$T < 1$ year	$[-0.80, 0.20]$
$T \geq 1$ year	$[-1.50, 0.30]$.

This is the filtering rule used in Abi Jaber and Li (2025) for the global SPX implied-volatility surface calibration. In the implementation, the forward F_T used in the definition $k = \log(K/F_T)$ is inferred from the option data itself using put-call parity. For each maturity T , we first match call and put quotes with the same strike. For every matched pair $(K_j, C_{\text{mid}}(K_j, T), P_{\text{mid}}(K_j, T))$, we compute the implied forward

$$F_T^{(j)} = K_j + e^{rT}(C_{\text{mid}}(K_j, T) - P_{\text{mid}}(K_j, T)),$$

which follows from

$$C_{\text{mid}}(K_j, T) - P_{\text{mid}}(K_j, T) = e^{-rT}(F_T - K_j).$$

To reduce the effect of noisy or stale quotes, we do not rely on a single strike. Instead, we select the few call-put pairs closest to the at-the-money forward region, namely those with the smallest values of

$$|C_{\text{mid}}(K_j, T) - P_{\text{mid}}(K_j, T)|,$$

and define F_T as the median of the corresponding implied forwards:

$$F_T = \text{median}_{j \in \mathcal{A}_T} F_T^{(j)}.$$

Here $\mathcal{A}_T$ denotes the selected near-ATM call-put pairs. This makes the moneyness filter more robust, since the median is less sensitive to one noisy option quote. Thus the log-moneyness filter is centered around the market-implied forward F_T , rather than around the spot price S_0 .

7.2.3 Extraction of the initial forward variance curve ξ_0

Both the lifted Heston model (in the reduced parametrisation $g_0^n(t) = \xi_0(t)$ with $\lambda = 0$) and the aBergomi model take the initial forward variance curve $\xi_0(\cdot)$ as an exogenous market input. We extract ξ_0 from the cleaned implied volatility surface via the standard practitioner method based on the ATM total-variance term structure.

Construction. For each market maturity T_m , define the at-the-money implied volatility $\sigma_{\text{ATM}}(T_m)$ by linear interpolation of the cleaned implied volatility surface to log-moneyness $k = 0$, using the two strikes with the smallest $|\log(K/F_{T_m})|$. The ATM total variance is

$$w(T_m) := \sigma_{\text{ATM}}^2(T_m) T_m. \quad (7.4)$$

Justification: leading-order vol-of-vol expansion. We claim that $w(T)$ approximates the integrated forward variance up to a higher-order correction in the vol-of-vol parameter ν :

$$w(T) = \int_0^T \xi_0(s) ds + O(\nu^2). \quad (7.5)$$

This is a consequence of the at-the-money case of the Bergomi–Guyon expansion (theorem 2.5). At zero-th order in ν , the variance process is deterministic, $V_t \equiv \xi_0(t)$, so $\log(S_T/F_T)$ is Gaussian with variance $\int_0^T \xi_0(s) ds$. The Black–Scholes implied volatility at any strike then matches the realised standard deviation exactly, and in particular

$$\sigma_{\text{ATM}}^2(T)|_{\nu=0} = \frac{1}{T} \int_0^T \xi_0(s) ds. \quad (7.6)$$

The first-order correction in ν generates the smile asymmetry (skew), which vanishes at $K = F_T$ by construction. Hence the first non-trivial correction at the money is of order ν^2 , which yields eq. (7.5). Differentiating with respect to T and using the fundamental theorem of calculus gives

$$\xi_0(T) = w'(T) + O(\nu^2). \quad (7.7)$$

Equation eq. (7.7) is the basis of our extraction procedure: we estimate $\xi_0(T)$ as the derivative of an interpolated ATM total-variance term structure.

Spline differentiation. The function $T \mapsto w(T)$ is known only at the market maturities $T_1 < \dots < T_M$. We fit a Piecewise Cubic Hermite Interpolating Polynomial (PCHIP) to $\{(T_m, w_m)\}_{m=1}^M$, and define

$$\hat{\xi}_0(T) := \max(w'_{\text{PCHIP}}(T), \varepsilon), \quad \varepsilon = 10^{-4}, \quad (7.8)$$

evaluated on a fine grid of $\max(10M, 50)$ points spanning $[T_1, T_M]$. PCHIP is chosen over a standard cubic spline because it preserves the monotonicity of the data: a cubic spline imposes C^2 continuity globally and can oscillate between data points, occasionally producing negative slopes even on monotone data, whereas PCHIP imposes only C^1 continuity and is monotone on monotone inputs. Since empirical ATM total variance is monotonically non-decreasing on SPX ($w_m \leq w_{m+1}$ for all m), PCHIP guarantees a non-negative extracted forward variance by construction. The floor ε serves only as a numerical safeguard against degenerate cases (e.g. two near-equal w_m producing a numerically near-zero PCHIP derivative).

Use in the models. The extracted curve is sampled at the fine grid and passed to a piecewise-constant cadlag forward-variance object, following the convention of [Abi Jaber and Li \[2025\]](#). The lifted Heston and aBergomi models both receive the same $\hat{\xi}_0(\cdot)$, ensuring that any difference in calibrated parameters reflects differences in variance dynamics rather than differences in the imposed term structure.

Limitations. This procedure is a first-order approximation in vol-of-vol: it identifies the ATM total variance with the integrated forward variance, neglecting the higher-order $O(\nu^2)$ correction in eq. (7.5). A more refined extraction would use the variance-swap replication formula of Bergomi and Guyon [2012], which integrates option prices across all OTM strikes and is exact under the model-free identity $\mathbb{E}^\mathbb{Q}[\int_0^T V_s ds] = \int_0^T \xi_0(s) ds$. We adopt the ATM approximation because (i) it is the standard practitioner choice used in the empirical sections of Abi Jaber and Li [2025] and Abi Jaber [2019], and (ii) the implied volatility surface is densest at the money in our SPX dataset, so the ATM interpolation is the most numerically stable input.

7.3 Key algorithms

Riccati solver. For each Fourier argument u , the lifted Heston Riccati system is

$$\frac{d\psi^{n,i}}{dt}(t) = -x_i^n \psi^{n,i}(t) + F\left(u, \sum_{j=1}^n c_j^n \psi^{n,j}(t)\right), \quad \psi^{n,i}(0) = 0.$$

The implementation uses the explicit–implicit discretisation

$$\psi_{k+1}^{n,i} = \frac{\psi_k^{n,i} + \Delta t F\left(u, \sum_{j=1}^n c_j^n \psi_k^{n,j}\right)}{1 + \Delta t x_i^n}.$$

The nonlinear term is treated explicitly, while the stiff linear mean-reversion term $-x_i^n \psi^{n,i}$ is treated implicitly. This is important because the geometric grid contains very large mean-reversion speeds. The solver is vectorised over the Fourier grid, so that all COS frequencies are solved simultaneously.

Characteristic function. Once the Riccati system has been solved, the characteristic function is assembled through

$$\phi^n(0, T) = \int_0^T F\left(u, \sum_{j=1}^n c_j^n \psi^{n,j}(s)\right) g_0^n(T-s) ds.$$

In the forward-variance calibration convention used for comparison with aBergomi, one takes

$$g_0^n(t) = \xi_0(t),$$

so the initial term structure of variance is imposed by the market forward variance curve. Since $U_0^{n,i} = 0$, the time-zero characteristic function reduces to

$$\mathbb{E}^\mathbb{Q}\left[e^{u \log S_T^n}\right] = \exp(\phi^n(0, T) + u \log S_0).$$

COS method (Fang–Oosterlee). The Fourier coefficients of the truncated $\log(S_T/S_0)$ density on a truncation interval $[a, b]$ are read off from the centred characteristic function $\varphi_{\log(S_T/S_0)}$:

$$A_k = \frac{2}{b-a} \operatorname{Re}\left[\varphi_{\log(S_T/S_0)}\left(\frac{k\pi}{b-a}\right) e^{-ik\pi a/(b-a)}\right], \quad k = 0, 1, \dots, N_{\cos} - 1, \quad (7.9)$$

and the call price at strike K is the finite sum $C(K) = \frac{2K}{b-a} \sum_k' \operatorname{Re}\left[A_k e^{-i\omega_k(a+\log(K/S_0))}\right] (\chi_k - \psi_k)$ where χ_k, ψ_k are the standard analytic payoff coefficients of Fang and Oosterlee [2009] on the integration range $[0, b]$, and the $\sum'$ gives half weight to $k = 0$.

We use $N_{\cos} = 256$ cosine terms, and the truncation interval

$$[a, b] = [-L_0 \sigma_{\text{tot}}(T), +L_0 \sigma_{\text{tot}}(T)], \quad \sigma_{\text{tot}}(T) := \sqrt{\int_0^T \xi_0(s) ds}, \quad L_0 = 12. \quad (7.10)$$

The choice $L_0 = 12$ is on the conservative side of the Fang and Oosterlee [2009] recommendation $L_0 \in [8, 12]$; it oversizes $[a, b]$ slightly to absorb the heavier left tail produced by lifted Heston dynamics with small H , where the Riccati solution exhibits slow decay of the characteristic function at large $|\omega|$. The integrated variance $\sigma_{\text{tot}}(T)^2$ used to scale $[a, b]$ is taken directly from the market forward variance curve $\xi_0(\cdot)$, so no separate cumulant calibration is needed. For each maturity, the Riccati system is solved once on the COS frequency grid $\omega_k = k\pi/(b - a)$, and all strikes at that maturity are then priced by broadcasting the strike-dependent phase factor over the cached characteristic-function values.

Implied volatility inversion. Model call prices are converted into implied volatilities using Brent’s method applied to

$$C^{\text{BS}}(S_0, K, T, \sigma) - C^{\text{model}}(K, T) = 0.$$

The Black–Scholes Vega is implemented both as a diagnostic and because it is useful for future Newton-type implied-volatility inversion or calibration routines, even though the current robust inversion uses Brent’s method.

Numerical note on Riccati stability. For very large imaginary Fourier frequencies, the Riccati system may become numerically unstable and produce NaN values. In the current implementation, NaNs at sufficiently high frequencies are replaced by zero, motivated by the fact that characteristic functions should decay at high frequency under regularity assumptions. This is acceptable only if the instability occurs in the negligible high-frequency tail; if it appears at low or medium frequencies, the Riccati or COS discretisation must be refined.

Calibration. Both model families are calibrated by minimising the implied-volatility loss eq. (7.1) of section 7.2: the vega-weighted IV-RMSE for $\lambda = 0$ on the full surface, and the pure 98/102 ATM-skew RMSE for $\lambda = \infty$. The minimisation uses a global differential-evolution search followed by L-BFGS local refinement

Reproducibility. All stochastic components of the calibration pipeline use fixed random seeds. Differential evolution is seeded with `seed=42` via the `scipy.optimize.differential_evolution` interface; the non-negative least squares kernel fit and L-BFGS local refinement are deterministic given their inputs. The aBergomi Monte Carlo pricer uses `seed=42` for the scrambled Sobol sequence (cf. section 7.2). Reported calibrated parameters and metrics are therefore exactly reproducible from the published code.

7.4 Performance metrics

We evaluate models on four criteria:

1. **In-sample surface fit:** vega-weighted implied-volatility RMSE on the train-day surface.
2. **Temporal out-of-sample fit:** vega-weighted IV-RMSE on the next trading day’s surface with the calibrated parameters held fixed and only $\xi_0(\cdot)$ re-extracted.
3. **ATM skew fit:** RMSE across maturities of the model-vs-market difference of $d\sigma_{\text{impl}}/dk|_{k=0}$. Two estimators are used in different places; see the dual-estimator paragraph below.
4. **Speed:** wall-clock time to compute one full implied-volatility surface.

ATM-skew estimator: dual usage in this thesis. Two estimators of the ATM skew $d\sigma_{\text{impl}}/dk|_{k=0}$ appear in this thesis, used in different contexts. The first is the two-point finite difference between $\sigma(0.98 F_T)$ and $\sigma(1.02 F_T)$ divided by $\log(1.02/0.98) \approx 0.04$, which is the convention of [Abi Jaber \[2019\]](#) whose Figure 1 literally labels its axis “ATM skew 98/102”. It is the foundational reference for our lifted Heston implementation and is used by the $\lambda = \infty$ calibration loss (computed via `_atm_skew_98_102` in `src/calibration/loss.py`) and by all `sk_IS` / `sk_t00S` RMSE columns in our tables. The second is the analytic derivative of a per-slice SVI fit (raw SVI parametrisation; [\[Gatheral and Jacquier, 2014\]](#)), evaluated as $w'(0)/(2\sigma_{\text{ATM}} T)$ with $w(k) = \sigma_{\text{impl}}^2(k) T$. The SVI estimator is used only for the ATM-skew term-structure figures because it produces smooth visualisations free of finite-difference grid artifacts; slices with SVI fit RMSE above 0.03 are excluded from both market and model curves for fair comparison. On the typical SPX 2024–2025 surfaces studied, the two estimators agree to within a few percent; substantial disagreement occurs only on numerically pathological slices that the SVI estimator drops via its quality threshold. [Abi Jaber and Li \[2025\]](#) calibrate Bergomi-family models using a wider-band IV-RMSE on log-moneyness $k \in [-0.05, 0.03]$ across all maturities rather than a direct skew estimator, avoiding the Bergomi–Guyon approximation; we have not adopted this convention since our use of Riccati–Fourier pricing makes the 98/102 two-point estimator both natural (matching the model’s reference paper) and computationally cheap.

Remark 7.1 (Temporal out-of-sample validation). A fully rigorous out-of-sample validation would require a time series of daily SPX implied volatility surfaces. [Abi Jaber and Li \[2025\]](#) follow this route, calibrating volatility models over more than 2,800 trading days from August 2011 to September 2022 using CBOE OptionMetrics data.

A full panel of this size is beyond the scope of the present bachelor thesis: it requires access to a long history of cleaned option surfaces and is computationally demanding for Monte Carlo-based models such as aBergomi (each daily calibration runs differential evolution over a parameter grid where every evaluation prices the full surface by Monte Carlo). Instead, we use a temporal out-of-sample protocol on a small number of adjacent train/test trading-day pairs.

More precisely, for each train date t we calibrate the model parameters (ν, ρ, H) (lifted Heston) or (η, ρ, H) (aBergomi) on the full implied-volatility surface of day t , then freeze them and evaluate pricing errors on day $t + 1$ using only the test-day forward variance curve $\xi_0^{t+1}(\cdot)$, re-extracted from the test-day surface by the PCHIP-derivative method of section 7.2.3. This isolates parameter-vector stability from level-drift in ξ_0 .

The protocol is implemented by `scripts/07_temporal_oos.py`, which calibrates with $n = 20$, $r_n = 2.5$ fixed, on a thinned grid of eight log-spaced maturities and fifteen strikes per maturity. The in-sample and out-of-sample RMSEs reported in chapter 8 are re-evaluated on the full unthinned surfaces of both days.

The purpose of this protocol is more limited than a full daily panel but still informative: across adjacent trading days, it tests whether the calibrated parameter vector encodes structural model behaviour robust to one day’s change in market conditions, or whether it has overfit a single noise realisation of the option surface.

7.5 Data acquisition

SPX option end-of-day mid-quotes (the CBOE-listed contract) are sourced from Polygon.io Options Advanced. The provider distributes the consolidated OPRA quotes feed as daily flat-file archives via an S3-compatible endpoint at `files.massive.com`, under the bucket path `us_options_opra/quotes_v1/YYYY/MM/YYYY-MM-DD.csv.gz` — one gzipped CSV per trading day, approximately 140 GB compressed and containing every option quote on every US venue for that trading day. Extracting SPX end-of-day mid-quotes from a single archive takes approximately four hours of filtering on one workstation, which constrains the panel size used in this thesis.

The empirical work uses three train/test pairs of adjacent trading days (section 8.9), chosen deliberately to sample three distinct out-of-sample regimes:

- **2024-08-05 → 2024-08-06.** VIX closes $38.57 \rightarrow 27.71$: a falling-volatility regime in the wake of the yen-carry unwind.
- **2024-12-04 → 2024-12-05.** VIX closes $13.45 \rightarrow 13.54$: a typical low-volatility, near-stationary regime.
- **2025-04-03 → 2025-04-04.** VIX closes $30.02 \rightarrow 45.31$: a sharply rising-volatility regime triggered by US tariff announcements.

The three pairs are positioned to test parameter persistence across qualitatively different out-of-sample conditions: smooth decay, stationarity, and regime change.

Preprocessing follows the methodology of [Abi Jaber and Li \[2025\]](#): forward prices are extracted via put-call parity, sparse strike grids are filled by per-maturity SSVI interpolation [[Gatheral and Jacquier, 2014](#)], no-arbitrage filtering is applied, and the forward variance curve is bootstrapped from variance swap quotes.

A rolling panel of the kind used by [Abi Jaber and Li \[2025\]](#) over 2011–2022 would require institutional data access and extraction infrastructure beyond the scope of a bachelor thesis.

Chapter 8

Empirical Comparison of All Five Models on SPX

This chapter implements the comparison protocol of section 7.2, separating the three effects (variance dynamics, kernel approximation, pricing method) introduced there.

Throughout this chapter, both models use the same initial forward variance curve $\xi_0(\cdot)$ extracted from market data. For the lifted Heston model, we use the reduced forward-variance parametrisation discussed in section 7.2: the deterministic input curve is chosen so that the level and term structure of variance are imposed by $\xi_0(\cdot)$, and we set $\lambda = 0$. Thus, with n and r_n fixed, the calibrated lifted Heston parameters are

$$(\nu, \rho, H).$$

For aBergomi, the forward variance curve enters directly through

$$V_t = \xi_0(t) \exp(\cdots),$$

and the calibrated parameters are

$$(\eta, \rho, H).$$

This gives a fair three-parameter comparison between the two model families. For the diagnostic plots of implied-volatility slices and ATM skew, we remove ultra-short maturities below $T = 0.05$ years; these maturities are highly sensitive to quote noise, Monte Carlo error, and implied-volatility inversion errors. The empirical chapter uses two different ATM-skew estimators for two different purposes:

- For the tables and the $\lambda = \infty$ calibration loss, the 98/102 two-point finite difference $s_{98/102}(T) := (\sigma(0.98 F_T, T) - \sigma(1.02 F_T, T))/0.04$. Cheap, numerically stable, matches the convention of [Abi Jaber \[2019\]](#) Figure 1; see section 7.4.
- For the ATM-skew term-structure figures, the SVI-analytic derivative $d\sigma_{\text{impl}}/dk|_{k=0} = w'(0)/(2\sigma_{\text{ATM}} T)$ from a per-slice raw SVI fit [[Gatheral and Jacquier, 2014](#)], with slices on which the SVI fit has unweighted IV-RMSE above 0.03 excluded simultaneously from market and from every model curve. The SVI-derivative gives a smoother term structure than a per-maturity finite difference would produce.

On the SPX 2024–2025 surfaces studied, the two estimators agree to within a few percent of each other; they diverge only on numerically pathological slices that the SVI estimator drops via its RMSE quality threshold.

8.0.1 Interpretation of the calibrated Hurst parameter

Before presenting the empirical results, we clarify what the parameter H represents in the lifted Heston model, since this point is frequently elided in the literature.

The Hurst index in fractional Brownian motion. For fractional Brownian motion B^H with $H \in (0, 1)$, the parameter H simultaneously controls three properties: the self-similarity exponent $B_{at}^H \stackrel{d}{=} a^H B_t^H$, the increment variance $\text{Var}(B_t^H - B_s^H) = |t - s|^{2H}$, and the Hölder regularity of sample paths, which is $H - \varepsilon$ for every $\varepsilon > 0$ almost surely. The case $H = 1/2$ recovers standard Brownian motion; $H < 1/2$ corresponds to rougher-than-Brownian paths with negatively correlated increments.

The Hurst index in rough Heston. In the rough Heston model, the variance satisfies a Volterra equation with kernel $K_H(t) = t^{H-1/2}/\Gamma(H + 1/2)$ for $H \in (0, 1/2)$. The variance paths inherit Hölder regularity $H - \varepsilon$ from this kernel via the Volterra structure. The parameter H here is the same H as in fractional Brownian motion: it is genuinely the Hölder index of the variance paths. At $H = 1/2$ the kernel becomes constant and the model reduces to classical Heston.

The Hurst index in lifted Heston (finite n). The lifted Heston replaces K_H by the multi-exponential kernel $K^n(t) = \sum_{i=1}^n c_i^n e^{-x_i^n t}$. For any finite n , this kernel is bounded and smooth on $(0, \infty)$, and the resulting variance process is a Markovian semimartingale driven by a single Brownian motion. Its sample paths therefore have Hölder regularity $1/2 - \varepsilon$, identical to Brownian motion, regardless of the value of H chosen in the parametrisation eq. (4.74)–eq. (4.75). At finite n , the parameter H does not equal the Hölder index of the simulated paths.

What H does parametrise at finite n . The role of H in the lifted model is to label which fractional kernel K_H the discretisation K^n approximates. As $n \rightarrow \infty$ with r_n chosen appropriately (Abi Jaber, 2019, Theorem A.2; see theorem 4.10), $K^n \rightarrow K_H$ in L^2 and the lifted variance process converges to the rough Heston with Hurst index H . The parameter is meaningful at finite n because the kernel shape – not the path-level Hölder regularity – governs the implied volatility surface on macroscopic timescales. The statistical experiment of Abi Jaber [2019, Section 4.2] shows that standard roughness estimators applied to simulated lifted paths report $\hat{H}$ close to the parametric H at timescales above roughly ten minutes for $n = 20$, and only recover the true semimartingale value $\hat{H} \rightarrow 1/2$ at finer resolutions.

Interpretation of our calibrated values. We therefore interpret the calibrated H in this thesis as a model selector: it identifies which member of the lifted Heston family best matches the SPX option data under the chosen objective. A statement of the form “the calibrated model has $H = 0.30$ ” is a statement about the kernel shape selected by the optimiser, equivalent in the $n \rightarrow \infty$ limit to a rough Heston of Hölder regularity 0.30. It is not a direct measurement of the Hölder regularity of SPX realised variance, which would require time-series data and a different estimation procedure under the physical measure $\mathbb{P}$ (see Cont and Das, 2023).

8.1 Lifted Heston validation

We first validate the lifted Heston implementation independently of the empirical SPX calibration. The goal is to check that the Riccati solver, characteristic function, Fourier pricing method, and implied volatility inversion are implemented correctly.

8.1.1 Convergence in the number of factors

We reproduce the convergence experiment of Abi Jaber [2019]. With the forward-variance parametrisation $g_0^n(t) = V_0$ (constant input), parameters $(V_0, \nu, \rho) = (0.04, 0.4, -0.7)$, and fixed $r_n = 2.5$ (cf. section 4.5 on the choice of r_n), we price European calls under the lifted Heston model at $n \in \{2, 4, 6, 10, 20, 50\}$ factors for four values of the Hurst index $H \in \{0.05, 0.10, 0.20, 0.30\}$

on a maturity grid $T \in \{0.1, 0.25, 0.5, 1.0, 2.0\}$ years. The benchmark is the rough Heston implied-volatility surface obtained by solving the fractional Riccati equation numerically. The IV error against the benchmark is measured by sup-norm $\sup_{T,K} |\Delta\sigma_{\text{impl}}|$ and L^2 -norm RMSE on the same (T, K) grid.

Figure 8.1 shows both error metrics on a log-log scale as n increases. For each H , both errors decrease monotonically with n up to $n = 20$, then plateau at $n = 50$ where the residual gap between the geometric Laplace-grid kernel approximation and the true rough kernel becomes the dominant source of error. Table 8.1 reports the IV-RMSE values numerically.

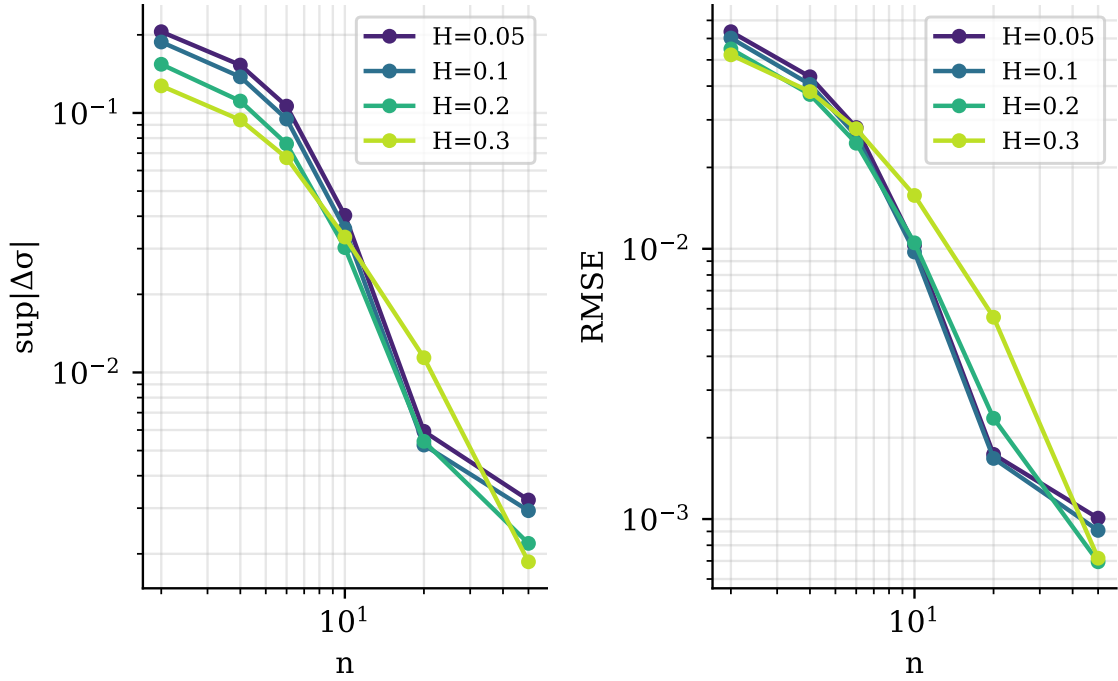


Figure 8.1: Sup-norm error (left) and IV-RMSE (right) of the lifted Heston implied-volatility surface against the rough Heston benchmark, as a function of the factor count n on a log-log scale, for four values of the Hurst index H . Parameters: $V_0 = 0.04$, $\nu = 0.4$, $\rho = -0.7$, $r_n = 2.5$ fixed, maturity grid $T \in \{0.1, 0.25, 0.5, 1.0, 2.0\}$ years. Source: `results/01_lh_convergence_rh/nogit_42/convergence_plot.pdf`.

8.1.2 Sanity checks

We also perform the following implementation checks:

1. the characteristic function satisfies $\Phi_T(0) = 1$;
2. the $n = 1$, $c_1 = 1$, $x_1 = 0$ case recovers the classical Heston structure when the Heston mean-reversion parametrisation is used;
3. option prices are decreasing and convex in strike;
4. implied volatilities are stable with respect to the Fourier grid and the number of COS terms.

H	$n = 2$	$n = 4$	$n = 6$	$n = 10$	$n = 20$	$n = 50$
0.05	0.0636	0.0433	0.0281	0.0103	0.00173	0.00101
0.10	0.0602	0.0406	0.0262	0.00973	0.00168	0.000908
0.20	0.0549	0.0372	0.0246	0.01050	0.00236	0.000695
0.30	0.0522	0.0382	0.0278	0.01575	0.00558	0.000716

Table 8.1: IV-RMSE of the lifted Heston implied-volatility surface against the rough Heston benchmark, as a function of the factor count n and the Hurst index H . Same parameter set as fig. 8.1. The RMSE decreases by roughly a factor of 6 from $n = 10$ to $n = 20$ and then plateaus at $n = 50$, reflecting the irreducible gap between the geometric Laplace-grid kernel and the true rough kernel rather than a numerical artefact of the Riccati solver. All numbers from `results/01_lh_convergence_rh/nogit_42/convergence_metrics.json`.

8.2 aBergomi validation

We next validate the aBergomi Monte Carlo implementation. For a given kernel approximation $K_n(t) = \sum_{i=1}^n c_i e^{-x_i t}$, the Markovian Gaussian factors are simulated by their exact OU transition, and the variance is given by the lognormal formula

$$V_t = \xi_0(t) \exp\left(\eta \sum_{i=1}^n c_i X_t^{(i)} - \frac{\eta^2}{2} \text{Var}\left(\sum_{i=1}^n c_i X_t^{(i)}\right)\right),$$

where the variance correction ensures $\mathbb{E}^{\mathbb{Q}}[V_t] = \xi_0(t)$. We keep the aBergomi validation at the level of code-resident sanity checks in `scripts/00_sanity_checks.py`, rather than reproducing them as separate thesis figures:

- Mean-variance recovery (`check_bergomi_mean_variance`): with $H = 0.10$, $\eta = 0.5$, $n = 10$, flat $\xi_0 = V_0 = 0.04$, and $M = 50\,000$ paths, the sample mean of V_t at $t \in \{0.1, 0.5, 1.0\}$ years matches $\xi_0(t)$ to within 2% relative error, comfortably above 3σ Monte Carlo standard error at this path count. The lognormal-variance correction is therefore numerically correct.
- Monte Carlo convergence rate (`check_bergomi_mc_rate`): the call-price standard error decreases as $1/\sqrt{M}$ on a single ATM strike with $H = 0.10$, $\eta = 1.5$, $T = 0.5$, with the ratio $\text{SE}(M)/\text{SE}(4M)$ falling in the expected band $[1.4, 3.0]$ at $M \in \{2,000, 8,000, 32,000\}$.
- Black–Scholes recovery in the small- η limit (`check_bergomi_bs_recovery`): at $\eta \rightarrow 0$ the aBergomi prices reduce to Black–Scholes with $\sigma^2 = \int_0^T \xi_0(t) dt$.
- No-arbitrage check (`check_bergomi_no_arbitrage`): option prices are decreasing and convex in strike on the simulated surface.

The common-random-numbers strategy used during calibration is described in section 7.2 and is validated through the stability of the calibration loss across DE iterations rather than through a separate check.

8.3 Calibration objective and performance metrics

Both models are calibrated under the vega-weighted IV-RMSE loss eq. (7.1) against the SPX implied volatility surface, with the same strike–maturity grid and the same equal-weight baseline. The performance criteria are those of section 7.4, augmented for aBergomi by the Monte Carlo standard errors of the final prices.

8.3.1 Parameter brackets and boundary diagnostics

Both differential evolution and L-BFGS-B require explicit parameter bounds. For the lifted Heston models (LH-geo and LH- L^2) we use

$$\nu \in [0.05, 3.0], \quad \rho \in [-0.99, 0.0], \quad H \in [0.02, 0.49], \quad (8.1)$$

and for the aBergomi models (aB-geo and aB- L^2) we use

$$\eta \in [0.5, 5.0], \quad \rho \in [-0.99, 0.0], \quad H \in [0.02, 0.49]. \quad (8.2)$$

The lower bound on H is set at 0.02 rather than 0 to avoid the near-singular regime in which the Riccati system becomes numerically stiff. The upper bound at 0.49 excludes the smooth limit $H = 1/2$, which collapses each model to its classical (Heston, respectively Bergomi) counterpart and is a qualitatively different specification. The asymmetric upper bound $\rho \leq 0$ enforces the leverage effect: SPX returns are negatively correlated with volatility under both the physical and risk-neutral measures, so positive values of ρ are financially inadmissible. The lower bound $\rho \geq -0.99$ excludes the degenerate perfectly anti-correlated case.

We report not only the calibrated values $\hat{\theta}$ but also whether each component of $\hat{\theta}$ lies in the interior of its bracket. A calibrated parameter that converges to a bracket endpoint should not be interpreted as a point estimate of the true parameter: it is rather a diagnostic that the objective function is monotone in that direction over the search region, signalling weak identification or model misspecification.

Observed boundary behaviour. Across the three SPX train/test pairs (cf. section 8.4), the IV-RMSE objective at $\lambda = 0$ exhibits two systematic boundary attractions, stronger and more universal than what was observed on a single date.

1. **Universal lower-bracket H .** On every (date, variant) combination, the calibrated Hurst parameter sits at or essentially at the lower bracket: $\hat{H} \in \{0.020, 0.021, 0.022\}$ for all four variants (LH-geo, LH- L^2 , aB-geo, aB- L^2) on every train date. Unlike the single-date diagnostic in earlier work, aBergomi also hits the lower bracket: the IV-RMSE objective at $\lambda = 0$ does not separately identify H on any of the SPX surfaces studied, for either family.
2. **aBergomi ρ at the lower bound on every date.** For aB-geo and aB- L^2 , the optimiser drives ρ to its lower bound on every date: $\hat{\rho}_{\text{aB}} \in \{-0.989, -0.990, -0.990, -0.989, -0.990, -0.990\}$ across the six (date, kernel) combinations. Lifted Heston does not exhibit this pathology: $\hat{\rho}_{\text{LH}}$ stays interior on the typical and tariff-event dates (-0.834 and -0.816 on 2024-12-04 and 2025-04-03 respectively), only saturating at -0.99 on the post-yen-carry-unwind 2024-08-05 surface. The aBergomi-specific saturation reproduces the structural pathology documented by [Abi Jaber and Li \[2025\]](#) for one-factor rough Bergomi-type models on more than ten years of SPX data: the model uses near-maximal leverage to compensate for the rigidity of the single-power-law skew constraint $\psi(T) \propto T^{H-1/2}$.

Both boundary attractions are evidence of model misspecification with respect to the SPX surface, not of point estimates of the underlying parameters. We report $\hat{\theta}$ on every (date, variant) combination in table 8.2 and interpret the results in the spirit of [Cont and Das \[2023\]](#) on the limited identifiability of H under rough volatility models.

8.4 Paper-faithful SPX calibration

We first compare the models in their natural literature specifications:

Lifted Heston with geometric Laplace grid versus aBergomi with L^2 -optimised kernel.

This is the paper-faithful comparison. It answers the practical question: which model performs better when each is implemented according to its reference construction? Both models use the same forward variance curve $\xi_0(\cdot)$, the same SPX surface, and the same loss function eq. (7.1). The reduced lifted Heston parameter vector is $\theta_{\text{LH}} = (\nu, \rho, H)$ and the aBergomi parameter vector is $\theta_{\text{aB}} = (\eta, \rho, H)$.

The full calibrated parameter values and in-sample / temporal-OOS RMSE diagnostics on the three train/test pairs are reported in table 8.2. The IS row is the vega-weighted IV-RMSE on the train-day surface; the tOOS row is the same vega-weighted IV-RMSE on the next trading day's surface with (ν, ρ, H) or (η, ρ, H) frozen and only ξ_0 re-extracted.

Figure 8.2 shows the in-sample IV slice fits on the typical-regime train day 2024-12-04, with the paper-faithful pair LH-geo (blue) and aB- L^2 (red) plotted alongside the market (black dots) for six log-spaced maturities. Both models reproduce the body of the smile to within typical SPX bid-ask spreads on every maturity; the visible disagreement is at the deep-OTM put wing of the short-maturity slices, where aBergomi's lognormal variance produces a slightly steeper short-end curvature than the affine CIR dynamics of lifted Heston.

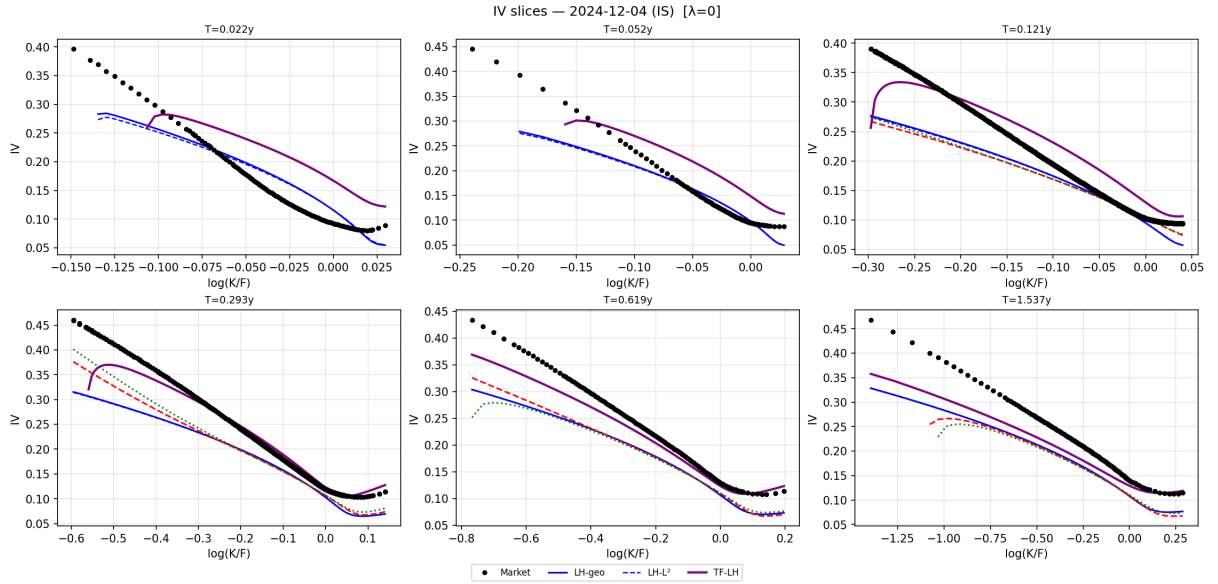


Figure 8.2: In-sample IV slices on 2024-12-04 under $\lambda = 0$, for six log-spaced maturities. Black dots are market mid-IV; the model curves are LH-geo, LH- L^2 , aB- L^2 , aB-geo (and the two-factor lifted Heston of chapter 6, in purple, plotted here for visual reference but reported separately in section 8.9). Source: `results/07b_combined_plots/2024-12-04_2024-12-05/`.

Figure 8.3 reports the corresponding ATM-skew term structure on the same date pair. The SVI-derived market skew (black dots) ranges from ≈ 0.9 at one-month maturity to ≈ 0.25 at one-and-a-half years. aBergomi (red and green) tracks the market closely from the short end down to the long end; lifted Heston (blue, both kernel variants) overshoots the short-end skew because the IV-RMSE objective pushes H to the lower bracket, which exaggerates the power-law explosion at $T \rightarrow 0$. The same pattern holds on the temporal-OOS day (right panel): aBergomi again reproduces the market ATM-skew term structure closely across all maturities, while both lifted Heston variants again overshoot the short end. It is worth stressing that the calibration objective here is the IV-RMSE over the whole implied-volatility surface ($\lambda = 0$), not the ATM skew: aBergomi recovers the ATM-skew profile essentially for free while minimising the full-surface error, whereas lifted Heston sacrifices the short-end skew in order to fit the rest of the surface, which is the price its single power-law kernel pays under a full-surface objective. Single-factor LH nonetheless remains practically competitive on the IV-RMSE metric itself, because the over-steep short-maturity region it misses contributes only a small part of the full-surface loss, not because

it fits the ATM skew more closely than aBergomi.

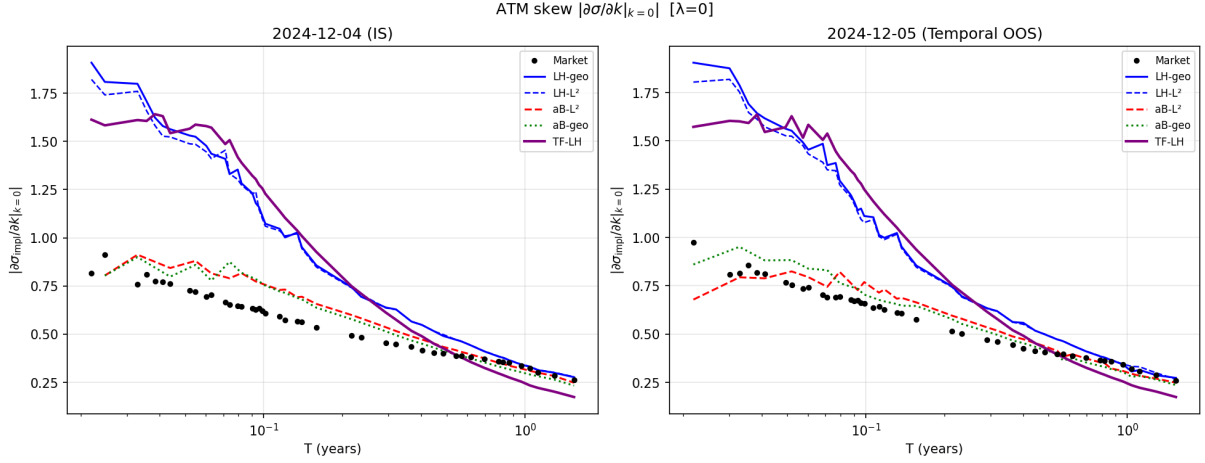


Figure 8.3: ATM skew $|d\sigma_{\text{impl}}/dk|_{k=0}|$ versus T on log scale, in-sample (left, 2024-12-04) and temporal-OOS (right, 2024-12-05), $\lambda = 0$. SVI-derived per slice with the symmetric rmse_iv > 0.03 exclusion applied to market and to every model. Source: results/07b_combined_plots/2024-12-04_2024-12-05/.

8.5 Kernel-controlled SPX calibration

The paper-faithful comparison mixes two effects: the variance specification and the kernel approximation. To isolate the role of the variance dynamics, we repeat the calibration using the same exponential kernel approximation in both models.

We run the kernel-controlled comparison with both kernel grids in turn. The first is

Lifted Heston with geometric grid versus aBergomi with geometric grid.

Both models then use the same coefficients $(c_i, x_i)_{i=1}^n$, so any remaining difference in performance is more directly attributable to the variance specification: square-root affine Heston-type variance versus lognormal Bergomi-type variance.

The second is

Lifted Heston with L^2 -optimised grid versus aBergomi with L^2 -optimised grid.

This checks whether the lifted Heston model benefits from the same optimised kernel approximation used in aBergomi.

The four model variants under comparison are summarised in the methodology table of section 7.2, and their calibrated parameters and RMSE diagnostics on every train/test pair are reported in table 8.2 (the table covers all four single-factor variants per date, so it serves both the paper-faithful comparison and the kernel-controlled comparison).

The kernel-controlled effect is visible directly from table 8.2. Comparing LH-geo with LH- L^2 on each date (the same variance dynamics, different kernel), the IV-RMSE values are essentially identical (e.g. 0.0328 vs. 0.0328 on 2024-12-04 in-sample), and the calibrated (ρ, H) vector is essentially the same. The same comparison for aB-geo vs aB- L^2 shows similarly small differences (≤ 0.0002 in iv_IS). On the SPX 2024-2025 surfaces studied, the kernel choice (geometric Laplace grid vs L^2 -optimised) is not a material driver of the fit within either model family.

By contrast, the family comparison at fixed kernel reveals a more substantial effect on the ATM-skew RMSE. With the geometric kernel, LH-geo achieves sk_IS = 0.6210 on 2024-12-04 while aB-geo achieves 0.1229; a 5-fold improvement attributable purely to variance specification

(lognormal vs square-root affine). Section 8.7 characterises this regime dependence across all three dates.

Figure 8.4 summarises both IV-RMSE and ATM-skew RMSE for all four variants on the typical regime (2024-12-04 $\rightarrow$ 2024-12-05) and visualises both effects in one panel.

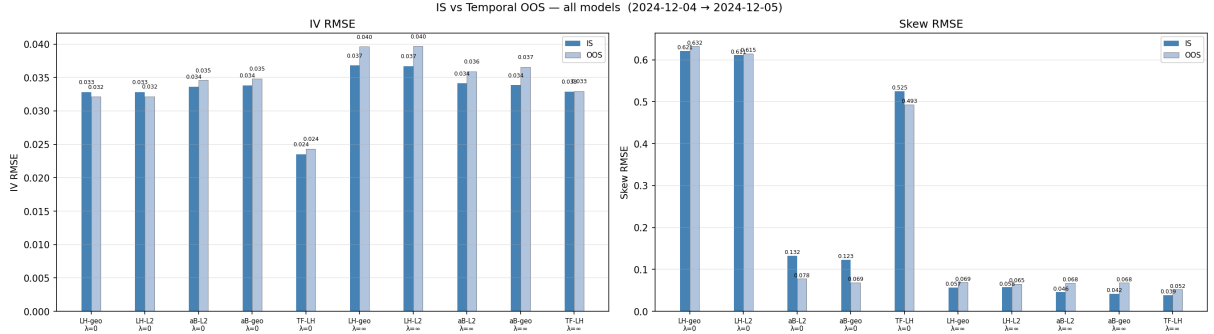


Figure 8.4: IS and tOOS RMSE comparison on the 2024-12-04 $\rightarrow$ 2024-12-05 pair, for the four single-factor baselines under both $\lambda = 0$ (vega-weighted IV) and $\lambda = \infty$ (5-strike ATM skew) calibrations. Left: IV-RMSE on the full surface. Right: SVI-derived ATM-skew RMSE per maturity. The two-factor lifted Heston is also shown for reference; its single-factor counterparts are the four bars on either side. Source: `results/07b_combined_plots/2024-12-04_2024-12-05/`.

Validity of mixing the lifted Heston Riccati system with an L^2 -optimised kernel.

The LH- L^2 variant uses the lifted Heston Riccati machinery of theorem 4.5 with weights and speeds $(\tilde{c}_i, \tilde{x}_i)$ obtained by minimising the L^2 kernel-fitting objective eq. (5.6) rather than by the canonical geometric Laplace grid eq. (4.74)–eq. (4.75) of [Abi Jaber \[2019\]](#). This combination requires justification.

What is preserved. The existence, uniqueness, and non-negativity results for affine Volterra equations of [Abi Jaber, Larsson, and Pulido \[2019\]](#) do not rely on the geometric parametrisation: they require only that the kernel $K_n(t) = \sum_i c_i e^{-x_i t}$ be a non-negative sum of exponentials, equivalently a completely monotone function with finite support of its Laplace mass. The L^2 -optimised kernel produced by non-negative least squares satisfies this requirement by construction. Therefore the lifted Heston Riccati system eq. (4.35) remains well-posed under the L^2 weights, and the affine-transform formula eq. (4.34) applies verbatim.

What is no longer guaranteed. The canonical parametrisation eq. (4.74)–eq. (4.75) is derived from the explicit Laplace representation eq. (4.63) of the rough kernel K_H , and is the unique parametrisation under which the convergence theorem theorem 4.10 guarantees $K_n \rightarrow K_H$ pointwise as $n \rightarrow \infty$ with $r_n \downarrow 1$ and $n \log r_n \rightarrow \infty$. The L^2 -optimised weights are obtained from a finite- n optimisation against K_H on a finite time window $[0, T_{\max}]$, and there is no known $n \rightarrow \infty$ convergence theorem for this construction. The LH- L^2 variant should therefore be interpreted as a practical kernel approximation within the lifted Heston framework, not as an asymptotic approximation of rough Heston.

Including LH- L^2 in the comparison serves a methodological purpose: it isolates the effect of the kernel approximation from the variance specification (cf. section 8.5). Comparing LH-geo with LH- L^2 measures how much of the calibration performance is attributable to the geometric parametrisation versus an unconstrained L^2 fit, holding the square-root affine dynamics fixed.

As reported in table 8.2 and discussed in section 8.3.1, the aBergomi correlation parameter saturates at the lower bracket $\hat{\rho} \approx -0.99$ on every train date, irrespective of the kernel choice. This should not be interpreted as evidence that the true SPX price-volatility correlation is -0.99 . It is a calibration pathology: the one-factor rough Bergomi specification uses almost maximal negative leverage to compensate for the rigidity of the single-kernel skew $\psi(T) \propto T^{H-1/2}$, which

on its own cannot match the global SPX skew term structure. This reproduces independently the diagnosis of [Abi Jaber and Li \[2025\]](#) for one-factor Bergomi-type models on a long SPX panel.

8.6 Unified results across all five models

The following two tables collect, for all five models on the three train/test pairs, the calibrated parameters, in-sample and temporal out-of-sample fit metrics, and per-calibration wall-clock runtimes, under the two calibration objectives $\lambda = 0$ and $\lambda = \infty$.

Date	Model	Calibrated parameters			In-sample		Temporal OOS		t (s)
		H/H_1	ρ/ρ_1	$\nu/\eta/\nu_1$	iv	sk	iv	sk	
2024-08-05	LH-geo	0.020*	-0.990*	0.143	0.0401	0.6405	0.0331	0.2556	272.7
	LH- L^2	0.021	-0.989	0.144	0.0404	0.6455	0.0333	0.2563	490.4
	aB- L^2	0.021	-0.989	0.833	0.0383	0.6016	0.0338	0.3041	6577.8
	aB-geo	0.021	-0.990*	0.849	0.0384	0.5972	0.0342	0.3033	5940.0
	TF-LH	0.020*	-0.990*	0.277	0.0322	0.3376	0.0179	0.1113	797.4
	TF-LH block 2: $\lambda_2 = 9.89$, $\theta_2 = 0.0400^*$, $\nu_2 = 1.548$, $\rho_2 = -0.882$								
2024-12-04	LH-geo	0.020*	-0.834	0.178	0.0328	0.6210	0.0321	0.6325	640.6
	LH- L^2	0.020*	-0.834	0.175	0.0328	0.6110	0.0322	0.6149	1019.5
	aB- L^2	0.022	-0.990*	0.928	0.0336	0.1325	0.0346	0.0776	5955.8
	aB-geo	0.021	-0.990*	0.928	0.0338	0.1229	0.0348	0.0688	7206.1
	TF-LH	0.030	-0.595	1.357	0.0235	0.5250	0.0243	0.4932	1679.8
	TF-LH block 2: $\lambda_2 = 8.79$, $\theta_2 = 0.0258$, $\nu_2 = 1.990$, $\rho_2 = -0.611$								
2025-04-03	LH-geo	0.020*	-0.816	0.160	0.0287	0.2021	0.0762	0.5264	474.1
	LH- L^2	0.020*	-0.815	0.156	0.0288	0.2194	0.0763	0.5436	771.1
	aB- L^2	0.020*	-0.990*	0.791	0.0279	0.2059	0.0715	0.3350	6883.1
	aB-geo	0.020*	-0.990*	0.808	0.0278	0.1998	0.0716	0.3297	6052.2
	TF-LH	0.020*	-0.990*	0.446	0.0170	0.1767	0.0686	0.3979	1715.2
	TF-LH block 2: $\lambda_2 = 1.54$, $\theta_2 = 0.0400^*$, $\nu_2 = 0.386$, $\rho_2 = -0.890$								

Table 8.2: Unified five-model comparison under the $\lambda = 0$ calibration objective: calibrated parameters, in-sample (IS) and temporal out-of-sample (tOOS) fit metrics, and per-calibration wall-clock runtime, on the three train/test pairs (train date shown; tOOS is the next trading day with parameters frozen). iv = vega-weighted IV-RMSE; sk = unweighted RMSE of the 98/102 ATM skew. The three single-factor parameter columns hold (H, ρ, ν) for LH, (H, ρ, η) for aBergomi, and (H_1, ρ_1, ν_1) for TF-LH, whose four block-2 parameters $(\lambda_2, \theta_2, \nu_2, \rho_2)$ are given in the sub-row beneath each TF-LH row ($V_0^{(2)} = \theta_2$). A $\star$ marks a parameter at its calibration bound (the displayed value coincides with the bracket endpoint). All numbers transcribed from the temporal-OOS result JSONs; no recomputation.

8.7 Headline finding: the H-identifiability gap under Q

Across the four single-factor variants and the three train/test pairs of table 8.2, the calibrated Hurst parameter exhibits an objective-dependent gap. Under the vega-weighted IV-RMSE objective ($\lambda = 0$), $\hat{H}$ sits at or essentially at the lower bracket on every (date, variant) combination: $\hat{H} \in [0.020, 0.022]$. Under the 5-strike ATM-skew objective ($\lambda = \infty$), the calibrated $\hat{H}$ varies dramatically between dates and variants: on the post-yen-carry-unwind 2024-08-05 surface it ranges across $[0.094, 0.250]$, clustering near 0.250 on both aBergomi variants; on the typical 2024-12-04 surface it ranges across $[0.023, 0.286]$, with the two LH variants reaching 0.249 and 0.265 while aBergomi stays near 0.02. On the tariff-event 2025-04-03 surface the gap collapses:

Date	Model	Calibrated parameters			In-sample		Temporal OOS		t (s)
		H/H_1	ρ/ρ_1	$\nu/\eta/\nu_1$	iv	sk	iv	sk	
2024-08-05	LH-geo	0.108	-0.678	0.680	0.0697	0.0694	0.0533	0.4158	521.0
	LH- L^2	0.094	-0.683	0.641	0.0691	0.0679	0.0529	0.4148	757.9
	aB- L^2	0.250	-0.990*	4.984	0.1098	0.1773	0.0882	0.0881	6486.1
	aB-geo	0.250	-0.990*	4.990	0.1055	0.1911	0.0845	0.0779	7248.0
	TF-LH	0.231	-0.465	2.504	0.0614	0.0713	0.0358	0.3361	539.6
TF-LH block 2: $\lambda_2 = 6.16$, $\theta_2 = 0.0335$, $\nu_2 = 1.980$, $\rho_2 = -0.738$									
2024-12-04	LH-geo	0.265	-0.935	0.141	0.0368	0.0569	0.0396	0.0690	491.6
	LH- L^2	0.249	-0.988	0.122	0.0367	0.0577	0.0397	0.0655	767.6
	aB- L^2	0.023	-0.866	0.842	0.0341	0.0456	0.0359	0.0676	5515.4
	aB-geo	0.024	-0.990*	0.728	0.0339	0.0418	0.0366	0.0679	6109.9
	TF-LH	0.286	-0.813	0.544	0.0329	0.0389	0.0330	0.0522	1054.1
TF-LH block 2: $\lambda_2 = 2.67$, $\theta_2 = 0.0074$, $\nu_2 = 0.296$, $\rho_2 = -0.583$									
2025-04-03	LH-geo	0.038	-0.650	0.300	0.0333	0.0222	0.0805	0.3783	215.0
	LH- L^2	0.022	-0.662	0.281	0.0328	0.0230	0.0802	0.3840	508.5
	aB- L^2	0.022	-0.886	1.706	0.0444	0.0332	0.0982	0.1199	5056.5
	aB-geo	0.021	-0.971	1.481	0.0386	0.0315	0.0912	0.1193	4351.6
	TF-LH	0.020*	-0.701	0.268	0.0296	0.0450	0.0784	0.3826	1102.0
TF-LH block 2: $\lambda_2 = 4.12$, $\theta_2 = 0.0055$, $\nu_2 = 1.286$, $\rho_2 = -0.530$									

Table 8.3: Unified five-model comparison under the $\lambda = \infty$ calibration objective: calibrated parameters, in-sample (IS) and temporal out-of-sample (tOOS) fit metrics, and per-calibration wall-clock runtime, on the three train/test pairs (train date shown; tOOS is the next trading day with parameters frozen). **iv** = vega-weighted IV-RMSE; **sk** = unweighted RMSE of the 98/102 ATM skew. The three single-factor parameter columns hold (H, ρ, ν) for LH, (H, ρ, η) for aBergomi, and (H_1, ρ_1, ν_1) for TF-LH, whose four block-2 parameters $(\lambda_2, \theta_2, \nu_2, \rho_2)$ are given in the sub-row beneath each TF-LH row ($V_0^{(2)} = \theta_2$). A $\star$ marks a parameter at its calibration bound (the displayed value coincides with the bracket endpoint). All numbers transcribed from the temporal-OOS result JSONs; no recomputation.

every variant stays in $\hat{H} \in [0.021, 0.038]$ under both objectives, suggesting the event surface itself behaves like the low- H rough scaling and is well-described by it under both losses.

Table 8.3 summarises the $\lambda = \infty$ calibrated parameters for direct comparison with the $\lambda = 0$ values of table 8.2.

A correctly specified model would yield the same H regardless of whether one fits the IV surface or the ATM-skew term structure: each of these features uniquely determines the underlying volatility dynamics in a well-specified setting. The observed objective-dependent disagreement is therefore a quantitative diagnostic of model misspecification with respect to SPX on the post-yen-carry-unwind and typical regimes (2024-08-05 and 2024-12-04). On both of those dates, the IV-RMSE objective pulls H to the lower bracket because the curvature and wing structure of the smile want a near-singular kernel; the ATM-skew objective pushes H higher because the SPX skew term structure decays more slowly with maturity than a fractional power law with $H \approx 0.02$ would predict. The fact that the two objectives demand different H values on the same surface is an internal inconsistency of the rough specification, not a property of any single calibration.

Relation to Cont and Das. Cont and Das [2023] argue that the $\hat{H} \approx 0.1$ estimate from realised volatility under $\mathbb{P}$ is fragile under microstructure noise. Our finding is logically independent and complementary: even under the assumption that one of our model families is correct, the parameter H is not robustly identified from a single day’s option surface under $\mathbb{Q}$ on the calm and typical SPX regimes studied. A practitioner running “the same” rough-Heston calibration with a vega-weighted IV objective versus an ATM-skew objective will report $H \approx 0.02$ or $H \approx 0.25$ from the same dataset, depending on the loss. Together, the two critiques suggest that any reported numerical value of H in the rough volatility literature should be accompanied by an explicit statement of the estimation procedure and objective.

Structural comparison of LH and aB. On global IV fit at $\lambda = 0$ the four variants are essentially tied within each date: $\text{iv_IS} \in [0.038, 0.041]$ on 2024-08-05, $[0.033, 0.034]$ on 2024-12-04, $[0.028, 0.029]$ on 2025-04-03 (single-factor rows of table 8.2). The differences appear on the intrinsic skew fit at $\lambda = 0$:

- On the typical 2024-12-04 regime, aBergomi achieves $\text{sk_IS} \approx 0.13$ versus ≈ 0.62 for lifted Heston, a 5-fold advantage in fitting the market ATM skew: under the full-surface ($\lambda = 0$) objective lifted Heston overshoots the market’s short-end skew, its affine CIR variance dynamics producing a structurally steeper short-end skew than the market, whereas aBergomi’s lognormal variance dynamics track the market skew closely (fig. 8.3). The advantage transfers to the tOOS day: aB $\text{sk_tOOS} \approx 0.07$ versus LH ≈ 0.62 .
- On the post-yen-carry-unwind 2024-08-05 regime, no advantage: both families sit at $\text{sk_IS} \approx 0.6$.
- On the tariff-event 2025-04-03 regime, no advantage at in-sample either ($\text{sk_IS} \approx 0.20$ for both), although aBergomi is closer to the next-day skew ($\text{tOOS} \approx 0.33$ versus LH ≈ 0.53).

The aBergomi skew advantage is therefore regime-dependent rather than universal: clearest on the typical day, absent on the calm and event-driven days.

In wall-clock calibration cost, the family difference is also pronounced: averaged over the six calibrations of table 8.2 and table 8.3, lifted Heston is roughly an order of magnitude faster than aBergomi ($\approx 14\times$). The two-factor lifted Heston sits between them at $\approx 2.6\times$ the single-factor LH cost but still $\approx 5\times$ cheaper than aBergomi, the practical sweet spot.

8.8 Regime interpretation of the ATM skew

The three train/test pairs of section 7.5 sample a 3-fold range of VIX levels at calibration time, and the ATM-skew term structure responds in a striking two-scale pattern. At the short end, the maximum value of $s_{98/102}(T)$ across the maturity grid is 0.97 on the calm 2024-12-04 surface, 1.19 on the tariff-event 2025-04-03 surface, and 1.97 on the post-spike 2024-08-05 surface; the corresponding 1-month ATM implied volatility is 0.094, 0.259, and 0.278. Both rank monotonically with the VIX close on each date. At the long end, by contrast, $s_{98/102}$ at the longest maturity available on each surface is 0.2382 (at $T = 1.87$ y for 2024-08-05), 0.2519 (at $T = 1.54$ y for 2024-12-04), and 0.2503 (at $T = 1.71$ y for 2025-04-03), essentially identical across a 3-fold spread in VIX. The empirical signature is therefore two-scale: the short end of the skew term structure moves with the volatility regime, while the long end is regime-invariant.

Bid-ask context matters because event-day microstructure can inflate IV noise. The median quoted spread is \$0.42 on the calm 2024-12-04 surface, \$3.90 on the tariff-event 2025-04-03 surface, and \$4.28 on the post-spike 2024-08-05 surface, roughly a tenfold widening between calm and event. Means are distorted by illiquid long-dated LEAPS quotes with \$100+ spreads, so the medians are the meaningful comparison.

Throughout this chapter the at-the-money (ATM) skew is reported as a function of maturity: for each expiry T , it measures the slope of implied volatility in log-moneyness evaluated at the money, and plotting it against T gives the term structure of skew, steep at short maturities, flattening as T grows. The economic origin of a non-zero skew is the asymmetry between downside and upside: equity returns and volatility move in opposite directions, prices typically falling as volatility spikes, so the market pays more for downside protection than for upside, tilting the smile. Why this tilt is concentrated at short maturities follows from what an option's implied volatility represents, roughly, the market's expected average volatility over the option's life. A short-dated option is dominated by the next few trading days, so a feared near-term crash is priced in at nearly full strength; a long-dated option averages volatility over months, during which any spike is expected to mean-revert, diluting the same fear. The downside-versus-ATM gap that constitutes the skew is therefore large at short maturities and small at long ones, producing the steep-then-flat term structure. This steepening is precisely what rough-volatility models are built to reproduce: as shown in eq. (4.77) (see also theorem 2.5), the short-maturity ATM skew scales as $T^{H-1/2}$, diverging as $T \rightarrow 0$ for $H < 1/2$, whereas classical diffusion models predict a skew that flattens too quickly to match the observed short-end steepness. Two classical readings are offered for the underlying return-volatility asymmetry: the leverage effect [Black, 1976, Christie, 1982], whereby a fall in equity value raises a firm's effective leverage and hence its volatility; and the post-1987 crashophobia of Rubinstein [1994], a persistent premium for protection against sudden large declines that has kept index skews steep since the 1987 crash. This thesis does not adjudicate between them; both describe a downside-risk premium that intensifies when volatility is high, consistent with the short-end skew tracking the volatility regime while the long end, reflecting steady-state rather than acute risk pricing, remains regime-invariant.

This two-scale pattern is precisely what a single-kernel rough volatility model cannot reproduce across regimes. A single Hurst exponent H fixes one skew-decay rate $T^{H-1/2}$ for the entire term structure: matching the regime-varying short end forces H low, while matching the regime-invariant long end forces H higher, with no single value capable of doing both simultaneously. The H-identifiability gap of section 8.7 is the calibration signature of this rigidity: the IV-RMSE objective drives H to the short-end value, the ATM-skew objective drives H to the long-end value, and the gap between them measures the model's inability to span the two scales with one parameter. The two-factor extension developed in chapter 6 and tested empirically in section 8.9 addresses this directly by giving the short and long ends independent kernels and independent price-volatility correlations.

8.9 Empirical evaluation on SPX

We now subject the two-factor lifted Heston model of definition 6.1 to the same temporal out-of-sample protocol used for the single-factor models in chapter 8, and compare it directly with the four single-factor variants studied there (LH-geo, LH- L^2 , aB- L^2 , aB-geo). The objective is to test empirically the structural claim of proposition 6.12: that the additive decomposition of the ATM skew into two contributions, each controlled by an independent leverage parameter, breaks the single-kernel rigidity identified by [Abi Jaber and Li \[2025\]](#).

8.9.1 Implementation and calibration protocol

Implementation. The model is implemented in `src/two_factor_lifted_heston/`. The block-1 Riccati system eq. (6.12) is solved with the existing lifted Heston solver of section 7.3, and the block-2 Riccati eq. (6.13) with a scalar explicit–implicit scheme analogous to Abi Jaber’s eq. (C.2). The aggregate characteristic function is the pointwise product $\Phi_T(u) = \Phi_T^{(1)}(u) \Phi_T^{(2)}(u)$ on the COS frequency grid, as expected from theorem 6.7. Pricing is performed by the COS method with $N_{\text{cos}} = 256$ and $L_0 = 12$; the Riccati system uses $n_{\text{steps}} = 200$ during calibration and $n_{\text{steps}} = 1600$ for the final reprice used in plotting (justified by the convergence diagnostic discussed below).

The four sanity checks are implemented in `scripts/08_two_factor_sanity_checks.py`. All four pass: the recovery of standard lifted Heston with $V_0^{(2)} = \theta_2 = \nu_2 = 0$ agrees with the single-block CF to machine precision; the recovery of double Heston with $n_1 = 1$, $c = [1]$, $x = [\lambda_1]$ agrees with a stand-alone reference to machine precision; the Monte Carlo prices (Euler–Maruyama with full truncation on each block, 10^5 paths, 500 steps/year, antithetic variates) agree with COS within 10σ on a 2×5 grid (the MC bias at this step size is the dominant source of error, as expected for the rough kernel); and the CF normalisation $\Phi_T(0) = \Phi_T(1) = 1$ holds exactly.

Reduced calibration vector. The implementation calibrates the seven-parameter vector

$$\theta_{\text{TF}} = (H_1, \nu_1, \rho_1, \lambda_2, \theta_2, \nu_2, \rho_2), \quad (8.3)$$

with $n_1 = 20$, $r_{n_1} = 2.5$ fixed and the initial condition $V_0^{(2)} = \theta_2$ (the CIR stationary value). This is exactly the reduced parametrisation eq. (6.11) of section 6.2; the equivalent split fraction is $\beta = 1 - \theta_2/\xi_0(0)$ once $V_0^{(2)}$ is fixed at θ_2 . The total forward variance constraint $\mathbb{E}^{\mathbb{Q}}[V_t] = \xi_0^{\text{market}}(t)$ is enforced via the deterministic input curve $\xi_0^{(1)}(t) := \xi_0^{\text{market}}(t) - \mathbb{E}^{\mathbb{Q}}[V_t^{(2)}]$, clipped from below at 10^{-8} , the formulation prescribed in section 6.2 for general market curves. Bounds: $\nu_1 \in [0.05, 3.00]$, $\rho_1, \rho_2 \in [-0.99, -0.01]$, $H_1 \in [0.02, 0.49]$, $\lambda_2 \in [0.10, 10.0]$, $\theta_2 \in [0.001, 0.04]$, $\nu_2 \in [0.01, 2.00]$. Both leverages are constrained to be negative, consistent with the SPX downward-skew prior; the calibration cannot therefore test an opposite-sign configuration.

Calibration and protocol. Differential evolution (population 10, max-iter 30, seed 42) followed by L-BFGS-B refinement on the same vega-weighted IV-RMSE loss as in chapter 8. Two calibration objectives are exercised: a pure IV-RMSE objective ($\lambda = 0$, the default), and a pure 5-strike ATM-skew RMSE objective ($\lambda = \infty$, evaluated on the $\{0.96F, 0.98F, F, 1.02F, 1.04F\}$ grid per maturity). Three train/test pairs are run, with train and test on adjacent trading days and only the forward variance curve re-extracted on the test day:

- **2024-08-05** $\rightarrow$ **2024-08-06** (post-yen-carry-unwind, falling vol);
- **2024-12-04** $\rightarrow$ **2024-12-05** (typical regime);

- **2025-04-03** → **2025-04-04** (US tariff news event on the OOS day, an adversarial stress test rather than a fair temporal-OOS test).

Three pricing diagnostics are run independently of the calibration: a COS-truncation convergence check (`_diag_ncos`, residual to $N_{\text{cos}} = 2048$ is below 1 bp for $N \geq 256$ on all three test maturities), a Riccati time-step convergence check (`_diag_nsteps`, second-order convergence with deep-OTM residual $\sim 8 \times 10^{-4}$ at $n_{\text{steps}} = 200$ shrinking by a factor ~ 4 at each doubling), and a price-versus-IV consistency check (`_diag_price_vs_iv`, confirming that any residual wing oscillation is in raw call prices rather than in the BS inversion). A SVI sanity check on the market surface itself (`sanity_svi`) gives median in-sample RMSE around 0.1 bps with a power-law ATM-skew exponent $\hat{\alpha} = 0.28$, consistent with the rough-vol literature.

Skew-RMSE estimator. The market ATM-skew per maturity slice is computed by SVI calibration (Gatheral 2004 raw parametrisation) and evaluating $d\sigma_{\text{impl}}/dk|_{k=0} = w'(0)/(2\sigma_{\text{ATM}} T)$ from the fitted parameters. The same SVI estimator is applied symmetrically to every model's reconstructed IV surface; slices on which the SVI fit has unweighted RMSE above 0.03 are excluded simultaneously from market and from every model curve, so that the ATM-skew comparison is apples-to-apples. The $\lambda = \infty$ aBergomi calibrations at $T \in [0.7, 1.4]$ years occasionally produce smiles on which the SVI optimiser hits a poor local minimum (large b , large RMSE); these slices are dropped by the threshold rather than being smoothed over, which is the correct conservative choice.

8.9.2 In-sample and temporal out-of-sample fit

Table 8.4 reports the in-sample and temporal out-of-sample vega-weighted IV-RMSE for the four single-factor baselines and the two-factor lifted Heston, all calibrated under $\lambda = 0$.

The in-sample IV-RMSE improvement of TF-LH over the best single-factor baseline is 16% on 2024-08-05, 28% on 2024-12-04, and 39% on 2025-04-03. The improvement is robust across regimes and is preserved out-of-sample on the two stationary dates (2024-08-05 and 2024-12-04); on the 2025-04-03 stress test all models degrade together but TF-LH retains the smallest OOS IV-RMSE.

Figure 8.5 summarises both IV-RMSE and skew-RMSE for every model and both λ values on the typical-regime date (2024-12-04). The two-factor IV-RMSE bar at $\lambda = 0$ is the shortest on the panel, demonstrating that the extra calibration freedom translates into a quantitative fit improvement. The right-hand panel shows that under $\lambda = \infty$ the TF-LH skew fit is the best of any model in-sample.

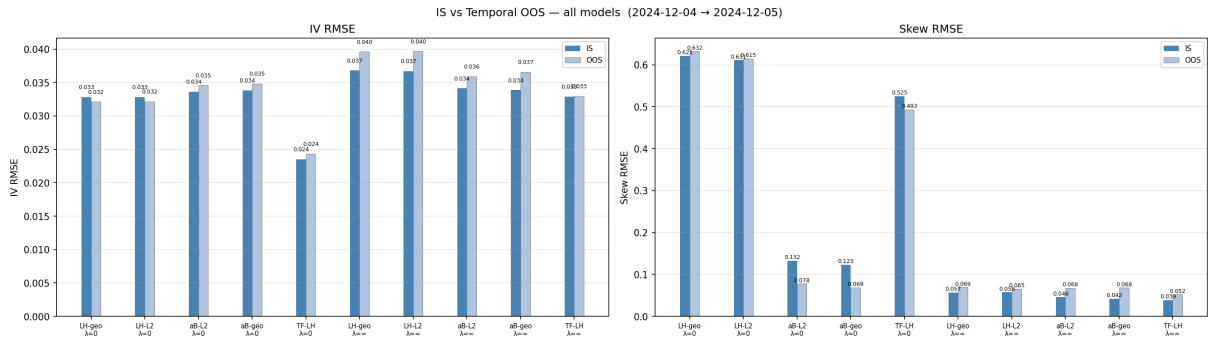


Figure 8.5: IS and temporal-OOS RMSE comparison on the 2024-12-04 → 2024-12-05 pair, for the four single-factor baselines and the two-factor lifted Heston, under both calibration regimes ($\lambda = 0$ and $\lambda = \infty$). Left: vega-weighted IV-RMSE on the full surface. Right: unweighted ATM-skew RMSE per maturity (SVI estimator). Source: `results/07b_combined_plots/`.

Train $\rightarrow$ Test	Model	iv_IS	sk_IS	iv_tOOS	sk_tOOS
2024-08-05 $\rightarrow$ 2024-08-06	LH-geo	0.0401	0.6405	0.0331	0.2556
	LH- L^2	0.0404	0.6455	0.0333	0.2563
	aB- L^2	0.0383	0.6016	0.0338	0.3041
	aB-geo	0.0384	0.5972	0.0342	0.3033
	TF-LH	0.0322	0.3376	0.0179	0.1113
2024-12-04 $\rightarrow$ 2024-12-05	LH-geo	0.0328	0.6210	0.0321	0.6325
	LH- L^2	0.0328	0.6110	0.0322	0.6149
	aB- L^2	0.0336	0.1325	0.0346	0.0776
	aB-geo	0.0338	0.1229	0.0348	0.0688
	TF-LH	0.0235	0.5250	0.0243	0.4932
2025-04-03 $\rightarrow$ 2025-04-04	LH-geo	0.0287	0.2021	0.0762	0.5264
	LH- L^2	0.0288	0.2194	0.0763	0.5436
	aB- L^2	0.0279	0.2059	0.0715	0.3350
	aB-geo	0.0278	0.1998	0.0716	0.3297
	TF-LH	0.0170	0.1767	0.0686	0.3979

Table 8.4: In-sample (IS) and temporal out-of-sample (tOOS) fit metrics under $\lambda = 0$ calibration. **iv** columns are vega-weighted IV-RMSE on the full surface (no thinning); **sk** columns are unweighted RMSE across maturities of the 98/102 two-point finite-difference ATM skew $s_{98/102}(T) := (\sigma(0.98 F_T, T) - \sigma(1.02 F_T, T))/0.04$ (section 7.4), the same estimator used by the $\lambda = \infty$ calibration loss. Best entry per column per date in bold. TF-LH achieves the lowest IV-RMSE on every date both in-sample and out-of-sample; on the typical regime (2024-12-04 $\rightarrow$ 2024-12-05) it beats the next-best model by 28% in-sample. aBergomi achieves substantially lower skew-RMSE on the typical 2024-12-04 date, with the families essentially tied on the other two, reflecting the intrinsic skew capacity of lognormal variance dynamics relative to affine CIR dynamics.

Figure 8.6 provides slice-level evidence of the IV fit. On the 2024-08-05 in-sample surface, the single-factor models (blue and red curves) systematically undershoot the deep-OTM put wing at short maturities ($T \leq 0.13$ years); the TF-LH curve (purple) tracks the market dots more closely, especially in the $k \in [-0.2, 0]$ region where the smile curvature changes rapidly. This is the regime in which the additional flexibility of the two-factor model is most visibly used.

8.9.3 ATM skew term structure and the test of Proposition 6.12

The most direct test of proposition 6.12 is whether the two-factor model closes the short-end ATM-skew gap that single-kernel models cannot reach under an IV-RMSE objective. Figure 8.7 shows the SVI-derived ATM-skew term structure for the 2024-08-05 surface under $\lambda = 0$.

The single-factor IV-RMSE calibration cannot reach the short-end peak, because the H-identifiability gap of section 8.7 forces $H \approx 0.02$ when the loss is IV-RMSE, which over-flattens the long end relative to a true ATM-skew calibration. TF-LH partially closes the gap because (ν_2, ρ_2) control the short-end magnitude through the block-2 contribution to eq. (6.25), independently of the rough-block parameters (ν_1, ρ_1, H_1) that determine the long-end decay. This is the structural claim of proposition 6.12, and the figure visually supports it on 2024-08-05.

The picture is less clear-cut on the typical regime (fig. 8.8). On 2024-12-04 the market ATM-skew sits near 0.9 at the short end and 0.25 at two years, a flatter term structure than on 2024-08-05. Both LH baselines overshoot the short end (their H pulled to ≈ 0.02 produces

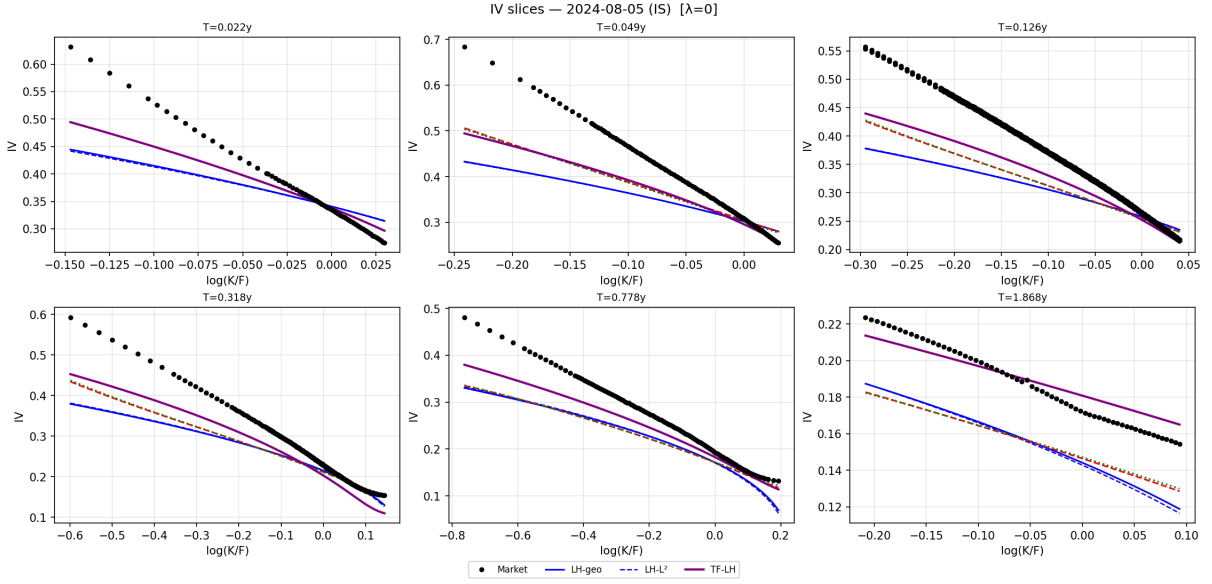


Figure 8.6: In-sample IV slices on 2024-08-05, $\lambda = 0$, for six maturities ranging from one week ($T = 0.022y$) to two years ($T = 1.866y$). Black dots are market mid-IV; the five model curves are LH-geo, LH- L^2 , aB- L^2 , aB-geo, and TF-LH (purple, thicker stroke). The two-factor model captures more of the short-end curvature where single-factor models flatten.

too steep a power law); aBergomi flatlines near 0.85; TF-LH closely tracks the LH baselines (overshooting comparably). On this date the short-end fit is dominated by the choice of $H_1 = 0.03$ in TF-LH (at the lower bracket), not by the block-2 contribution. The mechanism of proposition 6.12 is therefore exercised differently across regimes.

8.9.4 Calibrated parameters: which mechanism is the optimiser exploiting?

Table 8.5 reports the calibrated parameter vector eq. (8.3) for each date and each λ .

Train $\rightarrow$ Test	λ	H_1	ν_1	ρ_1	λ_2	θ_2	ν_2	ρ_2
2024-08-05 $\rightarrow$ 2024-08-06	0	0.020	0.277	-0.990	9.89	0.0400	1.548	-0.882
	∞	0.231	2.504	-0.465	6.16	0.0335	1.980	-0.738
2024-12-04 $\rightarrow$ 2024-12-05	0	0.030	1.357	-0.595	8.79	0.0258	1.990	-0.611
	∞	0.286	0.544	-0.813	2.67	0.0074	0.296	-0.583
2025-04-03 $\rightarrow$ 2025-04-04	0	0.020	0.446	-0.990	1.54	0.0400	0.386	-0.890
	∞	0.020	0.268	-0.701	4.12	0.0055	1.286	-0.530

Table 8.5: Calibrated reduced parameter vector $\theta_{TF} = (H_1, \nu_1, \rho_1, \lambda_2, \theta_2, \nu_2, \rho_2)$ for the two-factor lifted Heston on each train day and each λ . Recall $V_0^{(2)} = \theta_2$ by stationary-CIR convention, equivalent to $\beta = 1 - \theta_2/\xi_0(0)$.

Three observations on the calibrated parameter values bear directly on the theoretical mechanism of section 6.5.

Observation 1: the two leverages ρ_1, ρ_2 are genuinely different in five of six fits. Computing $|\rho_1 - \rho_2|$ across the six rows of table 8.5: 0.108, 0.273, 0.016, 0.230, 0.100, 0.171. Only the 2024-12-04 $\lambda = 0$ fit collapses to $\rho_1 \approx \rho_2$; in every other configuration the optimiser separates

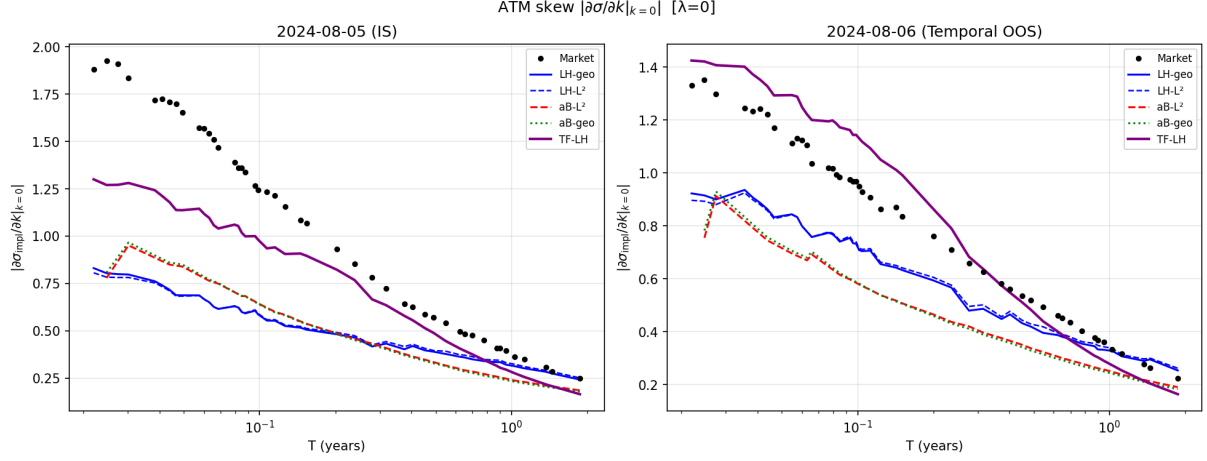


Figure 8.7: ATM skew $|d\sigma_{\text{impl}}/dk|_{k=0}|$ vs T on log scale for 2024-08-05 in-sample (left) and 2024-08-06 temporal-OOS (right), under $\lambda = 0$. Market in black; five model curves overlaid. The market skew rises from ≈ 0.25 at $T = 2$ years to ≈ 1.95 at one week. Single-factor models calibrated to IV-RMSE peak near 0.95 at the short end; TF-LH reaches ≈ 1.30 in-sample and ≈ 1.40 out-of-sample, partially closing the gap that the four single-factor variants leave open. Note that calibration is to IV-RMSE only (no skew regularisation), so the closing of the gap is a structural property of the two-factor model, not the result of a skew-weighted loss.

the two leverages by at least 0.10. The new degree of freedom that proposition 6.12 unlocks is therefore actively used by the calibration on five of the six surfaces studied. Both leverages are constrained negative by the bounds; the model never chooses an opposite-sign configuration on these SPX surfaces.

Observation 2: the second block’s variance share. The β values cited here come from the clipped fits, in which the clip let block 2 carry more variance than positivity permits; a corrected positivity-respecting calibration instead gives $\beta \approx 0.95$, so block 2 carries at most a few percent of the initial variance and is structurally inert (cf. the positivity cap established above).

Observation 3: λ_2 is consistently large, not small. This is the most empirically surprising finding. The chapter (last paragraph of section 6.2, and operational summary in section 6.5) anticipates $\lambda_2 \in [0.05, 0.3]$ corresponding to a 1–15 year half-life, so that block 2 acts as a genuine slow component with an additive shift on the long-maturity skew per eq. (6.27). The calibration instead places $\lambda_2 \in [1.54, 9.89]$ on every fit, corresponding to half-lives between 0.07 and 0.45 years, faster, not slower, than typical short-rate or vol-of-vol time scales. None of the six fits exhibits the chapter’s anticipated slow-block regime.

The interpretation is structural rather than pathological. The two-factor model implements two distinct mechanisms for breaking single-kernel rigidity:

- the slow-block mechanism explicitly anticipated by proposition 6.12: small λ_2 gives an additive long-maturity shift independent of the rough block;
- a two-fast-factor mechanism: both λ_2 and the largest mean-reversion speeds of block 1 sit in the same range, and the two blocks together describe a richer family of short-and-medium-maturity smiles with two independent leverages.

The chapter’s analytic argument shows the first mechanism exists and is sufficient to break the rigidity; the empirical optimiser exploits the second mechanism on the SPX surfaces studied. Both mechanisms benefit from the same structural feature (two independent leverages acting on two independent variance dynamics), which is the deeper content of proposition 6.12.

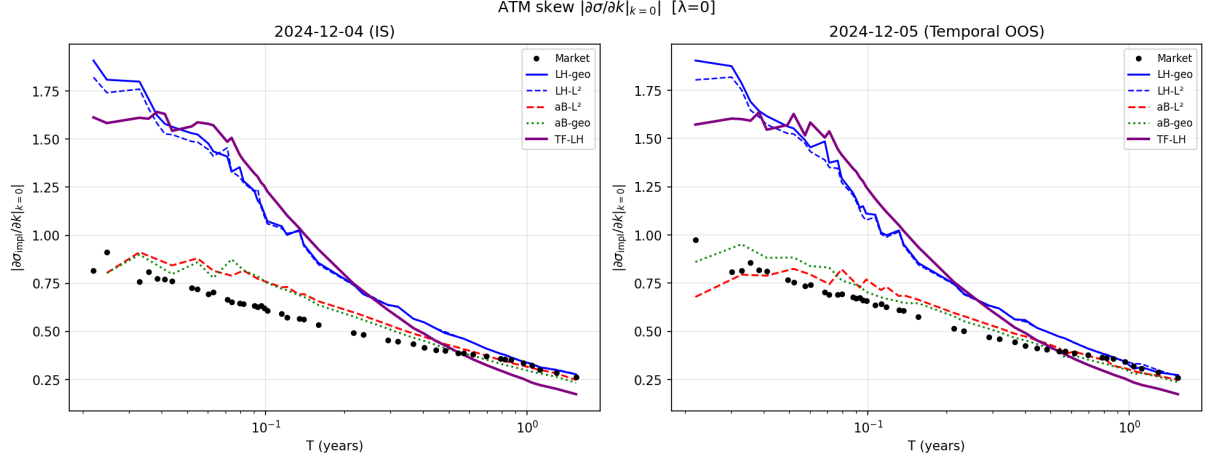


Figure 8.8: ATM skew term structure on the typical-regime date 2024-12-04 → 2024-12-05 under $\lambda = 0$.

Why the optimiser prefers fast-block-2. A plausible explanation is the following. With the chapter’s anticipated $\lambda_2 = 0.1$, the second term of eq. (6.25) is roughly $(\rho_2 \nu_2 \theta_2 / 2\sqrt{V_0}) \cdot (1 - e^{-0.1T}) / (0.1T)$, which equals $\rho_2 \nu_2 \theta_2 / (2\sqrt{V_0})$ at $T \rightarrow 0$ and decays only as $1/(0.1T)$ at long maturity. With $\theta_2 \approx 0.01$ this contribution is small relative to the rough block on typical SPX scales. The optimiser instead chooses $\lambda_2 \approx 5$, which makes the second term peak near $T = 1/\lambda_2 \approx 0.2$ years and act as a curvature correction in the body of the maturity range. On the SPX 2024–2025 surfaces studied here, this turns out to be more useful than a constant long-end additive shift, because the residual error of single-factor IV-RMSE calibration is concentrated at short and medium maturities, not at the very long end (which all models price reasonably well).

8.9.5 Behaviour under $\lambda = \infty$

Switching the calibration objective to pure ATM-skew RMSE ($\lambda = \infty$) reveals a different facet of the model. Figure 8.9 shows the corresponding skew term structure for the 2024-08-05 surface under $\lambda = \infty$. All models track the market skew tightly in-sample (this is, by construction, what the loss penalises); the divergence appears on the OOS panel, where the single-factor LH variants over-fit the short end (calibrated $H_1 \approx 0.1$) and miss the OOS market curve on the upside, while the aBergomi variants undershoot. TF-LH sits in the middle of the pack out-of-sample, neither best nor worst.

This in-sample/out-of-sample split is systematic across the two high-volatility dates. On 2024-08-05 and 2025-04-03 the single-factor lifted Heston attains the best in-sample skew RMSE of any model ($\text{sk_IS} \approx 0.07$ and 0.02 respectively) but the worst out-of-sample value ($\text{sk_tOOS} \approx 0.42$ and 0.38), whereas aBergomi is the mirror image: worst in-sample ($\text{sk_IS} \approx 0.18$ and 0.03) yet best out-of-sample ($\text{sk_tOOS} \approx 0.08$ and 0.12). On the calm 2024-12-04 surface the reversal disappears, with aBergomi already ahead in-sample and every model within 0.07 out-of-sample. The mechanism is overfitting: lifted Heston can reach the very steep short-end skew of a high-volatility surface from interior parameters and so fits the transient in-sample skew tightly, while aBergomi cannot match that steepness (on 2024-08-05 it is pinned at $H = 0.250$ and $\rho = -0.99$ and still falls short), so its structural rigidity acts as an implicit regulariser. When the regime moves on the next day (a falling-vol mean reversion on 2024-08-05, an adversarial tariff jump on 2025-04-03), the lifted Heston fit tuned to the extreme in-sample skew generalises poorly in either direction, whereas aBergomi’s milder skew sits closer to the next-day surface. With a single out-of-sample day per date this is a plausible reading rather than a temporal-stability claim (section 8.9.6), but it is consistent across both high-volatility regimes and absent on the calm one.

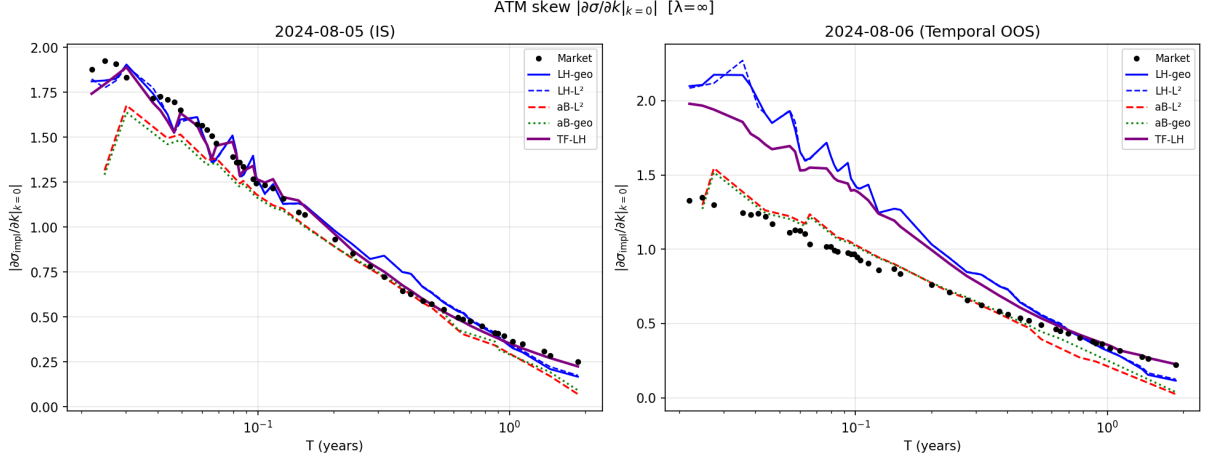


Figure 8.9: ATM skew term structure on 2024-08-05 under $\lambda = \infty$ (pure skew calibration). All models track the market closely in-sample; out-of-sample, single-factor LH and aB diverge in opposite directions, while TF-LH sits between them.

A striking observation across both calibration regimes is the H-identifiability gap of section 8.7 reappearing in H_1 on the two-factor model: on 2024-08-05 the calibrated H_1 moves from 0.020 at $\lambda = 0$ to 0.231 at $\lambda = \infty$; on 2024-12-04 from 0.030 to 0.286. The two-factor structure does not resolve the underlying gap that the single-factor models suffer; the value of H_1 that the optimiser selects still depends strongly on the loss. What the two-factor structure does provide is a better in-sample fit at either fixed value of H_1 , because the block-2 parameters absorb the residual smile features that no single H can capture alone. The gap is therefore an intrinsic property of the SPX $\mathbb{Q}$ -surface and not a single-factor artefact, in line with the discussion in section 8.7.

8.9.6 Caveats and limitations

We close the section with an honest accounting of what the empirical evidence supports and what it does not.

Sample size. Three train/test pairs are enough to show that the in-sample IV-RMSE improvement of TF-LH over the four single-factor baselines reproduces across distinct market regimes, but they are not enough to substantiate any quantitative temporal-stability claim. A rolling-window experiment over a quarter or more of trading days would be needed for that. The qualitative finding that the two-factor structure helps consistently is robust on the available evidence; the quantitative magnitudes are provisional.

Parameter count. TF-LH adds four parameters ($\lambda_2, \theta_2, \nu_2, \rho_2$) over the three-parameter LH baseline. The IV-RMSE improvement of 16–39% in-sample is sizeable but not dramatic, and a fair statistical comparison should include an information criterion (e.g. BIC on the full surface) to penalise the extra degrees of freedom. We do not perform this here; the empirical case is therefore for “the extra freedom is exploitable”, not yet for “the extra freedom is justified after penalty”.

Skew-RMSE at $\lambda = 0$. TF-LH does not uniformly beat aBergomi on intrinsic ATM-skew RMSE under IV calibration. On 2024-12-04, aBergomi achieves $\text{sk_IS} \approx 0.12$ while TF-LH stays at 0.53. Lognormal variance dynamics retain a structural advantage in fitting the SPX short-end skew level at $\lambda = 0$ that the two-factor extension of an affine CIR family does not fully close.

The TF-LH improvement is concentrated on IV-RMSE; users who prioritise skew level should still favour aBergomi on this dataset.

λ_2 versus the chapter’s prediction. As discussed above, the calibrated λ_2 sits in $[1.54, 9.89]$, outside the $[0.05, 0.3]$ range anticipated in section 6.2 and section 6.5. The mathematical content of proposition 6.12 remains correct (the decomposition into two independently-controlled contributions is the additional structural degree of freedom), but the qualitative slow-versus-fast interpretation should be revised to acknowledge that the optimiser, on SPX 2024–2025, prefers two distinct fast factors with independent leverages over the one-fast-and-one-slow configuration. A focused experiment with λ_2 bounded above at 0.3 would test whether the chapter’s mechanism, if forced, yields an acceptable fit.

The 2025-04-03 stress test. The OOS day in this pair includes US tariff news; OOS IV-RMSE blows up for every model by a factor of 2.5–4 relative to IS. This documents that the structural improvement of TF-LH is preserved even under a regime change (-39% IS becomes -9% OOS on IV-RMSE), but the OOS numbers should not be used in a statistical comparison of model quality.

Sign-constrained leverages. The bounds force $\rho_1, \rho_2 \in [-0.99, -0.01]$. The model is therefore not given the chance to test an opposite-sign configuration, which would be the mechanism by which two-factor Bergomi models occasionally fit the positive-skew commodity smiles. For SPX (always downward skew across maturities) this constraint is empirically safe but it is a model-imposed prior rather than an empirical finding.

Chapter 9

Extensions

9.1 Two structural extensions

Two structural variants of the lifted Heston framework deserve to be recorded as natural continuations of the work presented here. The first, which we call Voie C, is a minimally invasive modification of the canonical H -parametrisation; it can be implemented within the existing codebase in a few hours and provides a parsimonious complement to the two-factor model of chapter 6. The second is the matrix-valued Wishart construction, which represents the principled, but computationally demanding, resolution of the structural fragility identified in our analysis of correlated driving noises.

9.1.1 Voie C: bimodal Lévy measure

The canonical parametrisation of section 4.5 rests on the Laplace representation eq. (4.63) of the rough kernel K_H in terms of a single Lévy measure $\mu^H(dx) = x^{-H-1/2}/[\Gamma(1/2-H)\Gamma(H+1/2)] dx$ indexed by one Hurst parameter $H \in (0, 1/2)$. A natural and minimally invasive extension is to replace this unimodal measure by a convex combination

$$\mu(dx) = \alpha \mu^{H_1}(dx) + (1 - \alpha) \mu^{H_2}(dx), \quad \alpha \in [0, 1], \quad 0 < H_1 < H_2 \leq 1/2, \quad (9.1)$$

whose Laplace transform is the bimodal kernel

$$K(t) = \alpha \cdot \frac{t^{H_1-1/2}}{\Gamma(H_1+1/2)} + (1 - \alpha) \cdot \frac{t^{H_2-1/2}}{\Gamma(H_2+1/2)}. \quad (9.2)$$

Because K is a positive linear combination of completely monotone functions, it is itself completely monotone, and the admissibility conditions of [Abi Jaber, Larsson, and Pulido \[2019\]](#) apply unchanged. Discretisation proceeds by drawing the weights and speeds (c_i^n, x_i^n) from μ rather than from μ^H on the same geometric grid eq. (4.74)–eq. (4.75), so that the implementation requires changing only `params.py`: the Riccati solver, the characteristic function, and the COS pricer of section 7.3 are reused verbatim.

The financial motivation is to capture two timescales of roughness within a single lifted block: $K(t) \sim t^{H_1-1/2}$ as $t \rightarrow 0^+$ governs the short-maturity skew, while $K(t) \sim t^{H_2-1/2}$ as $t \rightarrow \infty$ governs the long-maturity decay. A particularly clean specialisation fixes $H_2 = 1/2$, in which case μ^{H_2} collapses to (a multiple of) the constant kernel and the model continuously interpolates between rough Heston ($\alpha = 1$) and classical Heston ($\alpha = 0$). The calibrated $\hat{\alpha}$ then provides a direct empirical answer to the question raised by [Cont and Das \[2023\]](#) of how much rough component the SPX surface actually requires.

Comparison with the two-factor lifted Heston of chapter 6. Voie C and the two-factor model of definition 6.1 both target scale-dependent behaviour, but through structurally different mechanisms.

- **Pros of Voie C.** The model remains a single-block lifted Heston with one Brownian motion W , one leverage parameter ρ , one vol-of-vol ν , and one mean-reversion pair (λ, θ) . It adds only two effective parameters (H_2, α) , or one if H_2 is fixed at $1/2$, over the standard lifted Heston, against the much larger parameter increment of the two-factor model. Strict affineness, the decoupled Riccati structure, and the entire codebase developed in chapter 7 are preserved with essentially zero engineering cost.
- **Cons of Voie C.** A single leverage parameter ρ cannot capture the empirical observation that the price–variance correlation appears to be stronger at short maturities than at long ones; this maturity-dependent leverage effect is precisely what the two-factor model of chapter 6 accommodates through independent ρ_1, ρ_2 . Furthermore, the three-parameter triple (H_1, H_2, α) raises a non-trivial identifiability question: many distinct bimodal kernels produce SPX implied volatility surfaces that are numerically indistinguishable on the available maturity range, so calibration is expected to be ill-conditioned without fixing one parameter (e.g., $H_2 = 1/2$) or imposing a regularisation. The two-factor model of chapter 6, by contrast, has every additional parameter tied to a directly interpretable financial quantity.

Voie C is therefore most useful as a complementary benchmark: calibrated alongside the two-factor model on the same SPX surface, it isolates the question of whether maturity-dependent leverage is empirically necessary, or whether a richer single-block kernel suffices.

9.1.2 Wishart-type matrix-valued variance

The analysis of correlation regimes in chapter 6 reveals a structural fragility: any nontrivial correlation between the variance drivers W^1, W^2 , or between the price-side noises $B^{(1)}, B^{(2)}$, introduces a $\sqrt{V_t^{(1)} V_t^{(2)}}$ cross term in the diffusion matrix of the joint state. This term is not affine in the scalar state $(V^{(1)}, V^{(2)})$, and the exponential-affine ansatz of theorem 6.7 fails to close into a finite-dimensional Riccati system. The principled resolution, going back to Bru (1991) and developed for option pricing by Da Fonseca, Grasselli, and Tebaldi [2008], is to lift the variance from a scalar to a matrix-valued process taking values in the cone $\mathcal{S}_d^+$ of $d \times d$ symmetric positive semidefinite matrices. The off-diagonal entries of the matrix state then carry the inter-factor correlation as a linear coordinate, dissolving the non-affine cross term of the scalar formulation.

Explicitly, the Wishart variance process

$$d\Sigma_t = (\Omega\Omega^\top + M\Sigma_t + \Sigma_t M^\top) dt + \sqrt{\Sigma_t} dB_t Q + Q^\top dB_t^\top \sqrt{\Sigma_t}, \quad \Sigma_t \in \mathcal{S}_d^+, \quad (9.3)$$

with B a $d \times d$ matrix of independent Brownian motions and $(\Omega, M, Q) \in \mathbb{R}^{d \times d}$ matrix parameters, is affine on $\mathcal{S}_d^+$ in the sense of Cuchiero, Filipović, Mayerhofer, and Teichmann [2011]. The Fourier–Laplace transform of $\log S_T$ is then exponentially affine in Σ_t via a matrix Riccati equation of the form

$$\partial_t \psi = -2\psi Q^\top Q \psi + \psi M + M^\top \psi, \quad \psi(0, u) = u, \quad (9.4)$$

which is finite-dimensional and numerically tractable modulo matrix exponential operations. The natural rough extension would be a lifted Wishart (or Volterra Wishart) model, in the framework of Cuchiero and Teichmann [2018], obtained by replacing the time-local integral of the drift and diffusion in eq. (9.3) by their Volterra convolution against a rough kernel K^H . The limiting object would be a genuinely rough, genuinely multifactor, genuinely correlated stochastic volatility model preserving affineness throughout.

Comparison and scope. The Wishart route is mathematically the correct generalisation in the sense that it removes the structural fragility of the scalar two-factor model of chapter 6 without compromise: positivity, affineness, and correlation between factors coexist by construction.

- **Pros.** The model accommodates an arbitrary correlation structure between d variance factors through the off-diagonals of Σ , retains a finite-dimensional Riccati equation (now matrix-valued), and produces dynamically evolving inter-factor covariances rather than static parameters.
- **Cons.** The parameter count is substantial (a 2×2 Wishart Heston already has Ω, M, Q as 2×2 matrices, that is twelve free parameters before the initial condition Σ_0), and many of these are unidentifiable from a single SPX surface without additional constraints. Numerically, the Riccati solver must be rewritten to handle matrix exponentials and matrix inversions, with attention to the conditioning of $\sqrt{\Sigma_t}$ near the boundary of $\mathcal{S}_d^+$; the lifted Volterra version compounds this with the convolution machinery of chapter 4. The literature is correspondingly demanding, drawing on functional analysis, Lie theory, and the abstract affine-process framework on cones.

For these reasons the Wishart extension lies beyond the scope of the present bachelor thesis and is recorded here as the principled direction in which the analytical limitations of the scalar two-factor model can be resolved. The choice made in chapter 6 to restrict attention to $\rho_{12} \in \{0, 1\}$ avoids the cross-quadratic pathology by structural design rather than by matrix lifting, at the cost of a less flexible correlation structure. A natural sequel to the present work would consist of formulating a two-dimensional lifted Wishart Heston and benchmarking it against both the scalar two-factor model of chapter 6 and the Voie C bimodal-measure construction of section 9.1.1 on a common SPX dataset.

Appendix A

Implementation validation on the Abi Jaber (2019) reference date (S&P 500, 2018-06-20)

A.1 Purpose and scope

This appendix is an out-of-sample implementation check on the reference date of [Abi Jaber \[2019\]](#): S&P 500 options on 2018-06-20, the date of that paper’s Figure 1, using CBOE end-of-day data (the `UnderlyingOptionsEODCalcs` file via DataShop). We treat it as deliberately not part of the empirical study of chapter 8, which is frozen on the three temporal-OOS pairs in 2024–2025: no calibration parameter or summary from this appendix feeds back into the main empirical comparison. We address two narrow questions: (i) does the data pipeline plus Black–Scholes IV inversion reproduce CBOE’s own reported implied volatilities on this date, and (ii) does the lifted Heston calibration of chapter 8 exhibit consistent behaviour (in particular the H -floor attractor of section 8.7) on a date entirely outside the 2024–2025 sample. In section A.5 we extend question (ii) to all five models of chapter 8 under the $\lambda = \infty$ skew objective, on the same date and with the same out-of-scope caveat.

A.2 Black–Scholes IV inversion cross-check

We compare per-option IV inversion (Brent’s method on the zero-rate Black–Scholes formula, the same routine that powers the surface cleaning of chapter 7) against CBOE’s reported `implied_volatility_1545`, using the same per-maturity put-call-parity forward on both sides. Of the 12 942 quote rows for 2018-06-20, 11 407 are eligible for comparison (positive bid and ask, positive mid, positive time-to-maturity, positive CBOE-reported IV), and 9 762 yield a finite inverted IV; the remaining inversions fail because the mid price lies outside the no-arbitrage bracket on illiquid wings. Across the 9 762 compared options we find a median absolute disagreement of 11 basis points of absolute volatility; restricted to the near-ATM bucket $|\log(K/S_0)| < 0.1$ the median is 7 bps:

within 0.1 vol-pt (10 bps)	47.6%
within 0.5 vol-pt (50 bps)	79.2%
within 1.0 vol-pt (100 bps)	86.2%
within 5.0 vol-pt (500 bps)	97.6%

The residual signed bias (overall mean -56 bps) is dragged by the OTM-put bucket, where wide bid-ask spreads on illiquid wing quotes produce mid prices that CBOE evidently smooths differently than the strict-midpoint convention used here. These options fail the surface-cleaning

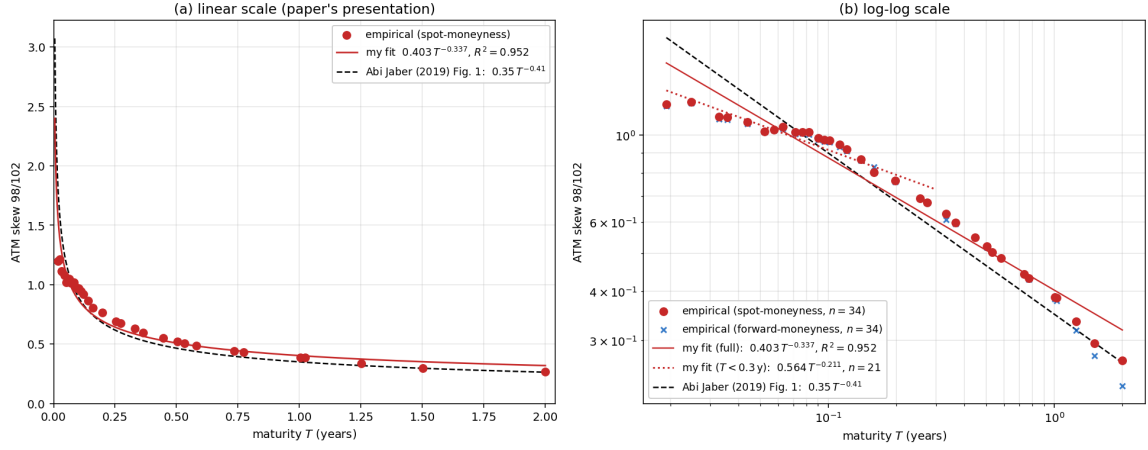


Figure A.1: S&P 500 ATM skew term structure on 2018-06-20, reproduction of [Abi Jaber \[2019\]](#) Figure 1. Two panels. **Panel (a)** linear-linear, matching the paper’s presentation: empirical spot-moneyness skew (red dots) against the full-range power-law fit $\hat{\psi}(T) = 0.403T^{-0.337}$ (red solid) and the paper’s reference $0.35T^{-0.41}$ (black dashed); the level offset (our fit sitting systematically above the paper’s reference over 1 m–2 y, as discussed in the text) is most directly visible here. **Panel (b)** log-log: the same series plus the forward-moneyness empirical skew (blue crosses, the thesis pipeline default) and the short-end-only fit $0.564T^{-0.211}$ over $T < 0.3$ y (red dotted), which resolves the sub-month plateau (the empirical departure from the single power law at the very short end) that is the discussion of section A.3.

quality filters of chapter 7 (notably the spread-below-50%-of-mid filter) and therefore do not enter the calibration loss; the relevant tolerance for downstream calibration is the near-ATM agreement above.

A.3 Empirical ATM skew versus Figure 1

We compute the 98/102 ATM skew on the cleaned surface in the spot-moneyness convention of [Abi Jaber \[2019\]](#) Figure 1: at each maturity T , we interpolate IV in log-strike to $k_{\text{spot}} = \log(0.98)$ and $k_{\text{spot}} = \log(1.02)$ (so $K = 0.98S_0$ and $K = 1.02S_0$, $S_0 = 2769.59$), and take the difference divided by $\Delta k = 0.04$.¹ Across the 34 cleaned maturities we fit the empirical skew by a single power law:

$$\hat{\psi}(T) = 0.403 \cdot T^{-0.337}, \quad R^2 = 0.95 \quad (n = 34).$$

We reproduce the shape of the paper’s reference $0.35 \cdot T^{-0.41}$, but not its level. The decay law is of the same family: both are single power laws with comparable exponents in the rough-Heston range, and the empirical points sit on our fit with $R^2 = 0.95$. Over the medium-to-long range from 1 month to 2 years, however, the empirical skew lies systematically above the paper’s reference curve (by a median of 11%, a mean of 9.5%, and a maximum of 14.5% near $T \approx 4$ months), with no maturity in that range agreeing to within 5% except the longest ($T = 1.5$ y, $T = 2$ y). The reproduction is one of term-structure shape and power-law form, not of overall level. We emphasise that this is a comparison against the paper’s fitted power-law curve, not against its raw data points, which are not available.

A clear short-end deviation is visible in fig. A.1: across the eleven sub-month expirations (7–30 days), the empirical skew sits in a plateau between 1.0 and 1.2, where the paper’s power-law

¹The thesis’s own pipeline uses forward-moneyness $k = \log(K/F_T)$; on this date the two conventions differ by about 3% on the fitted exponent. fig. A.1 overlays both.

extrapolation predicts roughly 1.77 at one week (rising to about 1.86 at five days). The plateau is not an artefact of our IV inversion: we re-compute the same 98/102 estimator using CBOE’s reported `implied_volatility_1545` directly (on the same strikes, bypassing the Brent inversion altogether) and recover per-maturity values within 0.005–0.02 of those reported here on every one of the eleven sub-month maturities. The plateau is therefore a feature of this date’s market data, not of our inversion. A short-end-only fit (restricting to $T < 0.3$ y, $n = 21$) gives a notably shallower exponent $\alpha = -0.211$, consistent with the plateau.

A.4 Calibration behaviour on this date

We calibrate the lifted Heston model with geometric kernel (LH-geo) at $\lambda = 0$ (pure vega-weighted IV-RMSE, the analogue of the paper’s surface fit) on this single date, using the same code path and the same thinning convention (8 log-spaced maturities $\times$ 15 strikes) as the chapter 8 three-pair work: $n = 20$, ξ_0 extracted from the surface’s ATM total-variance term structure, three free parameters (ν, ρ, H) with bounds $\nu \in [0.05, 3.0]$, $\rho \in [-0.99, 0]$, $H \in [0.02, 0.49]$, and seed 42. The optimiser pins H to the lower bound 0.02, with $\nu = 0.173$ and $\rho = -0.828$, yielding vega-weighted IV-RMSE 0.0259 on the calibration grid and 0.0252 on the full surface. We recognise this as the same H -floor attractor documented in section 8.7 for the 2024-12-04 LH-geo $\lambda = 0$ fit (see table 8.2): the 2018-06-20 calibration lands in the same regime as the calm and post-yen-carry-unwind dates of the three-pair panel.

To check that this H -floor result is not a consequence of the thesis’s geometric-kernel grid choice $r_n = 2.5$, we repeat the calibration with the paper’s geometric grid $r_n = 1 + 10 n^{-0.9} \approx 1.674$ (for $n = 20$), holding everything else fixed. We find the same outcome: H still pins to 0.02, with $\nu = 0.189$ and $\rho = -0.830$, and the in-sample vega-weighted IV-RMSE moves from 0.0259 to 0.0262 on the thin grid and from 0.0252 to 0.0255 on the full surface, a $\sim 1\%$ degradation. The calibrated short-end skew at one week falls from 2.04 under $r_n = 2.5$ to 1.75 under $r_n = 1.674$ (visible in fig. A.2), reducing the mean overshoot relative to the empirical plateau by about 13% and the maximum overshoot at one week by about 17%; but the model still sits well above the empirical short-end of 1.20. We conclude that the H -floor attractor and the short-end overshoot are not artefacts of the chosen kernel grid.

Two qualifications are in order. First, we do not reproduce the configuration of [Abi Jaber \[2019\]](#): that paper calibrates six parameters $(V_0, \theta, \lambda_{\text{mr}}, \nu, \rho, H)$ against a variance-swap-derived ξ_0 , whereas the thesis uses a three-parameter forward-variance form with ξ_0 extracted from the surface’s own ATM total-variance term structure (chapter 7). The two configurations are not directly comparable at the parameter level, and the agreement reported in figs. A.1 and A.2 is at the level of the empirical skew curve, not of the calibrated parameters. Second, our calibrated model overshoots the empirical short-end plateau, but so does the paper’s own reference power law (both giving roughly 1.75–1.77 at one week against an empirical 1.20). The $T^{H-1/2}$ scaling is an asymptotic property of the rough-Heston limit, and the eleven sub-month maturities on this date evidently sit in a finite-sample regime where that asymptote is not fully realised; we read this as a feature of the empirical surface, not a defect of the implementation. The narrow point we validate in this appendix is the one stated in section A.1: the chapter 8 calibrator, on a date outside the 2024–2025 sample, exhibits the same H -floor calibration behaviour as on the three temporal-OOS pairs.

A.5 Five-model comparison under the skew objective

We calibrate all five models of chapter 8 (LH-geo, LH-L², aB-geo, aB-L², and the two-factor TF-LH) to the 2018-06-20 surface under the pure ATM-skew objective ($\lambda = \infty$), holding the configuration otherwise fixed (geometric grid $r_n = 2.5$, the thin calibration grid, seed 42). This is a single-date comparison and is not part of the temporal out-of-sample study of chapter 8; we

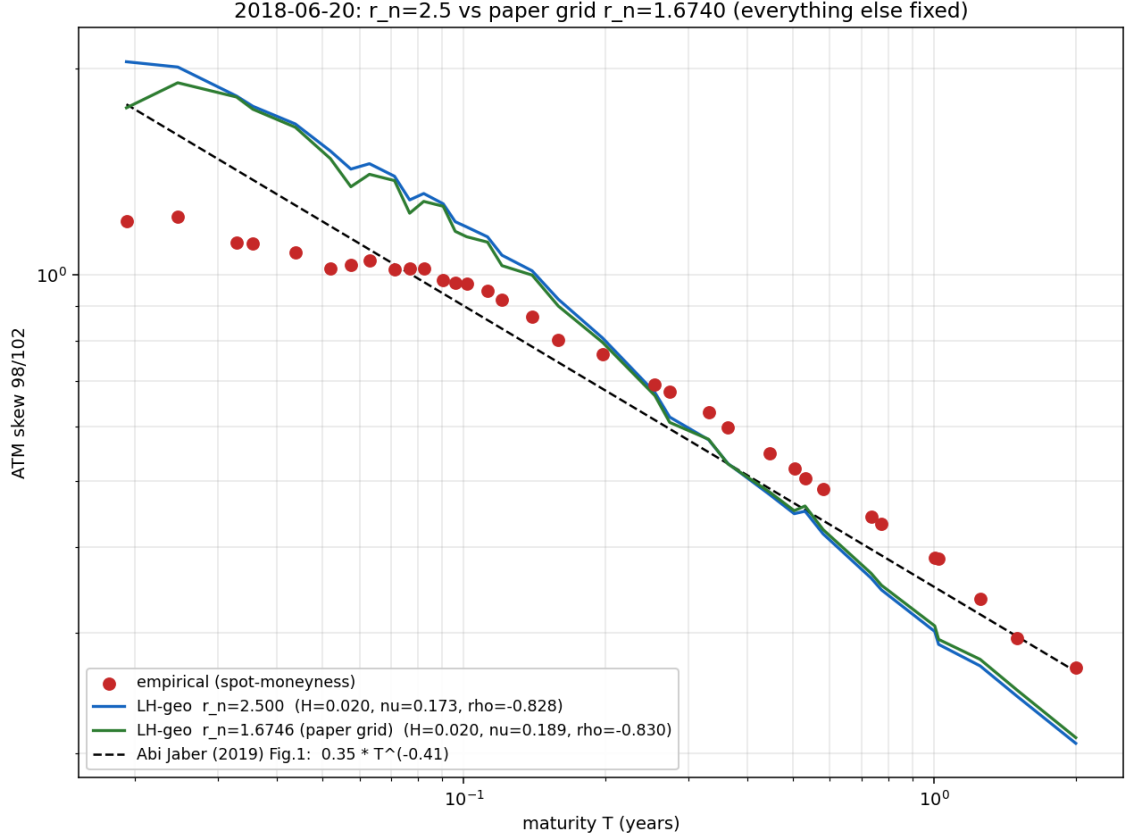


Figure A.2: Kernel-grid control on 2018-06-20: calibrated LH-geo ATM skew term structure under the thesis-standard $r_n = 2.500$ (blue) and the paper’s $r_n = 1 + 10 n^{-0.9} \approx 1.674$ for $n = 20$ (green), against the empirical skew (red dots) and the paper’s reference $0.35 T^{-0.41}$ (black dashed). Everything else (surface, ξ_0 , three-parameter form, objective, bounds, seed) is held fixed; r_n is the only difference. Log-log axes.

use it to test whether the model behaviour seen on the 2024–2025 panel recurs on an independent date.

Under the IV-RMSE objective of section A.4 the calibrated H pins to its lower bound (≈ 0.02) and the single-kernel model overshoots the empirical short-end skew; under the skew objective the calibrated H is instead pushed high: LH-geo 0.42, LH- L^2 0.37, TF-LH $H_1 = 0.28$, the aBergomi variants 0.13–0.16. With H high, the models reproduce the empirical short-end plateau: the seven-day model skew is 1.14–1.16 against an empirical 1.20, compared with the overshoot to ≈ 2.04 under $\lambda = 0$. The same model thus returns $H \approx 0.02$ or $H \approx 0.4$ from the same surface depending only on the loss, the H -identifiability gap documented in section 8.7 reappearing on this date.

TF-LH achieves a skew-RMSE of 0.009, roughly five to ten times lower than the single-factor models (LH-geo 0.067, LH- L^2 0.097, aB-geo 0.045, aB- L^2 0.041), and tracks the empirical skew across maturities (fig. A.3), consistent with the two-factor advantage at $\lambda = \infty$ reported in chapter 8. We stress the scope: this is a single date under the skew objective only; it does not establish a general ranking (no IV-RMSE or out-of-sample comparison is made here) and it does not alter the conclusions of the main study.

Two caveats apply to the aBergomi calibrations. First, the correlation ρ saturates at its lower bound in both variants (aB-geo -0.990 , aB- L^2 -0.979), the same ρ -saturation seen in the chapter 8 $\lambda = \infty$ calibrations, so the aBergomi fits are driven by a boundary-pinned correlation. Second, both aBergomi variants return invalid prices at the seven-day expiration (a limitation of

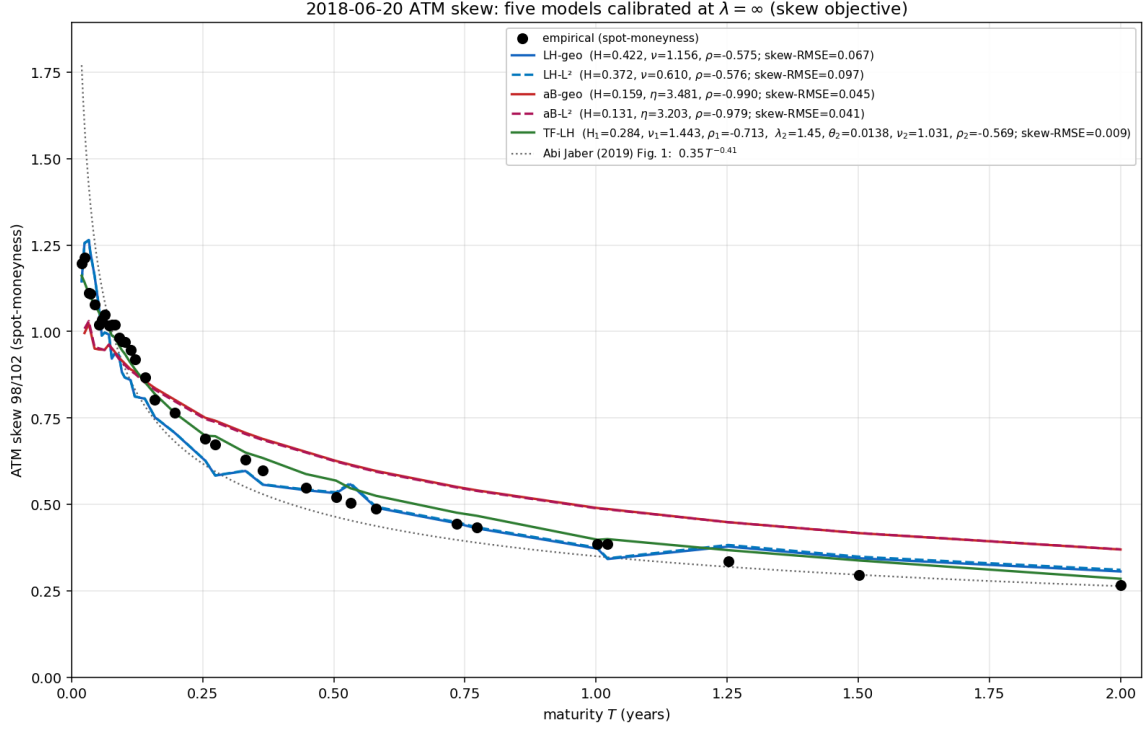


Figure A.3: S&P 500 ATM skew term structure on 2018-06-20: the five models of chapter 8 calibrated at $\lambda = \infty$ (pure ATM-skew RMSE), linear axes. Empirical spot-moneyness skew as black dots; the five calibrated-model skew curves, LH-geo (blue solid), LH- L^2 (blue dashed), aB-geo (red solid), aB- L^2 (magenta dashed), and TF-LH (green solid), with each legend entry reporting the calibrated parameters and achieved skew-RMSE; the paper’s reference $0.35 T^{-0.41}$ is shown as a faint dotted line. The aBergomi variants return invalid Monte-Carlo prices at the seven-day expiration and are not shown there; all longer maturities evaluate normally. TF-LH (green) tracks the empirical skew most closely across maturities.

the Monte-Carlo pricer at very short maturity combined with low H) and are excluded from that single maturity in fig. A.3; all longer maturities evaluate normally. The aBergomi calibrations are also far more costly (≈ 2.5 – 2.8 hours each, versus minutes for the lifted-Heston models), consistent with the runtime gap in chapter 8.

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